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Taming Transformers for High-Resolution Image Synthesis

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Figure 1. Our approach enables transformers to synthesize high-resolution images like this one, which contains 1280x460 pixels.

Abstract

Designed to learn long-range interactions on sequential data, transformers continue to show state-of-the-art results on a wide variety of tasks. In contrast to CNNs, they contain no inductive bias that prioritizes local interactions. This makes them expressive, but also computationally infeasible for long sequences, such as high-resolution images. We demonstrate how combining the effectiveness of the inductive bias of CNNs with the expressivity of transformers enables them to model and thereby synthesize high-resolution images. We show how to (i) use CNNs to learn a context-rich vocabulary of image constituents, and in turn (ii) utilize transformers to efficiently model their composition within high-resolution images. Our approach is readily applied to conditional synthesis tasks, where both non-spatial information, such as object classes, and spatial information, such as segmentations, can control the generated image. In particular, we present the first results on semantically-guided synthesis of megapixel images with transformers and obtain the state of the art among autoregressive models on class-conditional ImageNet. Code and pretrained models can be found at https://git.io/JnyvK.

1. Introduction

Transformers are on the rise—they are now the de-facto standard architecture for language tasks [74, 57, 58, 5] and are increasingly adapted in other areas such as audio [12] and vision [8, 16]. In contrast to the predominant vision architecture, convolutional neural networks (CNNs), the transformer architecture contains no built-in inductive prior on the locality of interactions and is therefore free to learn complex relationships among its inputs. However, this generality also implies that it *has to* learn all relationships, whereas CNNs have been designed to exploit prior knowledge about strong local correlations within images. Thus, the increased expressivity of transformers comes with quadratically increasing computational costs, because all pairwise interactions are taken into account. The resulting energy and time requirements of state-of-the-art transformer models thus pose fundamental problems for scaling them to high-resolution images with millions of pixels.

Observations that transformers tend to learn convolutional structures [16] thus beg the question: Do we have to re-learn everything we know about the local structure and regularity of images from scratch each time we train a vision model, or can we efficiently encode inductive image biases while still retaining the flexibility of transformers? We hypothesize that low-level image structure is well described by a local connectivity, i.e. a convolutional architecture, whereas this structural assumption ceases to be effective on higher semantic levels. Moreover, CNNs not only exhibit a strong locality bias, but also a bias towards spatial invariance through the use of shared weights across

驯服Transformer模型以实现高分辨率图像合成

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图1. 我们的方法 h使得变换器模型能够生成如这般分辨率高达1280x460像素的高清图像。 s.

摘要

Designed to learn long-range interactions on sequential data, transformers continue to show state-of-the-art results on a wide variety of tasks. In contrast to CNNs, they contain no inductive bias that prioritizes local interactions. This makes them expressive, but also computationally infeasible for long sequences, such as high-resolution images. We demonstrate how combining the effectiveness of the inductive bias of CNNs with the expressivity of transformers enables them to model and thereby synthesize high-resolution images. We show how to (i) use CNNs to learn a context-rich vocabulary of image constituents, and in turn (ii) utilize transformers to efficiently model their composition within high-resolution images. Our approach is readily applied to conditional synthesis tasks, where both non-spatial information, such as object classes, and spatial information, such as segmentations, can control the generated image. In particular, we present the first results on semantically-guided synthesis of megapixel images with transformers and obtain the state of the art among autoregressive models on class-conditional ImageNet. Code and pretrained models can be found at https://git.io/JnyvK.

1. 引言

Transformer模型正日益普及，如今它们已然成为语言处理任务领域的事实标准架构 [74, 57, 58, 5]。这类模型也越来越被应用于音频[12] 和视觉 [8, 16]等其他领域。与主流的视觉架构——卷积神经网络（CNNs）不同，变换器架构并未内置关于元素交互局部性的先验知识，因此能够自由学习输入数据之间复杂的关联关系。不过，这种通用性也意味着它必须自行学习所有的关联关系，而CNNs则是为利用图像中强烈的局部相关性这一先验知识而设计的。正因如此，尽管变换器模型的表达能力更强，但其计算成本却呈二次方增长趋势，因为所有成对交互都需要被纳入考虑。由此带来的高昂能耗和时间开销，给将这类最先进的变换器模型应用到拥有数百万像素的高分辨率图像上带来了根本性的挑战。

有研究观察到Transformer模型倾向于学习卷积结构 [16] ，这一现象引发了这样的疑问：每次训练视觉模型时，我们是否都必须从头重新学习关于图像局部结构与规律性的所有知识？或者，我们能否在保持Transformer模型灵活性的同时，高效地编码出具有归纳性的图像偏好？我们的假设是，低层图像结构可以通过局部连接性来很好地描述，也就是采用卷积架构；而这种结构假设在更高层次的语义表达上则不再适用。此外，卷积神经网络不仅具有强烈的局部性偏好，还通过在各位置之间共享权重的方式展现出对空间不变性的偏好，正因如此，当需要更全面地理解输入内容时，这类架构就显得不够有效了。

all positions. This makes them ineffective if a more holistic understanding of the input is required.

Our key insight to obtain an effective and expressive model is that, *taken together, convolutional and transformer architectures can model the compositional nature of our visual world [51]*: We use a convolutional approach to efficiently learn a codebook of context-rich visual parts and, subsequently, learn a model of their global compositions. The long-range interactions within these compositions require an expressive transformer architecture to model distributions over their consituent visual parts. Furthermore, we utilize an adversarial approach to ensure that the dictionary of local parts captures perceptually important local structure to alleviate the need for modeling low-level statistics with the transformer architecture. Allowing transformers to concentrate on their unique strength—modeling long-range relations—enables them to generate high-resolution images as in Fig. 1, a feat which previously has been out of reach. Our formulationgives control over the generated images by means of conditioning information regarding desired object classes or spatial layouts. Finally, experiments demonstrate that our approach retains the advantages of transformers by outperforming previous codebook-based state-of-the-art approaches based on convolutional architectures.

2. Related Work

The Transformer Family The defining characteristic of the transformer architecture [74] is that it models interactions between its inputs solely through attention [2, 36, 52] which enables them to faithfully handle interactions between inputs regardless of their relative position to one another. Originally applied to language tasks, inputs to the transformer were given by tokens, but other signals, such as those obtained from audio [41] or images [8], can be used. Each layer of the transformer then consists of an attention mechanism, which allows for interaction between inputs at different positions, followed by a position-wise fully connected network, which is applied to all positions independently. More specifically, the (self-)attention mechanism can be described by mapping an intermediate representation with three position-wise linear layers into three representations, query $Q \in \mathbb{R}^{N \times d_k}$, key $K \in \mathbb{R}^{N \times d_k}$ and value $V \in \mathbb{R}^{N \times d_v}$, to compute the output as

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^t}{\sqrt{d_k}}\right)V \in \mathbb{R}^{N \times d_v}. \quad (1)$$

When performing autoregressive maximum-likelihood learning, non-causal entries of QK^t, i.e. all entries below its diagonal, are set to $-\infty$ and the final output of the transformer is given after a linear, point-wise transformation to predict logits of the next sequence element. Since the attention mechanism relies on the computation of inner products between all pairs of elements in the sequence, its computational complexity increases quadratically with the sequence length. While the ability to consider interactions between *all* elements is the reason transformers efficiently learn long-range interactions, it is also the reason transformers quickly become infeasible, especially on images, where the sequence length itself scales quadratically with the resolution. Different approaches have been proposed to reduce the computational requirements to make transformers feasible for longer sequences. [55] and [76] restrict the receptive fields of the attention modules, which reduces the expressivity and, especially for high-resolution images, introduces assumptions on the independence of pixels. [12] and [26] retain the full receptive field but can reduce costs for a sequence of length n only from n^2 to $n\sqrt{n}$, which makes resolutions beyond 64 pixels still prohibitively expensive.

Convolutional Approaches The two-dimensional structure of images suggests that local interactions are particularly important. CNNs exploit this structure by restricting interactions between input variables to a local neighborhood defined by the kernel size of the convolutional kernel. Applying a kernel thus results in costs that scale linearly with the overall sequence length (the number of pixels in the case of images) and quadratically in the kernel size, which, in modern CNN architectures, is often fixed to a small constant such as 3×3. This inductive bias towards local interactions thus leads to efficient computations, but the wide range of specialized layers which are introduced into CNNs to handle different synthesis tasks [53, 80, 68, 85, 84] suggest that this bias is often too restrictive.

Convolutional architectures have been used for autoregressive modeling of images [70, 71, 10] but, for low-resolution images, previous works [55, 12, 26] demonstrated that transformers consistently outperform their convolutional counterparts. Our approach allows us to efficiently model high-resolution images with transformers while retaining their advantages over state-of-the-art convolutional approaches.

Two-Stage Approaches Closest to ours are two-stage approaches which first learn an encoding of data and afterwards learn, in a second stage, a probabilistic model of this encoding. [13] demonstrated both theoretical and empirical evidence on the advantages of first learning a data representation with a Variational Autoencoder (VAE) [38, 62], and then again learning its distribution with a VAE. [18, 78] demonstrate similar gains when using an unconditional normalizing flow for the second stage, and [63, 64] when using a conditional normalizing flow. To improve training efficiency of Generative Adversarial Networks (GANs), [43] learns a GAN [20] on representations of an autoencoder and [21] on low-resolution wavelet coefficients which are then

因此，当需要更全面的输入理解时，这些架构便不再具备有效性。

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2. 相关工作

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$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^t}{\sqrt{d_k}}\right)V \in \mathbb{R}^{N \times d_v}. \quad (1)$$

在执行自回归最大似然学习时， QK^t中非因果性的元素，即对角线以下的所有元素，会被设为 $-\infty$ ，随后通过对结果进行线性逐点变换，即可得到变换器模型的最终输出，用以预测下一个序列元素的logits值。由于注意力机制依赖于内积的计算

products between all pairs of elements in the sequence, its computational complexity increases quadratically with the sequence length. While the ability to consider interactions between *all* elements is the reason transformers efficiently learn long-range interactions, it is also the reason transformers quickly become infeasible, especially on images, where the sequence length itself scales quadratically with the resolution. Different approaches have been proposed to reduce the computational requirements to make transformers feasible for longer sequences. [55] and [76] restrict the receptive fields of the attention modules, which reduces the expressivity and, especially for high-resolution images, introduces assumptions on the independence of pixels. [12] and [26] retain the full receptive field but can reduce costs for a sequence of length n only from n^2 to $n\sqrt{n}$, which makes resolutions beyond 64 pixels still prohibitively expensive.

卷积方法 图像的二维结构表明局部交互尤为重要。卷积神经网络利用这一结构，将输入变量之间的交互限制在由卷积核大小所定义的局部邻域内。因此，应用卷积核时产生的计算成本会与整体序列长度（对于图像而言即像素数量）成线性关系，同时与卷积核大小成二次方关系；而在现代卷积神经网络架构中，卷积核大小通常被设定为 3×3这样的较小常数。这种偏向局部交互的归纳偏置虽然能够提升计算效率，但由于卷积神经网络为处理不同的合成任务而引入了众多各类专用层 [53, 80, 68, 85, 84]，这类偏置往往显得过于严格。

卷积架构已被用于图像的自回归建模 [70, 71, 10]，但对于低分辨率图像，先前的研究 [55, 12, 26] 表明，变换器模型在性能上始终优于传统的卷积模型。我们的方法能够利用变换器模型高效地对高分辨率图像进行建模，同时保留其相较于现有最先进卷积方法的各项优势。

两阶段方法与我们的方法最为接近的是两阶段方法，这类方法首先学习数据的编码方式，然后在第二阶段构建该编码方式的概率模型。 [13]通过理论和实证研究证明了，先使用变分自编码器（VAE）学习数据表示，再利用VAE学习其分布，能够带来显著优势。[18, 78]表明，若在第二阶段采用无条件归一化流，也能获得类似的优化效果； [63, 64] 而使用条件归一化流时同样如此。为提升生成对抗网络（GAN模型）的训练效率， [43]在自编码器的表示结果以及低分辨率小波系数基础上学习了aGAN [20]，随后

Figure 2. Our approach uses a convolutional *VQGAN* to learn a codebook of context-rich visual parts, whose composition is subsequently modeled with an autoregressive transformer architecture. A discrete codebook provides the interface between these architectures and a patch-based discriminator enables strong compression while retaining high perceptual quality. This method introduces the efficiency of convolutional approaches to transformer based high resolution image synthesis.

decoded to images with a learned generator.

[72] presents the Vector Quantised Variational Autoencoder (VQVAE), an approach to learn discrete representations of images, and models their distribution autoregressively with a convolutional architecture. [61] extends this approach to use a hierarchy of learned representations. However, these methods still rely on convolutional density estimation, which makes it difficult to capture long-range interactions in high-resolution images. [8] models images autoregressively with transformers in order to evaluate the suitability of generative pretraining to learn image representations for downstream tasks. Since input resolutions of 32×32 pixels are still quite computationally expensive [8], a VQVAE is used to encode images up to a resolution of 192×192. In an effort to keep the learned discrete representation as spatially invariant as possible with respect to the pixels, a shallow VQVAE with small receptive field is employed. In contrast, we demonstrate that a powerful first stage, which captures as much context as possible in the learned representation, is critical to enable efficient high-resolution image synthesis with transformers.

3. Approach

Our goal is to exploit the highly promising learning capabilities of transformer models [74] and introduce them to high-resolution image synthesis up to the megapixel range. Previous work [55, 8] which applied transformers to image generation demonstrated promising results for images up to a size of 64×64 pixels but, due to the quadratically increasing cost in sequence length, cannot simply be scaled to higher resolutions.

High-resolution image synthesis requires a model that understands the global composition of images, enabling it to generate locally realistic as well as globally consistent patterns. Therefore, instead of representing an image with pixels, we represent it as a composition of perceptually rich image constituents from a codebook. By learning an effective code, as described in Sec. 3.1, we can significantly reduce the description length of compositions, which allows us to efficiently model their global interrelations within images with a transformer architecture as described in Sec. 3.2. This approach, summarized in Fig. 2, is able to generate realistic and consistent high resolution images both in an unconditional and a conditional setting.

3.1. Learning an Effective Codebook of Image Constituents for Use in Transformers

To utilize the highly expressive transformer architecture for image synthesis, we need to express the constituents of an image in the form of a *sequence*. Instead of building on individual pixels, complexity necessitates an approach that uses a discrete codebook of learned representations, such that any image $x \in \mathbb{R}^{H \times W \times 3}$ can be represented by a spatial collection of codebook entries $z_{\mathbf{q}} \in \mathbb{R}^{h \times w \times n_z}$, where n_z is the dimensionality of codes. An equivalent representation is a sequence of $h \cdot w$ indices which specify the respective entries in the learned codebook. To effectively learn such a discrete spatial codebook, we propose to directly incorporate the inductive biases of CNNs and incorporate ideas from neural discrete representation learning [72]. First, we learn a convolutional model consisting of an encoder E and a decoder G, such that taken together, they learn to represent images with codes from a learned, discrete codebook $\mathcal{Z} = \{z_k\}_{k=1}^{K} \subset \mathbb{R}^{n_z}$ (see Fig. 2 for an overview). More

图2. 我们的方法采用卷积型*VQGAN*模型来学习包含丰富上下文信息的视觉元素构成的码本，随后再利用自回归变换器架构对这类元素的组合进行建模。离散码本充当了这两种架构之间的接口，而基于图像块的判别器则能够在保持高感知质量的同时实现强大的压缩效果。该方法将卷积式方法的效率引入到了基于变换器模型的高分辨率图像合成领域。

再通过学到的生成器将这些数据解码为图像。

[72] presents the Vector Quantised Variational Autoencoder (VQVAE), an approach to learn discrete representations of images, and models their distribution autoregressively with a convolutional architecture. [61] extends this approach to use a hierarchy of learned representations. However, these methods still rely on convolutional density estimation, which makes it difficult to capture long-range interactions in high-resolution images. [8] models images autoregressively with transformers in order to evaluate the suitability of generative pretraining to learn image representations for downstream tasks. Since input resolutions of 32×32 pixels are still quite computationally expensive [8], a VQVAE is used to encode images up to a resolution of 192×192. In an effort to keep the learned discrete representation as spatially invariant as possible with respect to the pixels, a shallow VQVAE with small receptive field is employed. In contrast, we demonstrate that a powerful first stage, which captures as much context as possible in the learned representation, is critical to enable efficient high-resolution image synthesis with transformers.

3. 方法

我们的目标是充分利用Transformer模型极具潜力的学习能力 [74]，并将其应用于高达百万像素级别的高分辨率图像合成领域。此前已有研究 [55, 8] 将Transformer模型用于图像生成，其在图像尺寸为 64×64 像素时已展现出良好的效果，但由于序列长度带来的成本呈二次方增长，这类方法无法直接扩展到更高的分辨率场景。

高分辨率图像合成需要一个能够实现该任务的模型能够理解图像的整体构成，从而生成在局部上真实且在全球范围内保持一致的模式。因此，我们不再用像素来表示图像，而是将其视为由码本中那些具有丰富感知特性的图像组件所构成的整体。正如第3.1节所述，通过学习有效的码本，3.1，我们可以大幅缩短这类组合的描述长度，进而借助第3.2节所述的变换器架构，高效地建模图像中这些组成部分之间的全局关联。3.2。如图2所示，2，这种方法无论是在无条件还是有条件的环境下，都能生成真实且一致的高分辨率图像。

3.1. 学习适用于Transformer模型的高效图像构成元素码本

To utilize the highly expressive transformer architecture for image synthesis, we need to express the constituents of an image in the form of a *sequence*. Instead of building on individual pixels, complexity necessitates an approach that uses a discrete codebook of learned representations, such that any image $x \in \mathbb{R}^{H \times W \times 3}$ can be represented by a spatial collection of codebook entries $z_{\mathbf{q}} \in \mathbb{R}^{h \times w \times n_z}$, where n_z is the dimensionality of codes. An equivalent representation is a sequence of $h \cdot w$ indices which specify the respective entries in the learned codebook. To effectively learn such a discrete spatial codebook, we propose to directly incorporate the inductive biases of CNNs and incorporate ideas from neural discrete representation learning [72]. First, we learn a convolutional model consisting of an encoder E and a decoder G, such that taken together, they learn to represent images with codes from a learned, discrete codebook $\mathcal{Z} = \{z_k\}_{k=1}^{K} \subset \mathbb{R}^{n_z}$ (see Fig. 2 for an overview). More

precisely, we approximate a given image x by $\hat{x} = G(z_{\mathbf{q}})$. We obtain $z_{\mathbf{q}}$ using the encoding $\hat{z} = E(x) \in \mathbb{R}^{h\times w\times n_z}$ and a subsequent element-wise quantization $\mathbf{q}(\cdot)$ of each spatial code $\hat{z}_{ij} \in \mathbb{R}^{n_z}$ onto its closest codebook entry z_k:

$$z_{\mathbf{q}} = \mathbf{q}(\hat{z}) \coloneqq \left(\underset{z_k\in\mathcal{Z}}{\arg\min} \|\hat{z}_{ij} - z_k\| \right) \in \mathbb{R}^{h\times w\times n_z}. \quad (2)$$

The reconstruction $\hat{x} \approx x$ is then given by

$$\hat{x} = G(z_{\mathbf{q}}) = G\left(\mathbf{q}(E(x))\right). \quad (3)$$

Backpropagation through the non-differentiable quantization operation in Eq. (3) is achieved by a straight-through gradient estimator, which simply copies the gradients from the decoder to the encoder [3], such that the model and codebook can be trained end-to-end via the loss function

$$\mathcal{L}_{\text{VQ}}(E, G, \mathcal{Z}) = \|x - \hat{x}\|^2 + \|\text{sg}[E(x)] - z_{\mathbf{q}}\|_2^2 + \|\text{sg}[z_{\mathbf{q}}] - E(x)\|_2^2. \quad (4)$$

Here, $\mathcal{L}_{\text{rec}} = \|x-\hat{x}\|^2$ is a reconstruction loss, $\text{sg}[\cdot]$ denotes the stop-gradient operation, and $\|\text{sg}[z_{\mathbf{q}}] - E(x)\|_2^2$ is the so-called "commitment loss" [72].

Learning a Perceptually Rich Codebook Using transformers to represent images as a distribution over latent image constituents requires us to push the limits of compression and learn a rich codebook. To do so, we propose *VQGAN*, a variant of the original VQVAE, and use a discriminator and perceptual loss [40, 30, 39, 17, 47] to keep good perceptual quality at increased compression rate. Note that this is in contrast to previous works which applied pixel-based [71, 61] and transformer-based autoregressive models [8] on top of only a shallow quantization model. More specifically, we replace the L_2 loss used in [72] for $\mathcal{L}_{\text{rec}}$ by a perceptual loss and introduce an adversarial training procedure with a patch-based discriminator D [28] that aims to differentiate between real and reconstructed images:

$$\mathcal{L}_{\text{GAN}}(\{E, G, \mathcal{Z}\}, D) = [\log D(x) + \log(1 - D(\hat{x}))] \quad (5)$$

The complete objective for finding the optimal compression model $\mathcal{Q}^* = \{E^*, G^*, \mathcal{Z}^*\}$ then reads

$$\mathcal{Q}^* = \underset{E,G,\mathcal{Z}}{\arg\min} \max_{D} \mathbb{E}_{x\sim p(x)} \Big[\mathcal{L}_{\text{VQ}}(E, G, \mathcal{Z}) + \lambda \mathcal{L}_{\text{GAN}}(\{E, G, \mathcal{Z}\}, D) \Big], \quad (6)$$

where we compute the adaptive weight λ according to

$$\lambda = \frac{\nabla_{G_L}[\mathcal{L}_{\text{rec}}]}{\nabla_{G_L}[\mathcal{L}_{\text{GAN}}] + \delta} \quad (7)$$

where $\mathcal{L}_{\text{rec}}$ is the perceptual reconstruction loss [81], $\nabla_{G_L}[\cdot]$ denotes the gradient of its input w.r.t. the last layer L of the decoder, and $\delta = 10^{-6}$ is used for numerical stability. To aggregate context from everywhere, we apply a single attention layer on the lowest resolution. This training procedure significantly reduces the sequence length when unrolling the latent code and thereby enables the application of powerful transformer models.

3.2. Learning the Composition of Images with Transformers

Latent Transformers With E and G available, we can now represent images in terms of the codebook-indices of their encodings. More precisely, the quantized encoding of an image x is given by $z_{\mathbf{q}} = \mathbf{q}(E(x)) \in \mathbb{R}^{h\times w\times n_z}$ and is equivalent to a sequence $s \in \{0, \dots, |\mathcal{Z}|-1\}^{h\times w}$ of indices from the codebook, which is obtained by replacing each code by its index in the codebook $\mathcal{Z}$:

$$s_{ij} = k \text{ such that } \left(z_{\mathbf{q}}\right)_{ij} = z_k. \quad (8)$$

By mapping indices of a sequence s back to their corresponding codebook entries, $z_{\mathbf{q}} = \left(z_{s_{ij}}\right)$ is readily recovered and decoded to an image $\hat{x} = G(z_{\mathbf{q}})$.

Thus, after choosing some ordering of the indices in s, image-generation can be formulated as autoregressive next-index prediction: Given indices $s_{<i}$, the transformer learns to predict the distribution of possible next indices, i.e. $p(s_i|s_{<i})$ to compute the likelihood of the full representation as $p(s) = \prod_i p(s_i|s_{<i})$. This allows us to directly maximize the log-likelihood of the data representations:

$$\mathcal{L}_{\text{Transformer}} = \mathbb{E}_{x\sim p(x)} \left[-\log p(s) \right]. \quad (9)$$

Conditioned Synthesis In many image synthesis tasks a user demands control over the generation process by providing additional information from which an example shall be synthesized. This information, which we will call c, could be a single label describing the overall image class or even another image itself. The task is then to learn the likelihood of the sequence given this information c:

$$p(s|c) = \prod_i p(s_i|s_{<i}, c). \quad (10)$$

If the conditioning information c has spatial extent, we first learn another *VQGAN* to obtain again an index-based representation $r \in \{0, \dots, |\mathcal{Z}_c|-1\}^{h_c\times w_c}$ with the newly obtained codebook $\mathcal{Z}_c$ Due to the autoregressive structure of the transformer, we can then simply prepend r to s and restrict the computation of the negative log-likelihood to entries $p(s_i|s_{<i}, r)$. This "decoder-only" strategy has also been successfully used for text-summarization tasks [44].

Generating High-Resolution Images The attention mechanism of the transformer puts limits on the sequence

具体而言，我们通过$\hat{x} = G(z_{\mathbf{q}})$来近似给定的图像 x。我们首先使用编码$\hat{z} = E(x) \in \mathbb{R}^{h\times w\times n_z}$获得 $z_{\mathbf{q}}$，随后对每个空间码$\hat{z}_{ij} \in \mathbb{R}^{n_z}$进行逐元素量化$\mathbf{q}(\cdot)$，将其映射到最接近的码本条目 z_k上。

$$z_{\mathbf{q}} = \mathbf{q}(\hat{z}) \coloneqq \left(\underset{z_k\in\mathcal{Z}}{\arg\min} \|\hat{z}_{ij} - z_k\| \right) \in \mathbb{R}^{h\times w\times n_z}. \quad (2)$$

此时，重构结果$\hat{x} \approx x$ 可由下式给出

$$\hat{x} = G(z_{\mathbf{q}}) = G\left(\mathbf{q}(E(x))\right). \quad (3)$$

通过直通梯度估计器来实现对方程(3)中不可微量化操作的反向传播，该估计器仅将梯度从解码器复制到编码器 [3]，，从而能够借助损失函数对模型与码本进行端到端的训练

$$\mathcal{L}_{\text{VQ}}(E, G, \mathcal{Z}) = \|x - \hat{x}\|^2 + \|\text{sg}[E(x)] - z_{\mathbf{q}}\|_2^2 + \|\text{sg}[z_{\mathbf{q}}] - E(x)\|_2^2. \quad (4)$$

此处，$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|^2$ 表示重构损失，$\text{sg}[\cdot]$ 代表梯度停止操作，而 $\|\text{sg}[z_{\mathbf{q}}] - E(x)\|_2^2$则是所谓的“承诺损失” [72]。

Learning a Perceptually Rich Codebook Using transformers to represent images as a distribution over latent image constituents requires us to push the limits of compression and learn a rich codebook. To do so, we propose *VQGAN*, a variant of the original VQVAE, and use a discriminator and perceptual loss [40, 30, 39, 17, 47] to keep good perceptual quality at increased compression rate. Note that this is in contrast to previous works which applied pixel-based [71, 61] and transformer-based autoregressive models [8] on top of only a shallow quantization model. More specifically, we replace the L_2 loss used in [72] for $\mathcal{L}_{\text{rec}}$ by a perceptual loss and introduce an adversarial training procedure with a patch-based discriminator D [28] that aims to differentiate between real and reconstructed images:

$$\mathcal{L}_{\text{GAN}}(\{E, G, \mathcal{Z}\}, D) = [\log D(x) + \log(1 - D(\hat{x}))] \quad (5)$$

寻找最优压缩模型 $\mathcal{Q}^* = \{E^*, G^*, \mathcal{Z}^*\}$ 的完整目标表述如下

$$\mathcal{Q}^* = \underset{E,G,\mathcal{Z}}{\arg\min} \max_{D} \mathbb{E}_{x\sim p(x)} \Big[\mathcal{L}_{\text{VQ}}(E, G, \mathcal{Z}) + \lambda \mathcal{L}_{\text{GAN}}(\{E, G, \mathcal{Z}\}, D) \Big], \quad (6)$$

在这里，我们根据情况计算自适应权重 λ。

$$\lambda = \frac{\nabla_{G_L}[\mathcal{L}_{\text{rec}}]}{\nabla_{G_L}[\mathcal{L}_{\text{GAN}}] + \delta} \quad (7)$$

其中 $\mathcal{L}_{\text{rec}}$ 表示感知重构损失 [81]，$\nabla_{G_L}[\cdot]$则表示该损失相对于最终层 L 的输入梯度。

$\delta = 10^{-6}$ 被用于提升数值稳定性。为从各个位置聚合上下文信息，我们会在最低分辨率层面应用一个注意力层。这种训练流程在展开潜码时能显著缩短序列长度，从而使得强大的变换器模型得以应用。

3.2. 利用Transformer模型学习图像的构图规律

潜空间变换器模型随着 E 和 G 的可用，我们现在能够用图像编码对应的码本索引来表示图像。更准确地说，图像的量化编码 x 由 $z_{\mathbf{q}} = \mathbf{q}(E(x)) \in \mathbb{R}^{h\times w\times n_z}$ 给出，它等价于一个由码本索引构成的序列 $s \in \{0, \dots, |\mathcal{Z}|-1\}^{h\times w}$，该序列是通过将每个码替换为它在码本中的索引而获得的 $\mathcal{Z}$:

$$s_{ij} = k \text{ such that } \left(z_{\mathbf{q}}\right)_{ij} = z_k. \quad (8)$$

通过将序列 s 中的索引映射回对应的码本条目，即可轻松获取 $z_{\mathbf{q}} = \left(z_{s_{ij}}\right)$，并将其解码为图像$\hat{x} = G(z_{\mathbf{q}})$。

因此，在为s中的索引选定某种排序方式后，图像生成可以被构建为自回归式下一个索引预测：给定索引 $s_{<i}$，变换器模型会学习预测可能的下一个索引的分布，即 $p(s_i|s_{<i})$，以此来计算完整表示 $p(s) = \prod_i p(s_i|s_{<i})$的概率密度。这样一来，我们就可以直接最大化数据表示的对数概率密度。

$$\mathcal{L}_{\text{Transformer}} = \mathbb{E}_{x\sim p(x)} \left[-\log p(s) \right]. \quad (9)$$

条件合成 在许多图像合成任务中，用户会通过提供用于生成示例的额外信息来要求对生成过程进行控制。这类信息我们将其称为 c，它可能是一个描述整体图像类别的单一标签，甚至可以是另一张图像本身。此时的任务就是学习在给定这些信息 c的情况下该序列出现的概率。

$$p(s|c) = \prod_i p(s_i|s_{<i}, c). \quad (10)$$

如果条件作用信息 c 具有空间范围，我们首先学习另一个 *VQGAN* 模型，利用新得到的码本 $\mathcal{Z}_c$ 再次生成基于索引的表示 $r \in \{0, \dots, |\mathcal{Z}_c|-1\}^{h_c\times w_c}$。由于变换器模型具有自回归结构，我们随后只需将 r 加到 s 的前面，并将负对数似然的计算限制在条目 $p(s_i|s_{<i}, r)$ 范围内。这种“仅解码器”策略也已成功应用于文本摘要任务 [44]中。

生成高分辨率图像 变换器模型的注意力机制会对序列长度设置限制

Figure 3. Sliding attention window.

length $h \cdot w$ of its inputs s. While we can adapt the number of downsampling blocks m of our *VQGAN* to reduce images of size $H \times W$ to $h = H/2^m \times w = W/2^m$, we observe degradation of the reconstruction quality beyond a critical value of m, which depends on the considered dataset. To generate images in the megapixel regime, we therefore have to work patch-wise and crop images to restrict the length of s to a maximally feasible size during training. To sample images, we then use the transformer in a sliding-window manner as illustrated in Fig. 3. Our *VQGAN* ensures that the available context is still sufficient to faithfully model images, as long as either the statistics of the dataset are approximately spatially invariant or spatial conditioning information is available. In practice, this is not a restrictive requirement, because when it is violated, *i.e*. unconditional image synthesis on aligned data, we can simply condition on image coordinates, similar to [42].

4. Experiments

This section evaluates the ability of our approach to retain the advantages of transformers over their convolutional counterparts (Sec. 4.1) while integrating the effectiveness of convolutional architectures to enable high-resolution image synthesis (Sec. 4.2). Furthermore, in Sec. 4.3, we investigate how codebook quality affects our approach. We close the analysis by providing a quantitative comparison to a wide range of existing approches for generative image synthesis in Sec. 4.4. Based on initial experiments, we usually set $|\mathcal{Z}| = 1024$ and train all subsequent transformer models to predict sequences of length $16 \cdot 16$, as this is the maximum feasible length to train a GPT2-medium architecture (307 M parameters) [58] on a GPU with 12GB VRAM. More details on architectures and hyperparameters can be found in the appendix (Tab. 7 and Tab. 8).

4.1. Attention Is All You Need in the Latent Space

Transformers show state-of-the-art results on a wide variety of tasks, including autoregressive image modeling. However, evaluations of previous works were limited to transformers working directly on (low-resolution) pixels [55, 12, 26], or to deliberately shallow pixel encodings [8]. This raises the question if our approach retains the advantages of transformers over convolutional approaches.

To answer this question, we use a variety of conditional and unconditional tasks and compare the performance between our transformer-based approach and a convolutional approach. For each task, we train a *VQGAN* with $m = 4$ downsampling blocks, and, if needed, another one for the conditioning information, and then train both a transformer and a PixelSNAIL [10] model on the same representations, as the latter has been used in previous state-of-the-art two-stage approaches [61]. For a thorough comparison, we vary the model capacities between 85M and 310M parameters and adjust the number of layers in each model to match one another. We observe that PixelSNAIL trains roughly twice as fast as the transformer and thus, for a fair comparison, report the negative log-likelihood both for the same amount of training time (*P-SNAIL time*) and for the same amount of training steps (*P-SNAIL steps*).

Negative Log-Likelihood (NLL)

Data / # params	Transformer *P-SNAIL steps*	Transformer *P-SNAIL time*	PixelSNAIL *fixed time*
RIN / 85M	**4.78**	4.84	4.96
LSUN-CT / 310M	**4.63**	4.69	4.89
IN / 310M	**4.78**	4.83	4.96
D-RIN / 180 M	**4.70**	4.78	4.88
S-FLCKR / 310 M	**4.49**	4.57	4.64

Table 1. Comparing Transformer and PixelSNAIL architectures across different datasets and model sizes. For all settings, transformers outperform the state-of-the-art model from the PixelCNN family, PixelSNAIL in terms of NLL. This holds both when comparing NLL at fixed times (PixelSNAIL trains roughly 2 times faster) and when trained for a fixed number of steps. See Sec. 4.1 for the abbreviations.

Results Tab. 1 reports results for unconditional image modeling on *ImageNet* (IN) [14], *Restricted ImageNet* (RIN) [65], consisting of a subset of animal classes from ImageNet, *LSUN Churches and Towers* (LSUN-CT) [79], and for conditional image modeling of RIN conditioned on depth maps obtained with the approach of [60] (D-RIN) and of landscape images collected from Flickr conditioned on semantic layouts (S-FLCKR) obtained with the approach of [7]. Note that for the semantic layouts, we train the first-stage using a cross-entropy reconstruction loss due to their discrete nature. The results shows that the transformer consistently outperforms PixelSNAIL across all tasks when trained for the same amount of time and the gap increases even further when trained for the same number of steps. These results demonstrate that gains of transformers carry over to our proposed two-stage setting.

4.2. A Unified Model for Image Synthesis Tasks

The versatility and generality of the transformer architecture makes it a promising candidate for image synthesis. In the conditional case, additional information c such as class labels or segmentation maps are used and the goal is to learn the distribution of images as described in Eq. (10). Using the same setting as in Sec. 4.1 (i.e. image size 256×256, latent size 16×16), we perform various conditional image synthesis experiments:

图3. 滑动注意力窗口。

length $h \cdot w$ of its inputs s. While we can adapt the number of downsampling blocks m of our *VQGAN* to reduce images of size $H \times W$ to $h = H/2^m \times w = W/2^m$, we observe degradation of the reconstruction quality beyond a critical value of m, which depends on the considered dataset. To generate images in the megapixel regime, we therefore have to work patch-wise and crop images to restrict the length of s to a maximally feasible size during training. To sample images, we then use the transformer in a sliding-window manner as illustrated in Fig. 3. Our *VQGAN* ensures that the available context is still sufficient to faithfully model images, as long as either the statistics of the dataset are approximately spatially invariant or spatial conditioning information is available. In practice, this is not a restrictive requirement, because when it is violated, *i.e*. unconditional image synthesis on aligned data, we can simply condition on image coordinates, similar to [42].

4. 实验

This section evaluates the ability of our approach to retain the advantages of transformers over their convolutional counterparts (Sec. 4.1) while integrating the effectiveness of convolutional architectures to enable high-resolution image synthesis (Sec. 4.2). Furthermore, in Sec. 4.3, we investigate how codebook quality affects our approach. We close the analysis by providing a quantitative comparison to a wide range of existing approches for generative image synthesis in Sec. 4.4. Based on initial experiments, we usually set $|\mathcal{Z}| = 1024$ and train all subsequent transformer models to predict sequences of length $16 \cdot 16$, as this is the maximum feasible length to train a GPT2-medium architecture (307 M parameters) [58] on a GPU with 12GB VRAM. More details on architectures and hyperparameters can be found in the appendix (Tab. 7 and Tab. 8).

4.1. 隐空间中的IS指标就是你所需的一切

Transformer模型在包括自回归图像建模在内的各类任务中均展现出最先进的性能。然而，以往的研究评估仅限于直接作用于（低分辨率）像素的变换器模型[55, 12, 26]，或那些刻意设计得较为浅层的像素编码方式[8]。这就引发了这样一个问题：我们的方法是否依然保留着变换器模型相较于卷积方法的优势。

为回答这一问题，我们采用了多种有条件与无条件的任务，对比了基于变换器模型的方法与卷积方法的表现。对于每项任务，我们会训练一个包含 $m = 4$个下采样块的*VQGAN* 模型；如有需要，还会再训练一个用于通过进行条件作用处理，再分别使用这些表示结果来训练变换器模型与PixelSNAIL [10]模型，因为后者已被用于以往最先进的二阶段方法中 [61]。为了进行全面对比，我们将模型的参数量设置在8500万到3.1亿之间，并调整各模型的层数使其保持一致。实验结果表明，PixelSNAIL的训练速度大约是变换器模型的两倍；因此，为保证公平性，我们会针对相同训练时间（*P-SNAIL*时间）以及相同训练步长（*P-SNAIL*步骤）的情况分别报告负对数似然值。

负对数似然（NLL）

数据 / # 参数	变换器模型 *P-SNAIL*步骤	变换器模型 *P-SNAIL*时间	PixelSNAIL 固定时长
RIN / 85M	**4.78**	4.84	4.96
LSUN-CT / 310M	**4.63**	4.69	4.89
IN数据集 / 3.1亿	**4.78**	4.83	4.96
D-RIN / 1.8亿	**4.70**	4.78	4.88
S-FLCKR平台 / 3.1亿	**4.49**	4.57	4.64

表1. 不同数据集及模型规模下Transformer架构与PixelSNAIL架构的对比。在所有测试设置中，就负对数似然值而言，Transformer的性能均优于PixelCNN系列中的最先进模型PixelSNAIL。这一结论在固定训练时间进行比较时依然成立（此时PixelSNAIL的训练速度大约是Transformer的2倍），在固定训练步数进行比较时也同样适用。详情见第4.1 节，其中介绍了相关缩写含义。

Results Tab. 1 reports results for unconditional image modeling on *ImageNet* (IN) [14], *Restricted ImageNet* (RIN) [65], consisting of a subset of animal classes from ImageNet, *LSUN Churches and Towers* (LSUN-CT) [79], and for conditional image modeling of RIN conditioned on depth maps obtained with the approach of [60] (D-RIN) and of landscape images collected from Flickr conditioned on semantic layouts (S-FLCKR) obtained with the approach of [7]. Note that for the semantic layouts, we train the first-stage using a cross-entropy reconstruction loss due to their discrete nature. The results shows that the transformer consistently outperforms PixelSNAIL across all tasks when trained for the same amount of time and the gap increases even further when trained for the same number of steps. These results demonstrate that gains of transformers carry over to our proposed two-stage setting.

4.2. 用于图像合成任务的统一模型

变换器模型的通用性与强大适应性使其成为图像合成领域极具潜力的候选方案。在存在条件约束的情况下， c 会引入诸如类别标签或分割图之类的额外信息，其目标便是如方程（10）中所描述的那样，学习图像的分布规律。我们采用与第4.1 节相同的设置（即图像尺寸 256×256、潜在空间尺寸 16×16），开展了多种有条件图像合成实验：

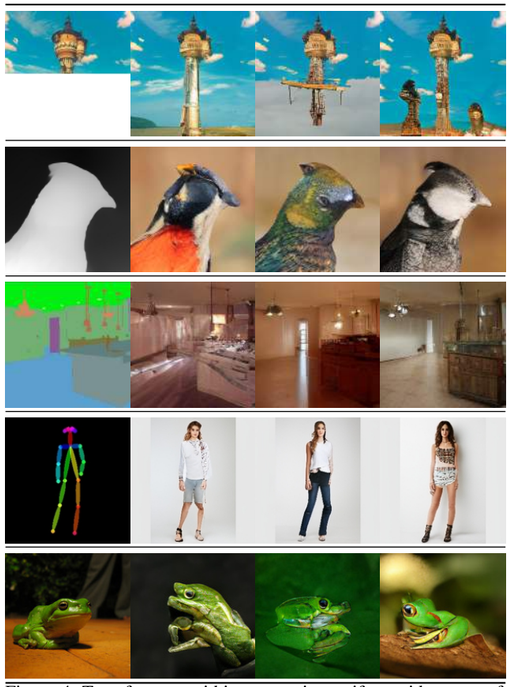


Figure 4. Transformers within our setting unify a wide range of image synthesis tasks. We show 256×256 synthesis results across different conditioning inputs and datasets, all obtained with the same approach to exploit inductive biases of effective CNN based *VQGAN* architectures in combination with the expressivity of transformer architectures. Top row: Completions from unconditional training on ImageNet. 2nd row: Depth-to-Image on RIN. 3rd row: Semantically guided synthesis on ADE20K. 4th row: Pose-guided person generation on DeepFashion. Bottom row: Class-conditional samples on RIN.

(i): **Semantic image synthesis**, where we condition on semantic segmentation masks of ADE20K [83], a web-scraped landscapes dataset (S-FLCKR) and COCO-Stuff [6]. Results are depicted in Figure 4, 5 and Fig. 6.

(ii): **Structure-to-image**, where we use either depth or edge information to synthesize images from both RIN and IN (see Sec. 4.1). The resulting depth-to-image and edge-to-image translations are visualized in Fig. 4 and Fig. 6.

(iii): **Pose-guided synthesis:** Instead of using the semantically rich information of either segmentation or depth maps, Fig. 4 shows that the same approach as for the previous experiments can be used to build a shape-conditional generative model on the DeepFashion [45] dataset.

(iv): **Stochastic superresolution**, where low-resolution images serve as the conditioning information and are thereby upsampled. We train our model for an upsampling factor of 8 on ImageNet and show results in Fig. 6.

(v): **Class-conditional image synthesis:** Here, the conditioning information c is a single index describing the class label of interest. Results for the RIN and IN dataset are demonstrated in Fig. 4 and Fig. 8, respectively.

All of these examples make use of the same methodology. Instead of requiring task specific architectures or modules, the flexibility of the transformer allows us to learn appropriate interactions for each task, while the *VQGAN* — which can be *reused* across different tasks — leads to short sequence lengths. In combination, the presented approach can be understood as an efficient, general purpose mechanism for conditional image synthesis. Note that additional results for each experiment can be found in the appendix, Sec. D.

High-Resolution Synthesis The sliding window approach introduced in Sec. 3.2 enables image synthesis beyond a resolution of 256×256 pixels. We evaluate this approach on unconditional image generation on LSUN-CT and FacesHQ (see Sec. 4.3) and conditional synthesis on D-RIN, COCO-Stuff and S-FLCKR, where we show results in Fig. 1, 6 and the supplementary (Fig. 29-39). Note that this approach can in principle be used to generate images of arbitrary ratio and size, given that the image statistics of the dataset of interest are approximately spatially invariant or spatial information is available. Impressive results can be achieved by applying this method to image generation from semantic layouts on S-FLCKR, where a strong *VQGAN* can be learned with $m = 5$, so that its codebook together with the conditioning information provides the transformer with enough context for image generation in the megapixel regime.

4.3. Building Context-Rich Vocabularies

How important are context-rich vocabularies? To investigate this question, we ran experiments where the transformer architecture is kept fixed while the amount of context encoded into the representation of the first stage is varied through the number of downsampling blocks of our *VQGAN*. We specify the amount of context encoded in terms of reduction factor in the side-length between image inputs and the resulting representations, *i.e.* a first stage encoding images of size $H \times W$ into discrete codes of size $H/f \times W/f$ is denoted by a factor f. For $f = 1$, we reproduce the approach of [8] and replace our *VQGAN* by a k-means clustering of RGB values with $k = 512$.

During training, we always crop images to obtain inputs of size 16×16 for the transformer, *i.e.* when modeling images with a factor f in the first stage, we use crops of size $16f \times 16f$. To sample from the models, we always apply them in a sliding window manner as described in Sec. 3.

Results Fig. 7 shows results for unconditional synthesis of faces on *FacesHQ*, the combination of *CelebA-HQ* [31] and

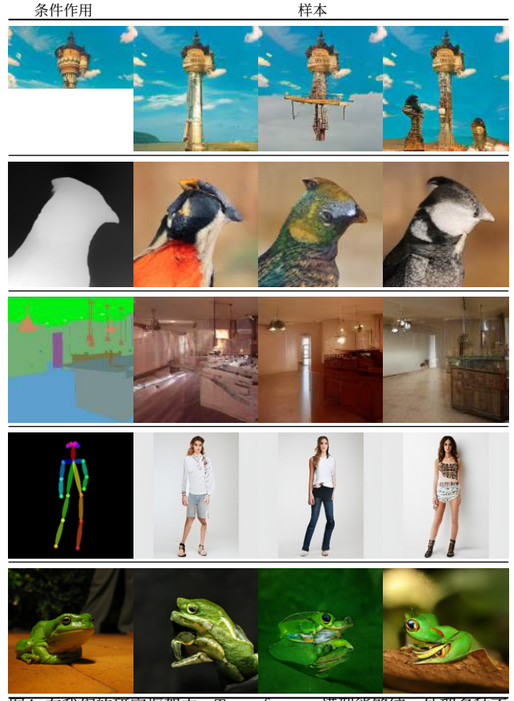


图4. 在我们的研究框架中，Transformer模型能够统一处理多种不同的图像合成任务。我们展示了在各种不同条件作用输入及数据集下获得的 256×256 合成结果，所有结果均是采用相同方法得出的——该方法充分利用了基于有效CNN结构的*VQGAN* 架构的归纳偏置，同时结合了变换器架构的表达能力。最上方一行：在ImageNet数据集上通过无条件训练得到的图像补全结果；第二行：在RIN模型上实现的深度转图像任务；第三行：在ADE20K数据集上进行的语义引导式合成；第四行：在DeepFashion数据集上实现的姿态引导式人物生成；最下方一行：在RIN模型上生成的带类别条件的样本。

(i): **语义图像合成**，其中我们以ADE20K数据集的语义分割掩码作为条件 [83]，以及通过网页爬取获得的风景数据集（S-FLCKR）和COCO-Stuff数据集[6]。实验结果展示在图4、图5以及图6中。

(ii): **结构转图像**，在该方法中我们利用深度信息或边缘信息，通过RIN和IN数据集来合成图像（详见第4.1节）。由此得到的深度转图像任务与边缘转图像任务的生成结果分别展示在图4和图6中。

(iii): **基于姿态引导的合成：** 图4并未采用分割图或深度图所具备的语义丰富信息，而是表明可以利用与先前实验相同的方法，在DeepFashion [45]数据集上构建一个基于形状条件的生成模型。

(iv): **随机超分辨率任务**，其中低分辨率图像作为条件作用信息被用于该过程。

对图像进行上采样处理。我们在ImageNet数据集上以8倍的上采样因子训练模型，并在图6中展示实验结果。

(v): **类别条件图像合成：** 此处，条件信息 c 是一个用于表示目标类别标签的单一索引。RIN与IN数据集的实验结果分别展示在图4和图8中。

所有这些示例都采用了相同的方法论。由于变换器模型的灵活性，我们无需依赖特定任务所需的架构或模块，即可为每项任务学习出合适的交互方式；而*VQGAN* ——这种模型可以被重复使用于不同的任务中——则使得序列长度较短。综合来看，所提出的方法可被视为一种用于有条件图像合成的高效、通用机制。请注意，各实验的更多结果可在附录的第D节中找到。

High-Resolution Synthesis The sliding window approach introduced in Sec. 3.2 enables image synthesis beyond a resolution of 256×256 pixels. We evaluate this approach on unconditional image generation on LSUN-CT and FacesHQ (see Sec. 4.3) and conditional synthesis on D-RIN, COCO-Stuff and S-FLCKR, where we show results in Fig. 1, 6 and the supplementary (Fig. 29-39). Note that this approach can in principle be used to generate images of arbitrary ratio and size, given that the image statistics of the dataset of interest are approximately spatially invariant or spatial information is available. Impressive results can be achieved by applying this method to image generation from semantic layouts on S-FLCKR, where a strong *VQGAN* can be learned with $m = 5$, so that its codebook together with the conditioning information provides the transformer with enough context for image generation in the megapixel regime.

4.3. 构建富含上下文的词汇表

How important are context-rich vocabularies? To investigate this question, we ran experiments where the transformer architecture is kept fixed while the amount of context encoded into the representation of the first stage is varied through the number of downsampling blocks of our *VQGAN*. We specify the amount of context encoded in terms of reduction factor in the side-length between image inputs and the resulting representations, *i.e.* a first stage encoding images of size $H \times W$ into discrete codes of size $H/f \times W/f$ is denoted by a factor f. For $f = 1$, we reproduce the approach of [8] and replace our *VQGAN* by a k-means clustering of RGB values with $k = 512$.

在训练过程中，我们始终会对图像进行裁剪，从而获得大小为 16×16 的输入数据，供变换器模型使用，也就是说。在第一阶段，使用 f 倍比例对图像进行建模时，我们采用的裁剪图像尺寸为 $16f \times 16f$。为了从这些模型中采样，我们始终按照第3节中所描述的滑动窗口方式来应用它们。

结果图。7 展示了在*FacesHQ*上无条件合成人脸的*CelebA-HQ* [31] 与

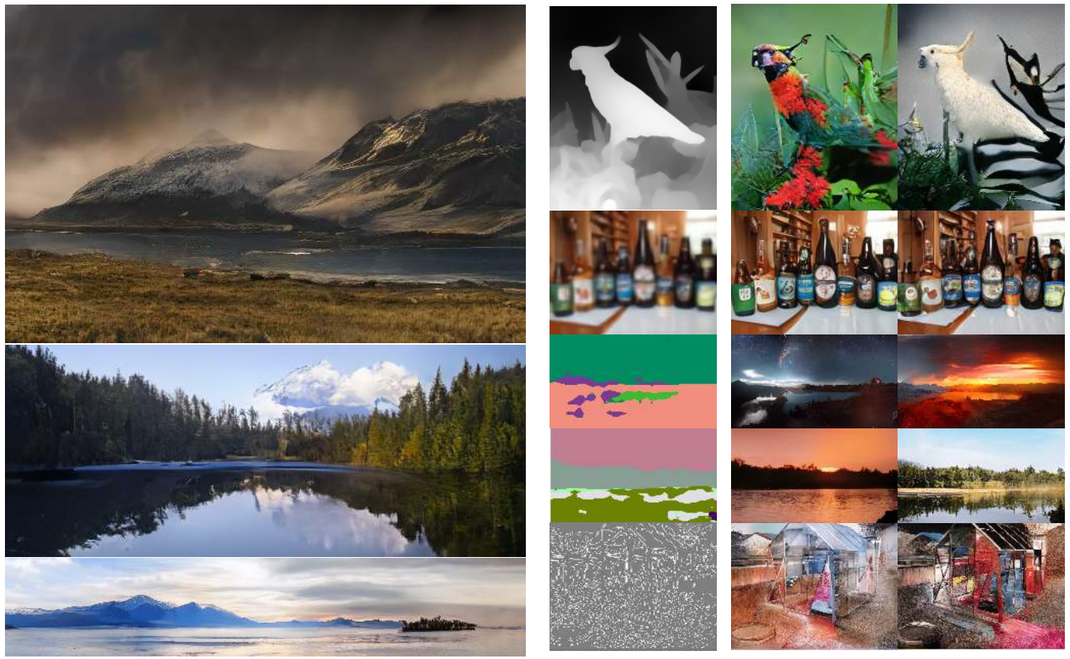


Figure 5. Samples generated from semantic layouts on S-FLCKR. Sizes from top-to-bottom: 1280×832, 1024×416 and 1280×240 pixels. Best viewed zoomed in. A larger visualization can be found in the appendix, see Fig 29.

Figure 6. Applying the sliding attention window approach (Fig. 3) to various conditional image synthesis tasks. Top: Depth-to-image on RIN, 2nd row: Stochastic superresolution on IN, 3rd and 4th row: Semantic synthesis on S-FLCKR, bottom: Edge-guided synthesis on IN. The resulting images vary between 368×496 and 1024×576, hence they are best viewed zoomed in.

FFHQ [33]. It clearly demonstrates the benefits of powerful *VQGAN*s by increasing the effective receptive field of the transformer. For small receptive fields, or equivalently small f, the model cannot capture coherent structures. For an intermediate value of $f = 8$, the overall structure of images can be approximated, but inconsistencies of facial features such as a half-bearded face and of viewpoints in different parts of the image arise. Only our full setting of $f = 16$ can synthesize high-fidelity samples. For analogous results in the conditional setting on S-FLCKR, we refer to the appendix (Fig. 13 and Sec. C).

To assess the effectiveness of our approach quantitatively, we compare results between training a transformer directly on pixels, and training it on top of a *VQGAN*'s latent code with $f = 2$, given a fixed computational budget. Again, we follow [8] and learn a dictionary of 512 RGB values on CIFAR10 to operate directly on pixel space and train the same transformer architecture on top of our *VQGAN* with a latent code of size $16 \times 16 = 256$. We observe improvements of 18.63% for FIDs and $14.08\times$ faster sampling of images.

Dataset	**ours**	SPADE [53]	Pix2PixHD (+aug) [75]	CRN [9]
COCO-Stuff	22.4	22.6/23.9(*)	111.5 (54.2)	70.4
ADE20K	35.5	33.9/35.7(*)	81.8 (41.5)	73.3

Table 2. FID score comparison for semantic image synthesis (256×256 pixels). (*): Recalculated with our evaluation protocol based on [50] on the validation splits of each dataset.

4.4. Benchmarking Image Synthesis Results

In this section we investigate how our approach quantitatively compares to existing models for generative image synthesis. In particular, we assess the performance of our model in terms of FID and compare to a variety of established models (GANs, VAEs, Flows, AR, Hybrid). The results on *semantic synthesis* are shown in Tab. 2, where we compare to [53, 75, 35, 9], and the results on *unconditional face synthesis* are shown in Tab. 3. While some task-specialized GAN models report better FID scores, our approach provides a unified model that works well across a wide range of tasks while retaining the ability to encode and reconstruct images. It thereby bridges the gap between purely adversarial and likelihood-based approaches.

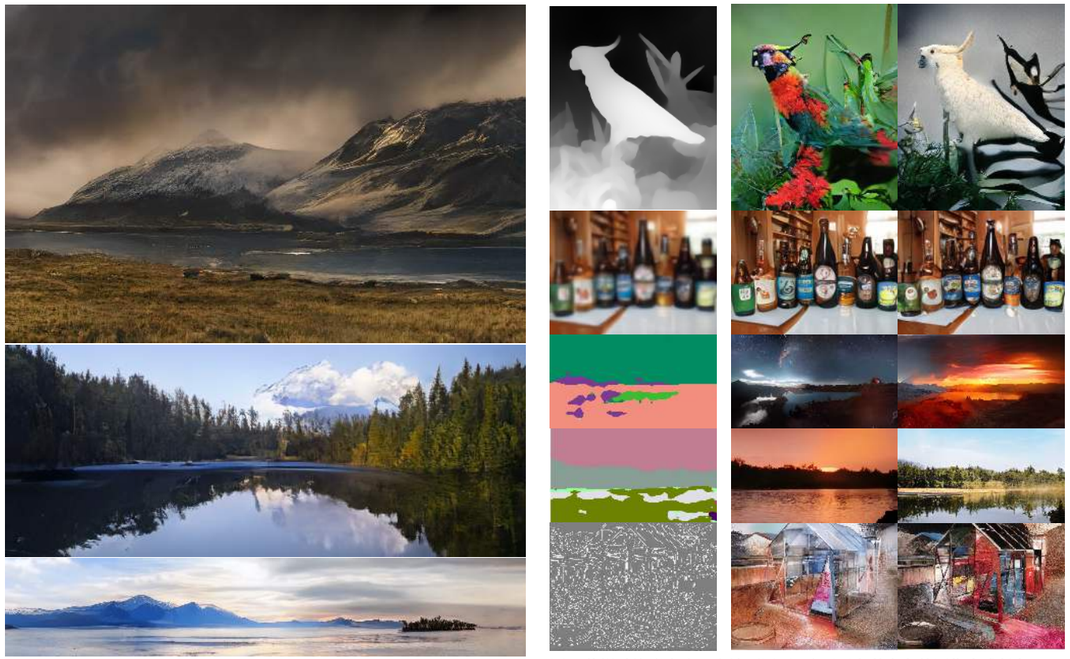


图5. 在S-FLCKR平台上基于语义布局生成的样本。从上到下的尺寸分别为：1280×832 1024×416 $1280 \times$，以及像素值。240
建议放大查看以获得最佳效果。更详细的可视化内容可见附录中的图29。

图6. 将滑动注意力窗口方法（图 3）应用于各种有条件图像合成任务。最上方：在RIN数据集上进行的深度转图像任务；第二行：在IN数据集上进行的随机超分辨率任务；第三行和第四行：在S-FLCKR平台上进行的语义合成任务；最下方：在IN数据集上基于边缘信息的合成任务。生成图像的分辨率在 368×496 与1024×576之间变化，因此建议放大查看以获得最佳效果。

FFHQ [33]. It clearly demonstrates the benefits of powerful *VQGAN*s by increasing the effective receptive field of the transformer. For small receptive fields, or equivalently small f, the model cannot capture coherent structures. For an intermediate value of $f = 8$, the overall structure of images can be approximated, but inconsistencies of facial features such as a half-bearded face and of viewpoints in different parts of the image arise. Only our full setting of $f = 16$ can synthesize high-fidelity samples. For analogous results in the conditional setting on S-FLCKR, we refer to the appendix (Fig. 13 and Sec. C).

To assess the effectiveness of our approach quantitatively, we compare results between training a transformer directly on pixels, and training it on top of a *VQGAN*'s latent code with $f = 2$, given a fixed computational budget. Again, we follow [8] and learn a dictionary of 512 RGB values on CIFAR10 to operate directly on pixel space and train the same transformer architecture on top of our *VQGAN* with a latent code of size $16 \times 16 = 256$. We observe improvements of 18.63% for FIDs and $14.08\times$ faster sampling of images.

数据集	**ours**	SPADE [53]	Pix2PixHD（+aug）[75]	CRN [9]
COCO-Stuff数据集	22.4	22.6/23.9(*)	111.5（54.2）	70.4
ADE20K数据集	35.5	33.9/35.7(*)	81.8（41.5）	73.3

表2. 语义图像合成的FID分数对比（256×256像素）。(*)：该数值是根据我们的评估方案，基于各数据集的验证数据集划分重新计算得出的。

4.4. 基准测试：图像合成结果评估

In this section we investigate how our approach quantitatively compares to existing models for generative image synthesis. In particular, we assess the performance of our model in terms of FID and compare to a variety of established models (GANs, VAEs, Flows, AR, Hybrid). The results on *semantic synthesis* are shown in Tab. 2, where we compare to [53, 75, 35, 9], and the results on *unconditional face synthesis* are shown in Tab. 3. While some task-specialized GAN models report better FID scores, our approach provides a unified model that works well across a wide range of tasks while retaining the ability to encode and reconstruct images. It thereby bridges the gap between purely adversarial and likelihood-based approaches.

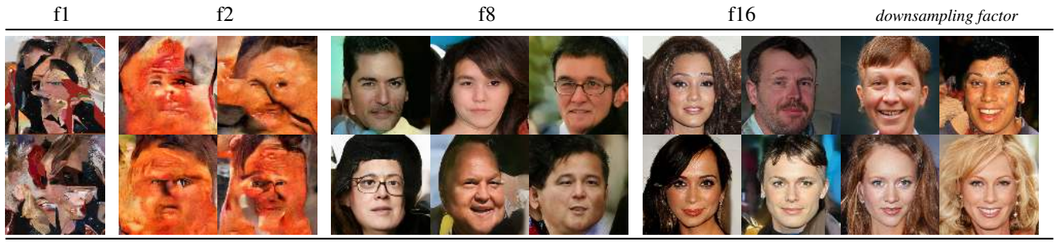


Figure 7. Evaluating the importance of effective codebook for HQ-Faces (CelebA-HQ and FFHQ) for a fixed sequence length $|s| = 16{\cdot}16 = 256$. Globally consistent structures can only be modeled with a context-rich vocabulary (right). All samples are generated with temperature $t = 1.0$ and top-k sampling with $k = 100$. Last row reports the speedup over the f1 baseline which operates directly on pixels and takes 7258 seconds to produce a sample on a NVIDIA GeForce GTX Titan X.

CelebA-HQ 256×256		FFHQ 256×256	
Method	FID ↓	Method	FID ↓
GLOW [37]	69.0	VDVAE ($t = 0.7$) [11]	38.8
NVAE [69]	40.3	VDVAE ($t = 1.0$)	33.5
PIONEER (B.) [23]	39.2 (25.3)	VDVAE ($t = 0.8$)	29.8
NCPVAE [1]	24.8	VDVAE ($t = 0.9$)	28.5
VAEBM [77]	20.4	*VQGAN*+P.SNAIL	21.9
Style ALAE [56]	19.2	BigGAN	12.4
DC-VAE [54]	15.8	**ours** (k=300)	9.6
ours (k=400)	10.2	U-Net GAN (+aug) [66]	10.9 (7.6)
PGGAN [31]	8.0	StyleGAN2 (+aug) [34]	3.8 (3.6)

Table 3. FID score comparison for face image synthesis. CelebA-HQ results reproduced from [1, 54, 77, 24], FFHQ from [66, 32].

Model	acceptance rate	FID	IS
mixed k, $p = 1.0$	1.0	17.04	70.6 ± 1.8
$k = 973$, $p = 1.0$	1.0	29.20	47.3 ± 1.3
$k = 250$, $p = 1.0$	1.0	15.98	78.6 ± 1.1
$k = 973$, $p = 0.88$	1.0	15.78	74.3 ± 1.8
$k = 600$, $p = 1.0$	0.05	5.20	280.3 ± 5.5
mixed k, $p = 1.0$	0.5	10.26	125.5 ± 2.4
mixed k, $p = 1.0$	0.25	7.35	188.6 ± 3.3
mixed k, $p = 1.0$	0.05	5.88	304.8 ± 3.6
mixed k, $p = 1.0$	0.005	6.59	402.7 ± 2.9
DCTransformer [48]	1.0	36.5	n/a
VQVAE-2 [61]	1.0	∼31	∼45
VQVAE-2	n/a	∼10	∼330
BigGAN [4]	1.0	7.53	168.6 ± 2.5
BigGAN-deep	1.0	6.84	203.6 ± 2.6
IDDPM [49]	1.0	12.3	n/a
ADM-G, no guid. [15]	1.0	10.94	100.98
ADM-G, 1.0 guid.	1.0	4.59	186.7
ADM-G, 10.0 guid.	1.0	9.11	283.92
val. data	1.0	1.62	234.0 ± 3.9

Table 4. FID score comparison for class-conditional synthesis on 256×256 ImageNet, evaluated between 50k samples and the training split. Classifier-based rejection sampling as in VQVAE-2 uses a ResNet-101 [22] classifier. BigGAN(-deep) evaluated via `https://tfhub.dev/deepmind` truncated at 1.0. "Mixed" k refers to samples generated with different top-k values, here $k \in \{100, 200, 250, 300, 350, 400, 500, 600, 800, 973\}$.

Autoregressive models are typically sampled with a decoding strategy [27] such as beam-search, top-k or nucleus sampling. For most of our results, including those in Tab. 2, we use top-k sampling with $k = 100$ unless stated otherwise. For the results on face synthesis in Tab. 3, we computed scores for $k \in \{100, 200, 300, 400, 500\}$ and report the best results, obtained with $k = 400$ for CelebA-HQ and $k = 300$ for FFHQ. Fig. 10 in the supplementary shows FID and Inception scores as a function of k.

Class-Conditional Synthesis on ImageNet To address a direct comparison with the previous state-of-the-art for autoregressive modeling of class-conditional image synthesis on ImageNet, VQVAE-2 [61], we train a class-conditional ImageNet transformer on 256×256 images, using a *VQGAN* with $\dim \mathcal{Z} = 16384$ and $f = 16$, and additionally compare to BigGAN [4], IDDPM [49], DCTransformer [48] and ADM [15] in Tab. 4. Note that our model uses $\simeq 10\times$ *less* parameters than VQVAE-2, which has an estimated parameter count of 13.5B (estimate based on [67]).

Samples of this model for different ImageNet classes are shown in Fig. 8. We observe that the adversarial training of the corresponding *VQGAN* enables sampling of high-quality images with realistic textures, of comparable or higher quality than existing approaches such as BigGAN and VQVAE-2, see also Fig. 14-17 in the supplementary.

Quantitative results are summarized in Tab. 4. We report FID and Inception Scores for the best k/p in top-k/top-p sampling. Following [61], we can further increase quality via classifier-rejection, which keeps only the best m-out-of-n samples in terms of the classifier's score, *i.e.* with an acceptance rate of m/n. We use a ResNet-101 classifier [22].

We observe that our model outperforms other autoregressive approaches (VQVAE-2, DCTransformer) in terms of FID and IS, surpasses BigGAN and IDDPM even for low rejection rates and yields scores close to the state of the art for higher rejection rates, see also Fig. 9.

How good is the *VQGAN*? Reconstruction FIDs obtained via the codebook provide an estimate on the achievable FID of the generative model trained on it. To quantify the per-

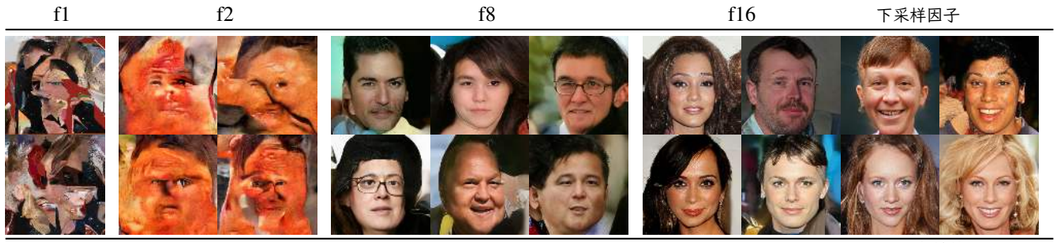


图 7 $|s| = 16{\cdot}16 =$ 。在固定序列长度下，评估高效码本对 256 HQ-Faces（CelebA-HQ数据集与FFHQ数据集）的重要性。仅有具备丰富上下文语义的词汇表才能用于构建全局一致的图像结构（右侧）。所有样本均是采用温度参数$t = 1.0$ 以及 $k = 100$设定的top-k s采样策略生成的。最后一行显示了相较于直接在像素层面处理的f1基准方法的加速比，后者在NVIDIA GeForce GTX Titan X显卡上生成一个样本需要7258秒。

CelebA-HQ 256×256		FFHQ 256×256	
方法	FID指标: ↓	方法	FID指标: ↓
GLOW算法: [37]	69.0	VDVAE算法（$t = 0.7$）[11]	38.8
NVAE算法: [69]	40.3	VDVAE ($t = 1.0$)	33.5
PIONEER (B.) [23]	39.2 (25.3)	VDVAE ($t = 0.8$)	29.8
NCPVAE [1]	24.8	VDVAE ($t = 0.9$)	28.5
VAEBM [77]	20.4	*VQGAN*模型+P.SNAIL	21.9
Style ALAE [56]	19.2	BigGAN	12.4
DC-VAE [54]	15.8	**我们的** (k=300)	9.6
我们的 (k=400)	10.2	U-Net GAN模型（+增强版）[66]	10.9（7.6）
PGGAN模型 [31]	8.0	StyleGAN2模型（+增强版）[34]	3.8（3.6）

表3. 人脸图像合成任务的FID分数对比。CelebA-HQ的测试结果引自 [1, 54, 77, 24], FFHQ数据集 [66, 32]。

模型	接受率	FID	IS
混合了 k和 $p = 1.0$	1.0	17.04	70.6 ± 1.8
$k = 973$, $p = 1.0$	1.0	29.20	47.3 ± 1.3
$k = 250$, $p = 1.0$	1.0	15.98	78.6 ± 1.1
$k = 973$, $p = 0.88$	1.0	15.78	74.3 ± 1.8
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IDDPM [49]模型	1.0	12.3	n/a
ADM-G模型，无引导参数 [15]	1.0	10.94	100.98
ADM-G模型，引导参数为1.0 {v1}	1.0	4.59	186.7
ADM-G模型，10.0引导值。	1.0	9.11	283.92
验证集数据	1.0	1.62	234.0 ± 3.9

表4. 在 256×256 ImageNet数据集上进行的条件分类合成任务的FID分数对比，评估对象为50k个样本与训练集之间的数据。与VQVAE-2类似，基于分类器的拒绝采样方法会使用ResNet-101 [22] 分类器。而BigGAN(-deep)则是通过 `https://tfhub.dev/deepmind` 截断至1.0的数值来进行的评估。"混合" k $k \in$ 指的是使用不同 $\{100, 200, 250, 300, 350, 400, 500, 600, 800, 973\}$ top-k值生成的样本，在此例中即为如此。

Autoregressive models are typically sampled with a decoding strategy [27] such as beam-search, top-k or nucleus sampling. For most of our results, including those in Tab. 2, we use top-k sampling with $k = 100$ unless stated otherwise. For the results on face synthesis in Tab. 3, we computed scores for $k \in \{100, 200, 300, 400, 500\}$ and report the best results, obtained with $k = 400$ for CelebA-HQ and $k = 300$ for FFHQ. Fig. 10 in the supplementary shows FID and Inception scores as a function of k.

Class-Conditional Synthesis on ImageNet To address a direct comparison with the previous state-of-the-art for autoregressive modeling of class-conditional image synthesis on ImageNet, VQVAE-2 [61], we train a class-conditional ImageNet transformer on 256×256 images, using a *VQGAN* with $\dim \mathcal{Z} = 16384$ and $f = 16$, and additionally compare to BigGAN [4], IDDPM [49], DCTransformer [48] and ADM [15] in Tab. 4. Note that our model uses $\simeq 10\times$ *less* parameters than VQVAE-2, which has an estimated parameter count of 13.5B (estimate based on [67]).

该模型针对不同ImageNet类别生成的样本展示在图中。8。我们发现，对相应的*VQGAN* 进行对抗训练后，能够生成具有逼真纹理的高质量图像，其质量可与BigGAN和VQVAE-2等现有方法相媲美甚至更优，详情参见附录中的14-17图。

定量分析结果汇总于表格中。4。我们报告了在top-k/top-p采样方式下最优 k/p 模型的FID指标与Inception分数。根据 [61], , 我们还可以通过分类器筛选进一步提升质量，该方法仅保留分类器评分最高的 m-out-of-n 样本，即。其接受率为 m/n。我们采用了ResNet-101分类器 [22]。

我们观察到，就FID指标和IS指标而言，我们的模型表现优于其他自回归方法（如VQVAE-2、DCTransformer模型）；即便在拒绝率较低的情况下，它也能超越BigGAN模型和IDDPM模型；而在拒绝率较高时，其得分则接近当前最先进水平，详情可见图表。9.

***VQGAN*模型的重构能力究竟有多强？** 通过码本获得的重构FID指标，能够为基于该码本训练的生成模型所能达到的FID值提供估算。为了量化每

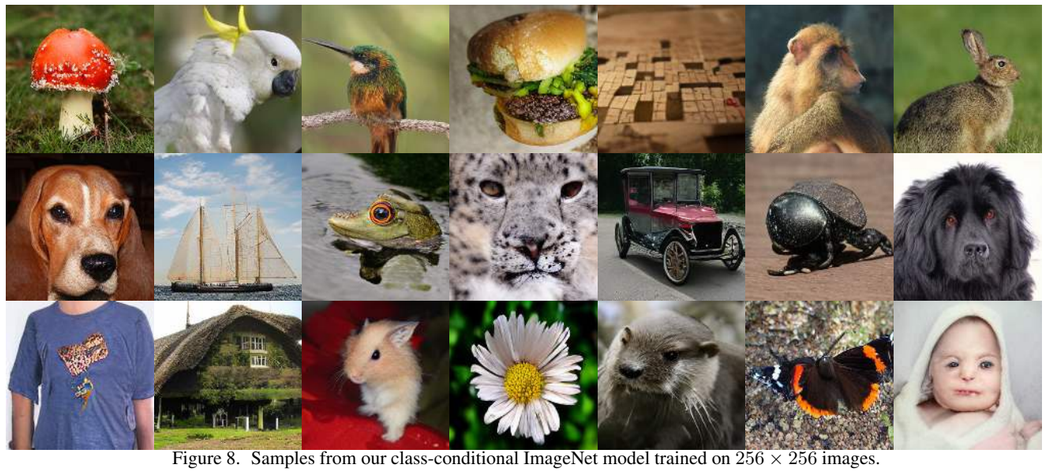


Figure 8. Samples from our class-conditional ImageNet model trained on 256×256 images.

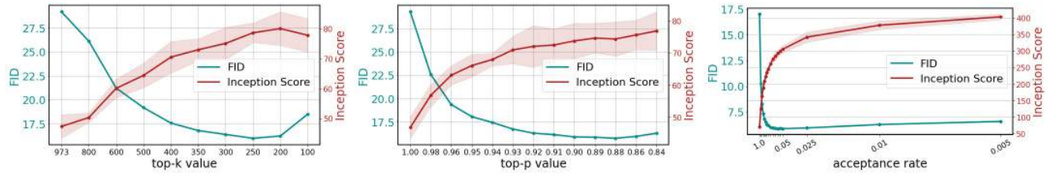


Figure 9. FID and Inception Score as a function of top-k, nucleus and rejection filtering.

Model	Codebook Size	dim $\mathcal{Z}$	FID/val	FID/train
VQVAE-2	64×64 & 32×32	512	n/a	~ 10
DALL-E [59]	32×32	8192	32.01	33.88
VQGAN	16×16	1024	7.94	10.54
VQGAN	16×16	16384	4.98	7.41
VQGAN*	32×32	8192	1.49	3.24
VQGAN	64×64 & 32×32	512	1.45	2.78

Table 5. FID on ImageNet between reconstructed validation split and original validation (FID/val) and training (FID/train) splits. *trained with Gumbel-Softmax reparameterization as in [59, 29].

formance gains of our *VQGAN* over discrete VAEs trained without perceptual and adversarial losses (*e.g.* VQVAE-2, DALL-E [59]), we evaluate this metric on ImageNet and report results in Tab. 5. Our *VQGAN* outperforms non-adversarial models while providing significantly more compression (seq. length of 256 vs. $5120 = 32^2 + 64^2$ for VQVAE-2, 256 vs 1024 for DALL-E). As expected, larger versions of *VQGAN* (either in terms of larger codebook sizes or increased code lengths) further improve performance. Using the same hierarchical codebook setting as in VQVAE-2 with our model provides the best reconstruction FID, albeit at the cost of a very long and thus impractical sequence. The qualitative comparison corresponding to the results in Tab. 5 can be found in Fig. 12.

5. Conclusion

This paper adressed the fundamental challenges that previously confined transformers to low-resolution images. We proposed an approach which represents images as a composition of perceptually rich image constituents and thereby overcomes the infeasible quadratic complexity when modeling images directly in pixel space. Modeling constituents with a CNN architecture and their compositions with a transformer architecture taps into the full potential of their complementary strengths and thereby allowed us to represent the first results on high-resolution image synthesis with a transformer-based architecture. In experiments, our approach demonstrates the efficiency of convolutional inductive biases and the expressivity of transformers by synthesizing images in the megapixel range and outperforming state-of-the-art convolutional approaches. Equipped with a general mechanism for conditional synthesis, it offers many opportunities for novel neural rendering approaches.

This work has been supported by the German Research Foundation (DFG) projects 371923335, 421703927 and a hardware donation from NVIDIA corporation.



图8：基于 256×256 张图像训练得到的带类别条件的ImageNet模型的样本。

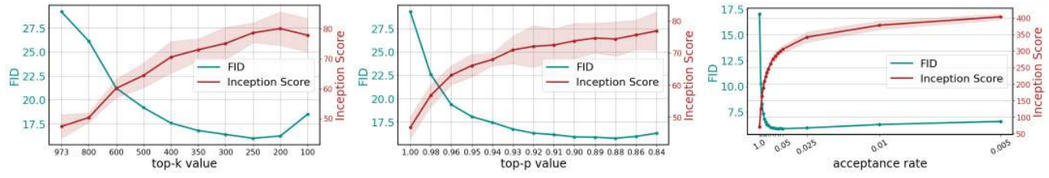


图9：不同top-k值、核过滤以及拒绝过滤条件下FID指标与InceptionScore指标的变化情况。

模型	码本大小	维度 $\mathcal{Z}$	验证集FID值	训练集FID值
VQVAE-2模型	64×64 与 32×32	512	n/a	~ 10
DALL-E [59]	32×32	8192	32.01	33.88
*VQGAN*模型	16×16	1024	7.94	10.54
*VQGAN*模型	16×16	16384	4.98	7.41
VQGAN*、*VQGAN*模型	32×32	8192	1.49	3.24
*VQGAN*模型	64×64 & 32×32	512	1.45	2.78

表5. ImageNet数据集上，重构后的验证集与原始验证集（FID/val）以及训练集（FID/train）对应的FID指标值。*是按照[59, 29]中的方法，通过Gumbel-Softmax重参数化进行训练的。

formance gains of our *VQGAN* over discrete VAEs trained without perceptual and adversarial losses (*e.g.* VQVAE-2, DALL-E [59]), we evaluate this metric on ImageNet and report results in Tab. 5. Our *VQGAN* outperforms non-adversarial models while providing significantly more compression (seq. length of 256 vs. $5120 = 32^2 + 64^2$ for VQVAE-2, 256 vs 1024 for DALL-E). As expected, larger versions of *VQGAN* (either in terms of larger codebook sizes or increased code lengths) further improve performance. Using the same hierarchical codebook setting as in VQVAE-2 with our model provides the best reconstruction FID, albeit at the cost of a very long and thus impractical sequence. The qualitative comparison corresponding to the results in Tab. 5 can be found in Fig. 12.

5. 结论

本文探讨了此前限制Transformer模型仅能用于低分辨率图像的根本性难题。我们提出了一种新方法，将图像表示为具有丰富感知特性的图像组成部分的组合，从而避免了在像素空间直接对图像建模时所面临的极高二次复杂度问题。通过使用CNN架构来建模这些组成元素，再借助Transformer架构来处理它们的组合，充分发挥了两者互补的优势，使我们得以首次基于Transformer模型的架构实现高分辨率图像合成的成果。实验结果表明，该方法不仅展现了卷积归纳偏置的高效率以及Transformer模型的强表达能力，还能生成百万级分辨率的图像，并且性能优于现有的最先进卷积方法。此外，由于具备条件合成的一般机制，该框架为开发全新的神经渲染技术提供了广阔的可能性。

本项研究得到了德国研究基金会（DFG）项目371923335、421703927的支持，同时还获得了NVIDIA公司的硬件捐赠助力。

Taming Transformers for High-Resolution Image Synthesis

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Supplementary Material

The supplementary material for our work *Taming Transformers for High-Resolution Image Synthesis* is structured as follows: First, Sec. A summarizes changes to a previous version of this paper. In Sec. B, we present hyperparameters and architectures which were used to train our models. Next, extending the discussion of Sec. 4.3, Sec. C presents additional evidence for the importance of perceptually rich codebooks and its interpretation as a trade-off between reconstruction fidelity and sampling capability. Additional results on high-resolution image synthesis for a wide range of tasks are then presented in Sec. D, and Sec. E shows nearest neighbors of samples. Finally, Sec. F contains results regarding the ordering of image representations.

A. Changelog

We summarize changes between this version [1] of the paper and its previous version [2].

In the previous version, Eq. (4) had a weighting term β on the commitment loss, and Tab. 8 reported a value of $\beta = 0.25$ for all models. However, due to a bug in the implementation, β was never used and all models have been trained with $\beta = 1.0$. Thus, we removed β in Eq. (4).

We updated class-conditional synthesis results on ImageNet in Sec. 4.4. The previous results, included here in Tab. 6 for completeness, were based on a slightly different implementation where the transformer did not predict the distribution of the first token but used a histogram for it. The new model has been trained for 2.4 million steps with a batch size of 16 accumulated over 8 batches, which took 45.8 days on a single A100 GPU. The previous model had been trained for 1.0 million steps. Furthermore, the FID values were based on 50k (18k) samples against 50k (18k) training examples (to compare with MSP). For better comparison with other works, the current version reports FIDs based on 50k samples against all training examples of ImageNet using `torch-fidelity` [50]. We updated all qualitative figures showing samples from this model and added visualizations of the effect of tuning top-k/p or rejection rate in Fig. 14-26.

To provide a better overview, we also include results from works that became available after the previous version of our work. Specifically, we include results on reconstruction quality of the VQVAE from [59] in Tab. 5 and Fig. 12 (which replaces the previous qualitative comparison), and results on class-conditional ImageNet sampling from [49, 48, 15] in Tab. 4. Note that with the exception of BigGAN and BigGAN-deep [4], no models or sampling results are available for the methods we compare to in Tab. 4. Thus, we can only report the numbers from the respective papers but cannot re-evaluate them with the same code. We follow the common evaluation protocol for class-conditional ImageNet synthesis from [4] and evaluate 50k samples from the model against the whole training split of ImageNet. However, it is not clear how different implementations resize the training images. In our code, we use the largest center-crop and resize it bilinearly with anti-aliasing to 256×256 using `Pillow` [73]. FID and Inception Scores are then computed with `torch-fidelity` [50].

We updated face-synthesis results in Tab. 3 based on a slightly different implementation as in the case of class-conditional ImageNet results and improve the previous results slightly. In addition, we evaluate the ability of our NLL-based training to detect overfitting. We train larger models (FFHQ (big) and CelebA-HQ (big) in Tab. 8) on the face datasets, and show nearest neighbors of samples obtained from checkpoints with the best NLL on the validation split and the training split in Sec. E. We also added Fig. 10, which visualizes the effect of tuning k in top-k sampling on FID and IS.

B. Implementation Details

The hyperparameters for all experiments presented in the main paper and supplementary material can be found in Tab. 8. Except for the *c-IN (big)*, *COCO-Stuff* and *ADE20K* models, these hyperparameters are set such that each transformer model can be trained with a batch-size of at least 2 on a GPU with 12GB VRAM, but we generally train on 2-4 GPUs with an accumulated VRAM of 48 GB. If hardware permits, 16-bit precision training is enabled.

[1] https://arxiv.org/abs/2012.09841v3
[2] https://arxiv.org/abs/2012.09841v2

驯服Transformer模型以实现高分辨率图像合成

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补充材料

The supplementary material for our work *Taming Transformers for High-Resolution Image Synthesis* is structured as follows: First, Sec. A summarizes changes to a previous version of this paper. In Sec. B, we present hyperparameters and architectures which were used to train our models. Next, extending the discussion of Sec. 4.3, Sec. C presents additional evidence for the importance of perceptually rich codebooks and its interpretation as a trade-off between reconstruction fidelity and sampling capability. Additional results on high-resolution image synthesis for a wide range of tasks are then presented in Sec. D, and Sec. E shows nearest neighbors of samples. Finally, Sec. F contains results regarding the ordering of image representations.

A. 变更日志

我们总结了本文的当前版本 [1] 与上一个版本 [2]之间的改动。

在之前的版本中，方程（4）中的承诺损失项包含一个 β 权重项，8 中报告称所有模型的该参数值均为 $\beta = 0.25$。然而由于实现上的缺陷，β这个权重项从未被使用，所有模型都是使用 $\beta = 1.0$来训练的。因此，我们移除了 β 方程（4）中的该权重项。

We updated class-conditional synthesis results on ImageNet in Sec. 4.4. The previous results, included here in Tab. 6 for completeness, were based on a slightly different implementation where the transformer did not predict the distribution of the first token but used a histogram for it. The new model has been trained for 2.4 million steps with a batch size of 16 accumulated over 8 batches, which took 45.8 days on a single A100 GPU. The previous model had been trained for 1.0 million steps. Furthermore, the FID values were based on 50k (18k) samples against 50k (18k) training examples (to compare with MSP). For better comparison with other works, the current version reports FIDs based on 50k samples against all training examples of ImageNet using `torch-fidelity` [50]. We updated all qualitative figures showing samples from this model and added visualizations of the effect of tuning top-k/p or rejection rate in Fig. 14-26.

To provide a better overview, we also include results from works that became available after the previous version of our work. Specifically, we include results on reconstruction quality of the VQVAE from [59] in Tab. 5 and Fig. 12 (which replaces the previous qualitative comparison), and results on class-conditional ImageNet sampling from [49, 48, 15] in Tab. 4. Note that with the exception of BigGAN and BigGAN-deep [4], no models or sampling results are available for the methods we compare to in Tab. 4. Thus, we can only report the numbers from the respective papers but cannot re-evaluate them with the same code. We follow the common evaluation protocol for class-conditional ImageNet synthesis from [4] and evaluate 50k samples from the model against the whole training split of ImageNet. However, it is not clear how different implementations resize the training images. In our code, we use the largest center-crop and resize it bilinearly with anti-aliasing to 256×256 using `Pillow` [73]. FID and Inception Scores are then computed with `torch-fidelity` [50].

We updated face-synthesis results in Tab. 3 based on a slightly different implementation as in the case of class-conditional ImageNet results and improve the previous results slightly. In addition, we evaluate the ability of our NLL-based training to detect overfitting. We train larger models (FFHQ (big) and CelebA-HQ (big) in Tab. 8) on the face datasets, and show nearest neighbors of samples obtained from checkpoints with the best NLL on the validation split and the training split in Sec. E. We also added Fig. 10, which visualizes the effect of tuning k in top-k sampling on FID and IS.

B. 实现细节

主论文及补充材料中介绍的所有实验的超参数均可在表 8 中找到。除了 *c-IN*（大型版本模型）、*COCO-Stuff* 以及*ADE20K* 模型之外，这些超参数的设置使得每个变换器模型都能够在拥有12GB显存的图形处理器上以至少2的批量大小进行训练；不过我们通常会在2到4台总显存为48GB的图形处理器上开展训练。在硬件条件允许的情况下，还会采用16位精度进行训练。

[1] https://arxiv.org/abs/2012.09841v3 [2]
https://arxiv.org/abs/2012.09841v2

Dataset	ours-previous (+R)	BigGAN (-deep)	MSP
IN 256, 50K	19.8 (11.2)	7.1 (7.3)	n.a.
IN 256, 18K	23.5	9.6 (9.7)	50.4

Dataset	ours-previous	ours-new
CelebA-HQ 256	10.7	10.2
FFHQ 256	11.4	9.6

Table 6. Results from a previous version of this paper, see also Sec. A. Left: Previous results on class-conditional ImageNet synthesis with a slightly different implementation and evaluated against 50k and 18k training examples instead of the whole training split. See Tab. 4 for new, improved results evaluated against the whole training split. Right: Previous results on face-synthesis with a slightly different implementation compared to the new implementation. See also Tab. 3 for comparison with other methods.

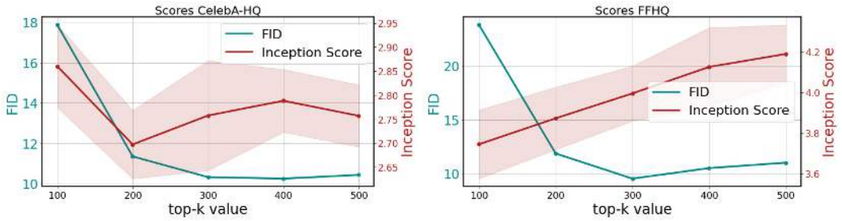


Figure 10. FID and Inception Score as a function of top-k for CelebA-HQ (left) and FFHQ (right).

Encoder	Decoder
$x \in \mathbb{R}^{H\times W\times C}$	$z_{\mathbf{q}} \in \mathbb{R}^{h\times w\times n_z}$
Conv2D $\to \mathbb{R}^{H\times W\times C'}$	Conv2D $\to \mathbb{R}^{h\times w\times C''}$
$m\times$ { Residual Block, Downsample Block} $\to \mathbb{R}^{h\times w\times C''}$	Residual Block $\to \mathbb{R}^{h\times w\times C''}$
Residual Block $\to \mathbb{R}^{h\times w\times C''}$	Non-Local Block $\to \mathbb{R}^{h\times w\times C''}$
Non-Local Block $\to \mathbb{R}^{h\times w\times C''}$	Residual Block $\to \mathbb{R}^{h\times w\times C''}$
Residual Block $\to \mathbb{R}^{h\times w\times C''}$	$m\times$ { Residual Block, Upsample Block} $\to \mathbb{R}^{H\times W\times C'}$
GroupNorm, Swish, Conv2D $\to \mathbb{R}^{h\times w\times n_z}$	GroupNorm, Swish, Conv2D $\to \mathbb{R}^{H\times W\times C}$

Table 7. High-level architecture of the encoder and decoder of our *VQGAN*. The design of the networks follows the architecture presented in [25] with no skip-connections. For the discriminator, we use a patch-based model as in [28]. Note that $h = \frac{H}{2^m}$, $w = \frac{W}{2^m}$ and $f = 2^m$.

VQGAN Architecture The architecture of our convolutional encoder and decoder models used in the *VQGAN* experiments is described in Tab. 7. Note that we adopt the compression rate by tuning the number of downsampling steps m. Further note that λ in Eq. 5 is set to zero in an initial warm-up phase. Empirically, we found that longer warm-ups generally lead to better reconstructions. As a rule of thumb, we recommend setting $\lambda = 0$ for at least one epoch.

Transformer Architecture Our transformer model is identical to the GPT2 architecture [58] and we vary its capacity mainly through varying the amount of layers (see Tab. 8). Furthermore, we generally produce samples with a temperature $t = 1.0$ and a top-k cutoff at $k = 100$ (with higher top-k values for larger codebooks).

C. On Context-Rich Vocabularies

Sec. 4.3 investigated the effect of the downsampling factor f used for encoding images. As demonstrated in Fig. 7, large factors are crucial for our approach, since they enable the transformer to model long-range interactions efficiently. However, since larger f correspond to larger compression rates, the reconstruction quality of the *VQGAN* starts to decrease after a certain point, which is analyzed in Fig. 11. The left part shows the reconstruction error (measured by LPIPS [81]) versus the negative log-likelihood obtained by the transformer for values of f ranging from 1 to 64. The latter provides a measure of the ability to model the distribution of the image representation, which increases with f. The reconstruction error on the other hand decreases with f and the qualitative results on the right part show that beyond a critical value of f, in this case $f = 16$, reconstruction errors become severe. At this point, even when the image representations are modeled faithfully, as suggested by a low negative log-likelihood, sampled images are of low-fidelity, because the reconstruction capabilities provide an upper bound on the quality that can be achieved.

Hence, Fig. 11 shows that we must learn perceptually rich encodings, *i.e.* encodings with a large f and perceptually faithful reconstructions. This is the goal of our *VQGAN* and Fig. 12 compares its reconstruction capabilities against the VQVAE [72]

数据集	之前的数据集（+R）	BigGAN模型（深度结构）	MSP
256维度、50K样本量	19.8（11.2）	7.1（7.3）	n.a.
256维度、18K样本量	23.5	9.6（9.7）	50.4

数据集	之前的数据集	新的数据集
CelebA-HQ 256	10.7	10.2
FFHQ 256版本	11.4	9.6

表6. 本文早期版本的研究结果，详见第A节。左侧：采用略有不同的实现方式在带类别条件的ImageNet数据集上获得的先前合成结果，评估时仅使用了5万和1.8万个训练样本，而非整个训练集。详情见表格4 。右侧：在人脸合成任务上获得的先前结果，其实现方式与新版本存在一定差异。亦可参见表格3，以便与其他方法进行对比。

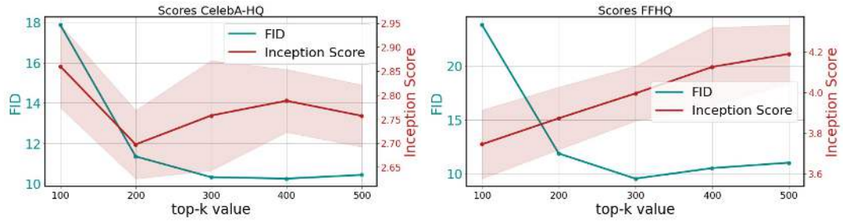


图10. CelebA-HQ数据集（左）与FFHQ数据集（右）的FID指标及InceptionScore指标随top-k值的变化情况。

编码器	解码器
$x \in \mathbb{R}^{H\times W\times C}$	$z_{\mathbf{q}} \in \mathbb{R}^{h\times w\times n_z}$
Conv2D $\to \mathbb{R}^{H\times W\times C'}$	Conv2D $\to \mathbb{R}^{h\times w\times C''}$
$m\times$ {残差块与下采样块} $\to \mathbb{R}^{h\times w\times C''}$	残差块 $\to \mathbb{R}^{h\times w\times C''}$
残差块 $\to \mathbb{R}^{h\times w\times C''}$	非局部块 $\to \mathbb{R}^{h\times w\times C''}$
非局部块 $\to \mathbb{R}^{h\times w\times C''}$	残差块 $\to \mathbb{R}^{h\times w\times C''}$
残差块 $\to \mathbb{R}^{h\times w\times C''}$	$m\times$ { 残差块、上采样块} $\to \mathbb{R}^{H\times W\times C'}$
GroupNorm、Swish、Conv2D $\to \mathbb{R}^{h\times w\times n_z}$	GroupNorm、Swish、Conv2D $\to \mathbb{R}^{H\times W\times C}$

表7. 我们所设计的 *VQGAN*的编码器与解码器的高层架构。这些网络的设计遵循 [25] 中提出的架构，且不包含跳接连接。至于判别器，我们采用了如 [28]中所描述的基于补丁的模型。需要注意的是 $h = \frac{H}{2^m}$、 $w = \frac{W}{2^m}$ 以及 $f = 2^m$。

VQGAN Architecture The architecture of our convolutional encoder and decoder models used in the *VQGAN* experiments is described in Tab. 7. Note that we adopt the compression rate by tuning the number of downsampling steps m. Further note that λ in Eq. 5 is set to zero in an initial warm-up phase. Empirically, we found that longer warm-ups generally lead to better reconstructions. As a rule of thumb, we recommend setting $\lambda = 0$ for at least one epoch.

变换器模型架构 我们的变换器模型与GPT2的架构完全相同 [58] 我们主要通过调整层数来改变其容量（详见表格）。8）。此外，我们在生成样本时通常会设置温度参数$t = 1.0$ 以及top-k 截断值在 $k = 100$<style id='19'〉（对于规模更大的码本，其top-k 值也会相应更高）。

C. 关于富含上下文的词汇表

Sec. 4.3 investigated the effect of the downsampling factor f used for encoding images. As demonstrated in Fig. 7, large factors are crucial for our approach, since they enable the transformer to model long-range interactions efficiently. However, since larger f correspond to larger compression rates, the reconstruction quality of the *VQGAN* starts to decrease after a certain point, which is analyzed in Fig. 11. The left part shows the reconstruction error (measured by LPIPS [81]) versus the negative log-likelihood obtained by the transformer for values of f ranging from 1 to 64. The latter provides a measure of the ability to model the distribution of the image representation, which increases with f. The reconstruction error on the other hand decreases with f and the qualitative results on the right part show that beyond a critical value of f, in this case $f = 16$, reconstruction errors become severe. At this point, even when the image representations are modeled faithfully, as suggested by a low negative log-likelihood, sampled images are of low-fidelity, because the reconstruction capabilities provide an upper bound on the quality that can be achieved.

因此，图11 表明我们必须学习具有丰富感知特性的编码方式，也就是说，那些具有较大 f 规模且能够实现高保真感知重构的编码。这正是我们*VQGAN* 所追求的目标，而图12 则将其重构能力与VQVAE进行了对比 [72]

Experiment	n_{layer}	# params $[M]$	n_z	$\|\mathcal{Z}\|$	dropout	length(s)	n_e	m
RIN	12	85	64	768	0.0	512	1024	4
c-RIN	18	128	64	768	0.0	257	768	4
D-RINv1	14	180	256	1024	0.0	512	768	4
D-RINv2	24	307	256	1024	0.0	512	1024	4
IN	24	307	256	1024	0.0	256	1024	4
c-IN	24	307	256	1024	0.0	257	1024	4
c-IN (big)	48	1400	256	16384	0.0	257	1536	4
IN-Edges	24	307	256	1024	0.0	512	1024	3
IN-SR	12	153	256	1024	0.0	512	1024	3
S-FLCKR, $f = 4$	24	307	256	1024	0.0	512	1024	2
S-FLCKR, $f = 16$	24	307	256	1024	0.0	512	1024	4
S-FLCKR, $f = 32$	24	307	256	1024	0.0	512	1024	5
(FacesHQ, $f = 1$)*	24	307	–	512	0.0	512	1024	–
FacesHQ, $f = 2$	24	307	256	1024	0.0	512	1024	1
FacesHQ, $f = 4$	24	307	256	1024	0.0	512	1024	2
FacesHQ, $f = 8$	24	307	256	1024	0.0	512	1024	3
FacesHQ**, $f = 16$	24	307	256	1024	0.0	512	1024	4
FFHQ**, $f = 16$	28	355	256	1024	0.0	256	1024	4
CelebA-HQ**, $f = 16$	28	355	256	1024	0.0	256	1024	4
FFHQ (big)	24	801	256	1024	0.0	256	1664	4
CelebA-HQ (big)	24	801	256	1024	0.0	256	1664	4
COCO-Stuff	32	651	256	8192	0.0	512	1280	4
ADE20K	28	405	256	4096	0.1	512	1024	4
DeepFashion	18	129	256	1024	0.0	340	768	4
LSUN-CT	24	307	256	1024	0.0	256	1024	4
CIFAR-10	24	307	256	1024	0.0	256	1024	1

Table 8. Hyperparameters. For every experiment, we set the number of attention heads in the transformer to $n_h = 16$. n_{layer} denotes the number of transformer blocks, # params the number of transformer parameters, n_z the dimensionality of codebook entries, $|\mathcal{Z}|$ the number of codebook entries, dropout the dropout rate for training the transformer, length(s) the total length of the sequence, n_e the embedding dimensionality and m the number of downsampling steps in the *VQGAN*. D-RINv1 is the experiment which compares to Pixel-SNAIL in Sec. 4.1. Note that the experiment (FacesHQ, $f = 1$)* does not use a learned *VQGAN* but a fixed k-means clustering algorithm as in [8] with $K = 512$ centroids. A prefix "c" refers to a class-conditional model. The models marked with a '**' are trained on the same *VQGAN*.

used in DALL-E [59]. We observe that for $f = 8$ and 8192 codebook entries, both the VQVAE and *VQGAN* capture the global structure faithfully. However, the textures produced by the VQVAE are blurry, whereas those of the *VQGAN* are crisp and realistic looking (*e.g*. the stone texture and the fur and tail of the squirrel). When we increase the compression rate of the *VQGAN* further to $f = 16$, we see that some reconstructed parts are not perfectly aligned with the input anymore (*e.g*. the paw of the squirrel), but, especially with slightly larger codebooks, the reconstructions still look realistic. This demonstrates how the *VQGAN* provides high-fidelity reconstructions at large factors, and thereby enables efficient high-resolution image synthesis with transformers.

To illustrate how the choice of f depends on the dataset, Fig. 13 presents results on S-FLCKR. In the left part, it shows, analogous to Fig. 7, how the quality of samples increases with increasing f. However, in the right part, it shows that reconstructions remain faithful perceptually faithful even for $f32$, which is in contrast to the corresponding results on faces in Fig. 11. These results might be explained by a higher perceptual sensitivity to facial features as compared to textures, and allow us to generate high-resolution landscapes even more efficiently with $f = 32$.

D. Additional Results

Qualitative Comparisons The qualitative comparison corresponding to Tab. 4 and Tab. 6 can be found in Fig. 14, 15, 16 and 17. Since no models are available for VQVAE-2 and MSP, we extracted results directly from the supplementary[3] and

[3] https://drive.google.com/file/d/1H2nr_Cu7OK18tRemsWn_6o5DGMNYentM/view?usp=sharing

实验	n_{layer}	# 参数 $[M]$	n_z	$\|\mathcal{Z}\|$	丢弃率	长度(s)	n_e	m
RIN	12	85	64	768	0.0	512	1024	4
c-RIN	18	128	64	768	0.0	257	768	4
D-RINv1	14	180	256	1024	0.0	512	768	4
D-RINv2	24	307	256	1024	0.0	512	1024	4
IN	24	307	256	1024	0.0	256	1024	4
c-IN	24	307	256	1024	0.0	257	1024	4
c-IN大型版本模型	48	1400	256	16384	0.0	257	1536	4
IN边	24	307	256	1024	0.0	512	1024	3
IN-SR	12	153	256	1024	0.0	512	1024	3
S-FLCKR, $f = 4$	24	307	256	1024	0.0	512	1024	2
S-FLCKR, $f = 16$	24	307	256	1024	0.0	512	1024	4
S-FLCKR, $f = 32$	24	307	256	1024	0.0	512	1024	5
(FacesHQ $f = 1$)*	24	307	–	512	0.0	512	1024	–
FacesHQ, $f = 2$	24	307	256	1024	0.0	512	1024	1
FacesHQ, $f = 4$	24	307	256	1024	0.0	512	1024	2
FacesHQ, $f = 8$	24	307	256	1024	0.0	512	1024	3
FacesHQ**, $f = 16$	24	307	256	1024	0.0	512	1024	4
FFHQ **, $f = 16$	28	355	256	1024	0.0	256	1024	4
CelebA-HQ **, $f = 16$	28	355	256	1024	0.0	256	1024	4
FFHQ（大型版本模型）	24	801	256	1024	0.0	256	1664	4
CelebA-HQ（大型版本模型）	24	801	256	1024	0.0	256	1664	4
COCO-Stuff数据集	32	651	256	8192	0.0	512	1280	4
ADE20K数据集	28	405	256	4096	0.1	512	1024	4
DeepFashion数据集	18	129	256	1024	0.0	340	768	4
LSUN-CT	24	307	256	1024	0.0	256	1024	4
CIFAR-10数据集	24	307	256	1024	0.0	256	1024	1

Table 8. Hyperparameters. For every experiment, we set the number of attention heads in the transformer to $n_h = 16$. n_{layer} denotes the number of transformer blocks, # params the number of transformer parameters, n_z the dimensionality of codebook entries, $|\mathcal{Z}|$ the number of codebook entries, dropout the dropout rate for training the transformer, length(s) the total length of the sequence, n_e the embedding dimensionality and m the number of downsampling steps in the *VQGAN*. D-RINv1 is the experiment which compares to Pixel-SNAIL in Sec. 4.1. Note that the experiment (FacesHQ $f = 1$)* does not use a learned *VQGAN* but a fixed k-means clustering algorithm as in [8] with $K = 512$ centroids. A prefix "c" refers to a class-conditional model. The models marked with a '**' are trained on the same *VQGAN*.

used in DALL-E [59]. We observe that for $f = 8$ and 8192 codebook entries, both the VQVAE and *VQGAN* capture the global structure faithfully. However, the textures produced by the VQVAE are blurry, whereas those of the *VQGAN* are crisp and realistic looking (*e.g*. the stone texture and the fur and tail of the squirrel). When we increase the compression rate of the *VQGAN* further to $f = 16$, we see that some reconstructed parts are not perfectly aligned with the input anymore (*e.g*. the paw of the squirrel), but, especially with slightly larger codebooks, the reconstructions still look realistic. This demonstrates how the *VQGAN* provides high-fidelity reconstructions at large factors, and thereby enables efficient high-resolution image synthesis with transformers.

To illustrate how the choice of f depends on the dataset, Fig. 13 presents results on S-FLCKR. In the left part, it shows, analogous to Fig. 7, how the quality of samples increases with increasing f. However, in the right part, it shows that reconstructions remain faithful perceptually faithful even for $f32$, which is in contrast to the corresponding results on faces in Fig. 11. These results might be explained by a higher perceptual sensitivity to facial features as compared to textures, and allow us to generate high-resolution landscapes even more efficiently with $f = 32$.

D. 其他结果

定性对比与表4 及表6 对应的定性对比可见于图14、15、16以及17。由于VQVAE-2和MSP没有可用的模型，我们直接从补充材料中提取了相关结果[3] 和

[3] https://drive.google.com/file/d/1H2nr_Cu7OK18tRemsWn_6o5DGMNYentM/view?usp=sharing

from the provided samples[4], respectively. For BigGAN, we produced the samples via the provided model[5]. Similarly, the qualitative comparison with the best competitor model (SPADE) for semantic synthesis on standard benchmarks (see Tab. 2) can be found in Fig. 40 (ADE20K) and Fig. 41 (COCO-Stuff)[6].

Comparison to Image-GPT To further evaluate the effectiveness of our approach, we compare to the state-of-the-art generative transformer model on images, ImageGPT [8]. By using immense amounts of compute the authors demonstrated that transformer models can be applied to the pixel-representation of images and thereby achieved impressive results both in representation learning and image synthesis. However, as their approach is confined to pixel-space, it does not scale beyond a resolution of 192×192. As our approach leverages a strong compression method to obtain context-rich representations of images and *then* learns a transformer model, we can synthesize images of much higher resolution. We compare both approaches in Fig. 27 and Fig. 28, where completions of images are depicted. Both plots show that our approach is able to synthesize consistent completions of dramatically increased fidelity. The results of [8] are obtained from https://openai.com/blog/image-gpt/.

Additional High-Resolution Results Fig. 29, 30, 31 and Fig. 32 contain additional HR results on the S-FLCKR dataset for both $f = 16$ ($m = 4$) and $f = 32$ ($m = 5$) (semantically guided). In particular, we provide an enlarged version of Fig. 5 from the main text, which had to be scaled down due to space constraints. Additionally, we use our sliding window approach (see Sec. 3) to produce high-resolution samples for the depth-to-image setting on RIN in Fig. 33 and Fig. 34, edge-to-image on IN in Fig. 35, stochastic superresolution on IN in Fig. 36, more examples on semantically guided landscape synthesis on S-FLCKR in Fig. 37 with $f = 16$ and in Fig. 38 with $f = 32$, and unconditional image generation on LSUN-CT (see Sec. 4.1) in Fig. 39. Moreover, for images of size 256×256, we provide results for generation from semantic layout on (i) ADE20K in Fig. 40 and (ii) COCO-Stuff in Fig. 41, depth-to-image on IN in Fig. 42, pose-guided person generation in Fig. 43 and class-conditional synthesis on RIN in Fig. 44.

E. Nearest Neighbors of Samples

One advantage of likelihood-based generative models over, *e.g.*, GANs is the ability to evaluate NLL on training data and validation data to detect overfitting. To test this, we trained large models for face synthesis, which can easily overfit them, and retained two checkpoints on each dataset: One for the best validation NLL (at the 10th and 13th epoch for FFHQ and CelebA-HQ, respectively), and another for the best training NLL (at epoch 1000). We then produced samples from both checkpoints and retrieved nearest neighbors from the training data based on the LPIPS similarity metric [81]. The results are shown in Fig. 45, where it can be observed that the checkpoints with best training NLL (best train NLL) reproduce the training examples, whereas samples from the checkpoints with best validation NLL (best val. NLL) depict new faces which are not found in the training data.

Based on these results, we can conclude that early-stopping based on validation NLL can prevent overfitting. Furthermore, the bottleneck for our approach on face synthesis is given by the dataset size since it has the capacity to almost perfectly fit the training data. Unfortunately, FID scores cannot detect such an overfitting. Indeed, the best train NLL checkpoints achieve FID scores of 3.86 on CelebA-HQ and 2.68 on FFHQ, compared to 10.2 and 9.6 for the best val. NLL checkpoints. While validation NLL provides a way to detect overfitting for likelihood-based models, it is not clear if early-stopping based on it is optimal if one is mainly interested in the quality of samples. To address this and the evaluation of GANs, new metrics will be required which can differentiate between models that produce new, high-quality samples and those that simply reproduce the training data.

Our class-conditional ImageNet model does not display overfitting according to validation NLL, and the nearest neighbors shown in Fig. 46 also provide evidence that the model produces new, high-quality samples.

F. On the Ordering of Image Representations

For the "classical" domain of transformer models, NLP, the order of tokens is defined by the language at hand. For images and their discrete representations, in contrast, it is not clear which linear ordering to use. In particular, our sliding-window approach depends on a row-major ordering and we thus investigate the performance of the following five different permutations of the input sequence of codebook indices: (i) **row major**, or *raster scan order*, where the image representation is

[4] https://bit.ly/2FJkvhJ
[5] https://tfhub.dev/deepmind/biggan-deep-256/1
[6] samples were reproduced with the authors' official implementation available at https://github.com/nvlabs/spade/

来自所提供的样本[4]，，分别对应。对于BigGAN模型，我们是通过给定的模型来生成这些样本的[5]。同样地，在标准基准测试上针对语义合成任务与最佳竞争模型（SPADE）的定性对比结果（见表2），可见于图40(ADE20K数据集)以及图41(COCO-Stuff数据集)[6]。

Comparison to Image-GPT To further evaluate the effectiveness of our approach, we compare to the state-of-the-art generative transformer model on images, ImageGPT [8]. By using immense amounts of compute the authors demonstrated that transformer models can be applied to the pixel-representation of images and thereby achieved impressive results both in representation learning and image synthesis. However, as their approach is confined to pixel-space, it does not scale beyond a resolution of 192×192. As our approach leverages a strong compression method to obtain context-rich representations of images and *then* learns a transformer model, we can synthesize images of much higher resolution. We compare both approaches in Fig. 27 and Fig. 28, where completions of images are depicted. Both plots show that our approach is able to synthesize consistent completions of dramatically increased fidelity. The results of [8] are obtained from https://openai.com/blog/image-gpt/.

Additional High-Resolution Results Fig. 29, 30, 31 and Fig. 32 contain additional HR results on the S-FLCKR dataset for both $f = 16$ ($m = 4$) and $f = 32$ ($m = 5$) (semantically guided). In particular, we provide an enlarged version of Fig. 5 from the main text, which had to be scaled down due to space constraints. Additionally, we use our sliding window approach (see Sec. 3) to produce high-resolution samples for the depth-to-image setting on RIN in Fig. 33 and Fig. 34, edge-to-image on IN in Fig. 35, stochastic superresolution on IN in Fig. 36, more examples on semantically guided landscape synthesis on S-FLCKR in Fig. 37 with $f = 16$ and in Fig. 38 with $f = 32$, and unconditional image generation on LSUN-CT (see Sec. 4.1) in Fig. 39. Moreover, for images of size 256×256, we provide results for generation from semantic layout on (i) ADE20K in Fig. 40 and (ii) COCO-Stuff in Fig. 41, depth-to-image on IN in Fig. 42, pose-guided person generation in Fig. 43 and class-conditional synthesis on RIN in Fig. 44.

E. 样本的最近邻

One advantage of likelihood-based generative models over, *e.g.*, GANs is the ability to evaluate NLL on training data and validation data to detect overfitting. To test this, we trained large models for face synthesis, which can easily overfit them, and retained two checkpoints on each dataset: One for the best validation NLL (at the 10th and 13th epoch for FFHQ and CelebA-HQ, respectively), and another for the best training NLL (at epoch 1000). We then produced samples from both checkpoints and retrieved nearest neighbors from the training data based on the LPIPS similarity metric [81]. The results are shown in Fig. 45, where it can be observed that the checkpoints with best training NLL (best train NLL) reproduce the training examples, whereas samples from the checkpoints with best validation NLL (best val. NLL) depict new faces which are not found in the training data.

Based on these results, we can conclude that early-stopping based on validation NLL can prevent overfitting. Furthermore, the bottleneck for our approach on face synthesis is given by the dataset size since it has the capacity to almost perfectly fit the training data. Unfortunately, FID scores cannot detect such an overfitting. Indeed, the best train NLL checkpoints achieve FID scores of 3.86 on CelebA-HQ and 2.68 on FFHQ, compared to 10.2 and 9.6 for the best val. NLL checkpoints. While validation NLL provides a way to detect overfitting for likelihood-based models, it is not clear if early-stopping based on it is optimal if one is mainly interested in the quality of samples. To address this and the evaluation of GANs, new metrics will be required which can differentiate between models that produce new, high-quality samples and those that simply reproduce the training data.

根据验证阶段的负对数似然值来看，我们的带类别条件的ImageNet模型并未出现过拟合现象，图46 中所展示的最近邻样本也进一步证明了该模型能够生成全新且高质量的样本。

F. 关于图像表示的排序问题

在变换器模型的“传统”应用领域——自然语言处理中，标记的顺序是由相应语言决定的。而针对图像及其离散表示，目前尚不清楚应采用何种线性排序方式。尤其需要注意的是，我们的滑动窗口方法依赖于行主序排序，因此我们研究了输入码本索引序列以下五种不同排列方式的性能：(i) **行主序**，或光栅扫描顺序，即图像表示所采用的排序方式。

[4] https://bit.ly/2FJkvhJ [5] https://tfhub.dev/deepmind/biggan-deep-256/1 [6] 这些样本是使用作者提供的官方实现复现的，该实现可访问地址为https://github.com/nvlabs/spade/

unrolled from top left to bottom right. (ii) **spiral out**, which incorporates the prior assumption that most images show a *centered* object. (iii) **z-curve**, also known as *z-order* or *morton curve*, which introduces the prior of *preserved locality* when mapping a 2D image representation onto a 1D sequence. (iv) **subsample**, where prefixes correspond to subsampled representations, see also [46]. (v) **alternate**, which is related to *row major*, but alternates the direction of unrolling every row. (vi) **spiral in**, a reversed version of *spiral out* which provides the most context for predicting the center of the image. A graphical visualization of these permutation variants is shown in Fig. 47. Given a *VQGAN* trained on ImageNet, we train a transformer for each permutation in a controlled setting, i.e. we fix initialization and computational budget.

Results Fig.47 depicts the evolution of negative log-likelihood for each variant as a function of training iterations, with final values given by (i) 4.767, (ii) 4.889, (iii) 4.810, (iv) 5.015, (v) 4.812, (vi) 4.901. Interestingly, *row major* performs best in terms of this metric, whereas the more hierarchical *subsample* prior does not induce any helpful bias. We also include qualitative samples in Fig. 48 and observe that the two worst performing models in terms of NLL (*subsample* and *spiral in*) tend to produce more textural samples, while the other variants synthesize samples with much more recognizable structures. Overall, we can conclude that the autoregressive codebook modeling is *not* permutation-invariant, but the common *row major* ordering [71, 8] outperforms other orderings.

unrolled from top left to bottom right. (ii) **spiral out**, which incorporates the prior assumption that most images show a *centered* object. (iii) **z-curve**, also known as *z-order* or *morton curve*, which introduces the prior of *preserved locality* when mapping a 2D image representation onto a 1D sequence. (iv) **subsample**, where prefixes correspond to subsampled representations, see also [46]. (v) **alternate**, which is related to *row major*, but alternates the direction of unrolling every row. (vi) **spiral in**, a reversed version of *spiral out* which provides the most context for predicting the center of the image. A graphical visualization of these permutation variants is shown in Fig. 47. Given a *VQGAN* trained on ImageNet, we train a transformer for each permutation in a controlled setting, i.e. we fix initialization and computational budget.

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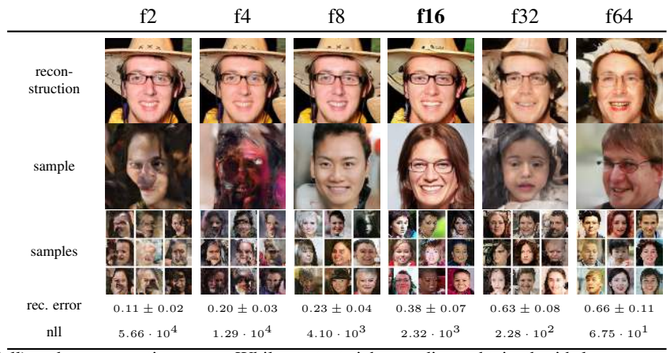


Figure 11. Trade-off between negative log-likelihood (nll) and reconstruction error. While context-rich encodings obtained with large factors f allow the transformer to effectively model long-range interactions, the reconstructions capabilities and hence quality of samples suffer after a critical value (here, $f = 16$). For more details, see Sec. C.

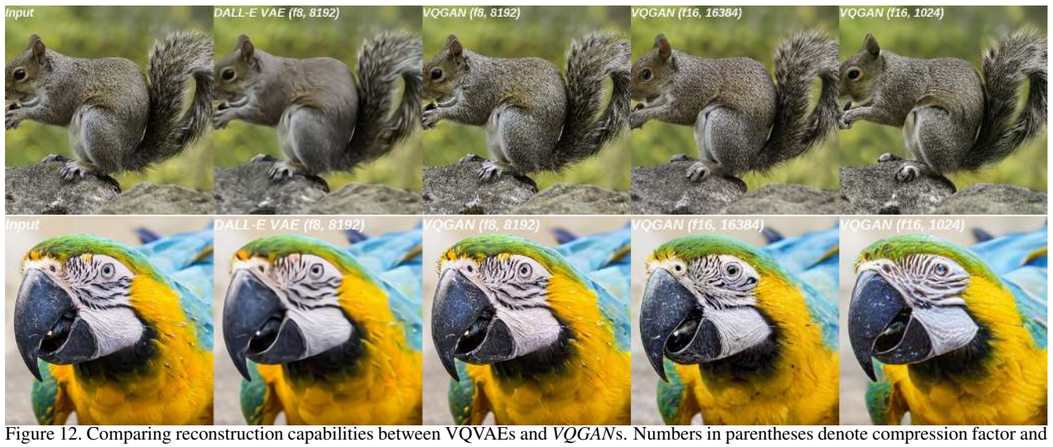


Figure 12. Comparing reconstruction capabilities between VQVAEs and *VQGAN*s. Numbers in parentheses denote compression factor and codebook size. With the same compression factor and codebook size, *VQGAN*s produce more realistic reconstructions compared to blurry reconstructions of VQVAEs. This enables increased compression rates for *VQGAN* while retaining realistic reconstructions. See Sec. C.

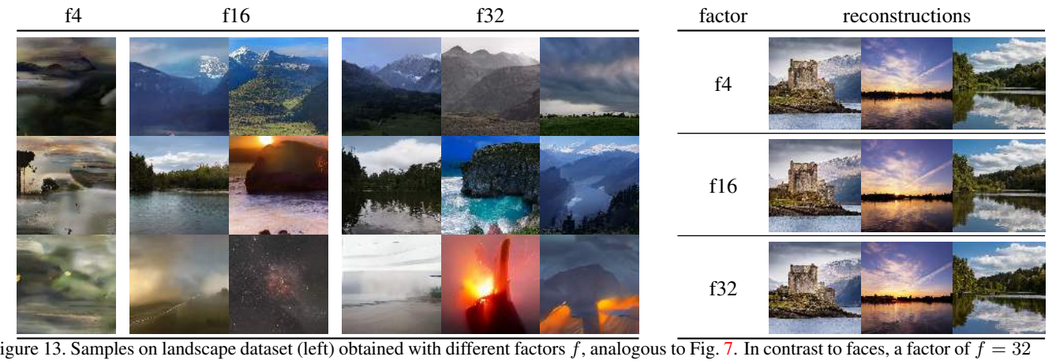


Figure 13. Samples on landscape dataset (left) obtained with different factors f, analogous to Fig. 7. In contrast to faces, a factor of $f = 32$ still allows for faithful reconstructions (right). See also Sec. C.

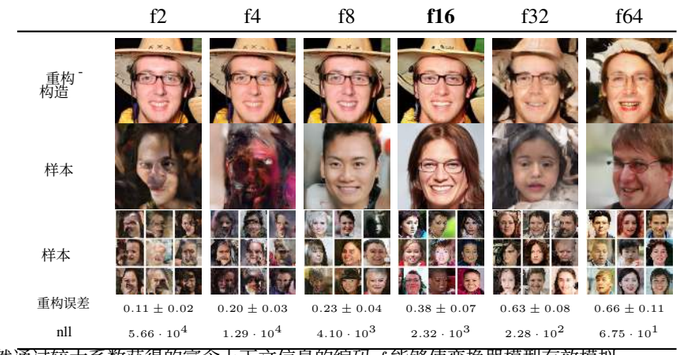


图11. 负对数似然与重建误差之间的权衡关系。虽然通过较大系数获得的富含上下文信息的编码 f 能够使变换器模型有效模拟长距离交互，但一旦超过某个临界值（此处为 $f = 16$），样本的重建能力及其质量便会下降。更多细节请参见第C节。



图12. VQVAE模型与*VQGAN*模型在重建能力上的对比。括号内的数字分别表示压缩因子和码本大小。在相同的压缩因子及码本大小条件下，*VQGAN*模型生成的重建结果比VQVAE模型产生的模糊重建结果更为逼真。这使得*VQGAN* 能够在保持高逼真度重建结果的同时实现更高的压缩率。详见第C章。

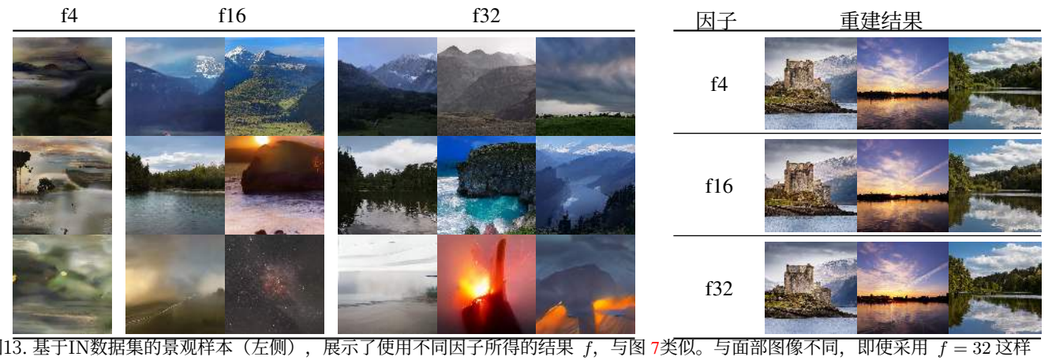


图13. 基于IN数据集的景观样本（左侧），展示了使用不同因子所得的结果 f，与图 7类似。与面部图像不同，即使采用 $f = 32$ 这样的因子，依然能够获得高度逼真的重构结果（右侧）。另请参阅第 C节。



Figure 14. Qualitative assessment of various models for class-conditional image synthesis on ImageNet. Depicted classes: *28: spotted salamander* (top) and *97: drake* (bottom). We report class labels as in VQVAE-2 [61].



图14. 定性评估 火蝾螈（顶部）以及97：龙 *ed*

图注：对不同模型在*ImageNet*数据集上实现类别条件图像合成能力的对比分析。 （底部）此处所示的类别标签与VQVAE-2 [61]中的标注一致。

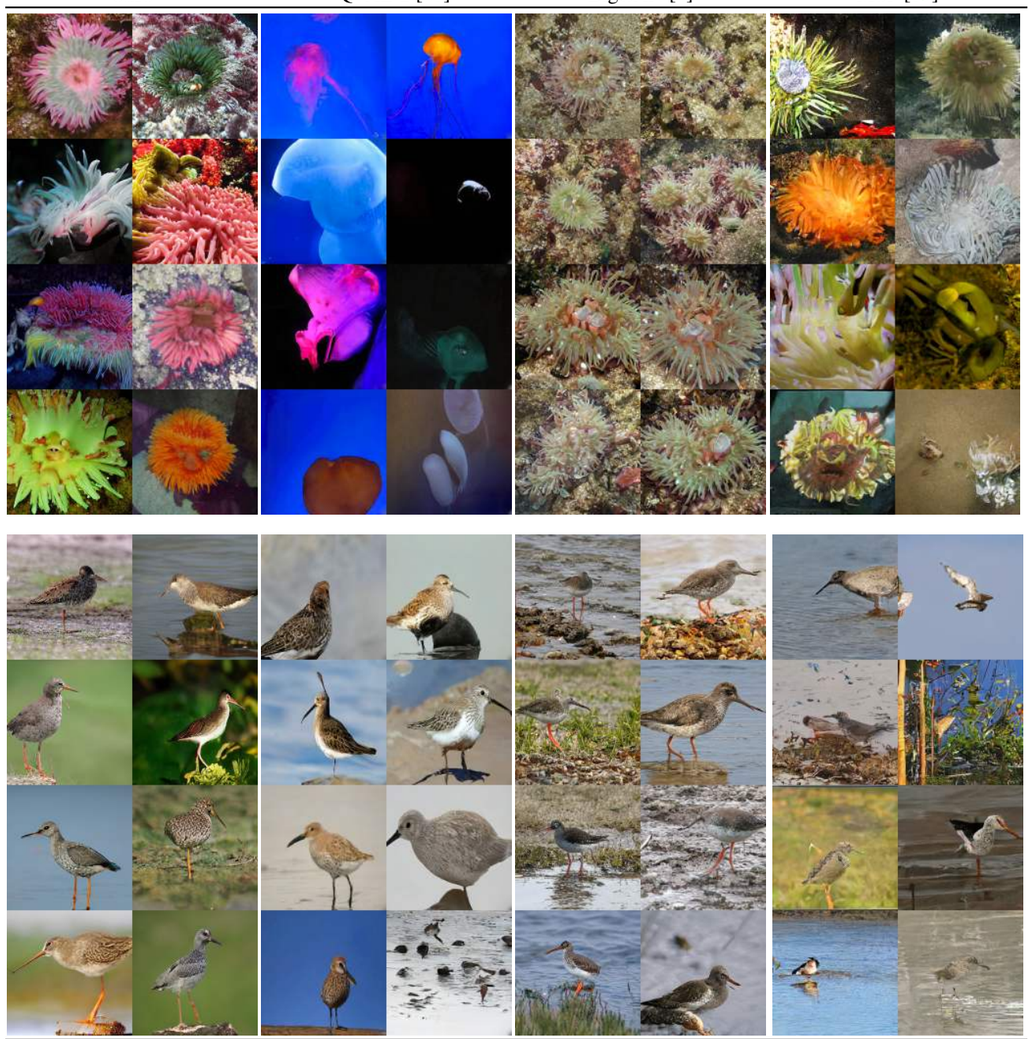


Figure 15. Qualitative assessment of various models for class-conditional image synthesis on ImageNet. Depicted classes: *108: sea anemone* (top) and *141: redshank* (bottom). We report class labels as in VQVAE-2 [61].

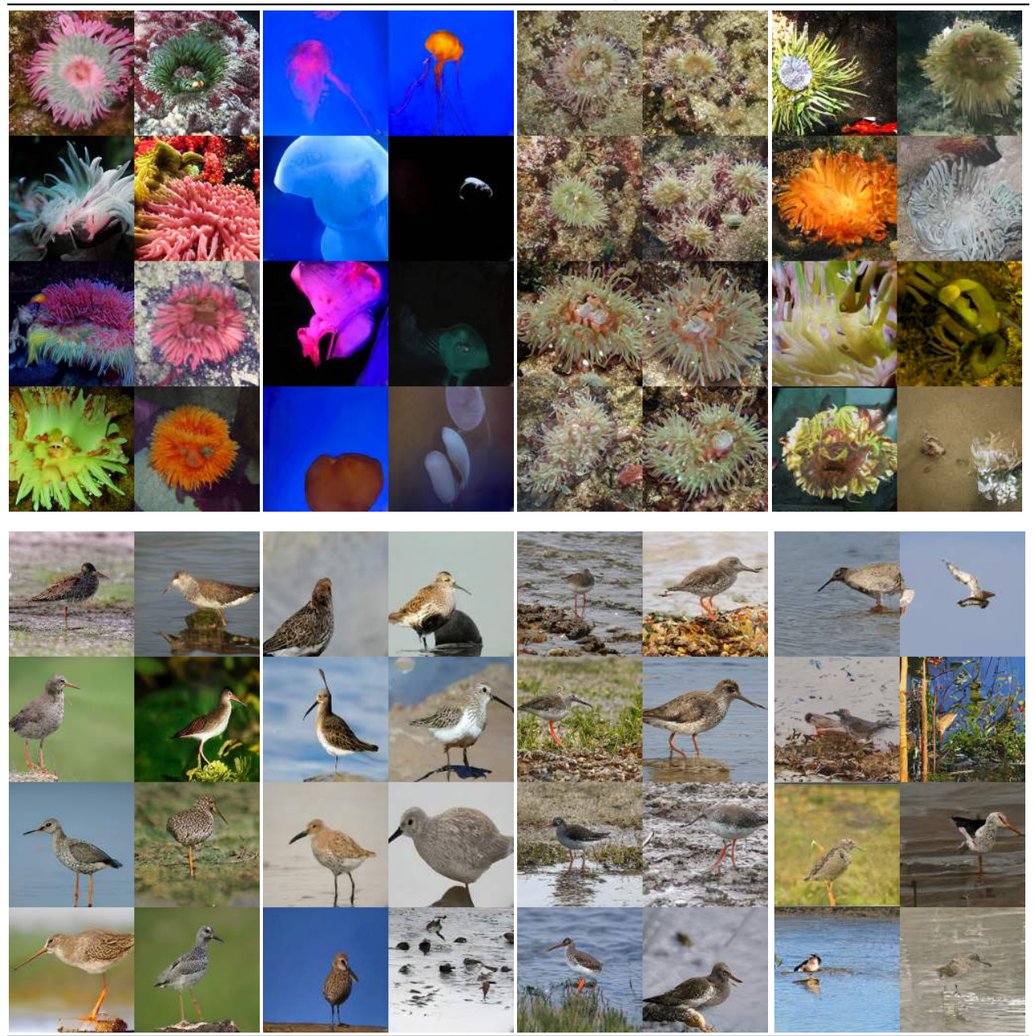


图15. 定性评估 海葵（顶部）以及141：redsha *ea*

*"Qualitative assess"*应为"定性评估"的笔误***nk***（底部）。此处呈现的类别标签与*VQVAE*-中的完全一致。

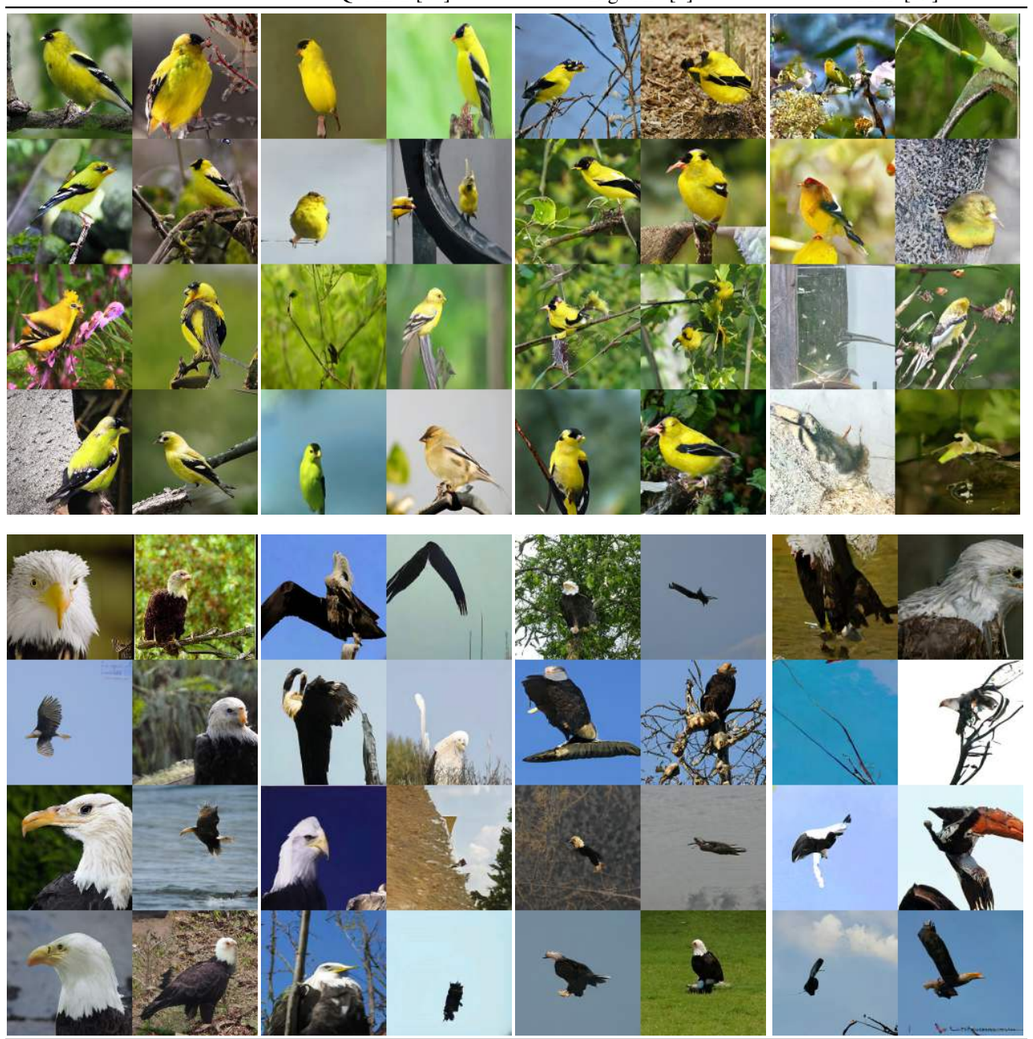


Figure 16. Qualitative assessment of various models for class-conditional image synthesis on ImageNet. Depicted classes: *11: goldfinch* (top) and *22: bald eagle* (bottom).

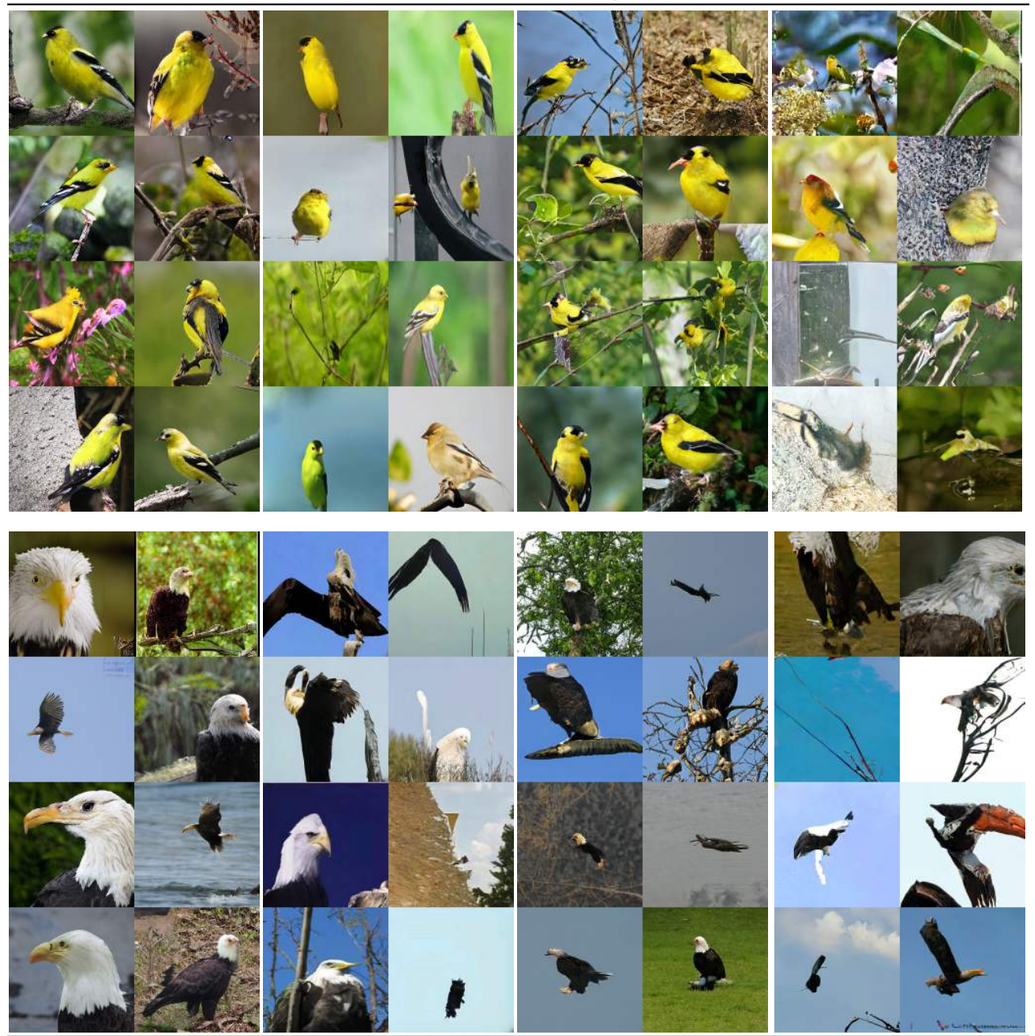


图16. 定性评估 (顶部)以及*22*：白头海雕（底部） *ch*
（上方）和*22*：白头海雕（下方）m).

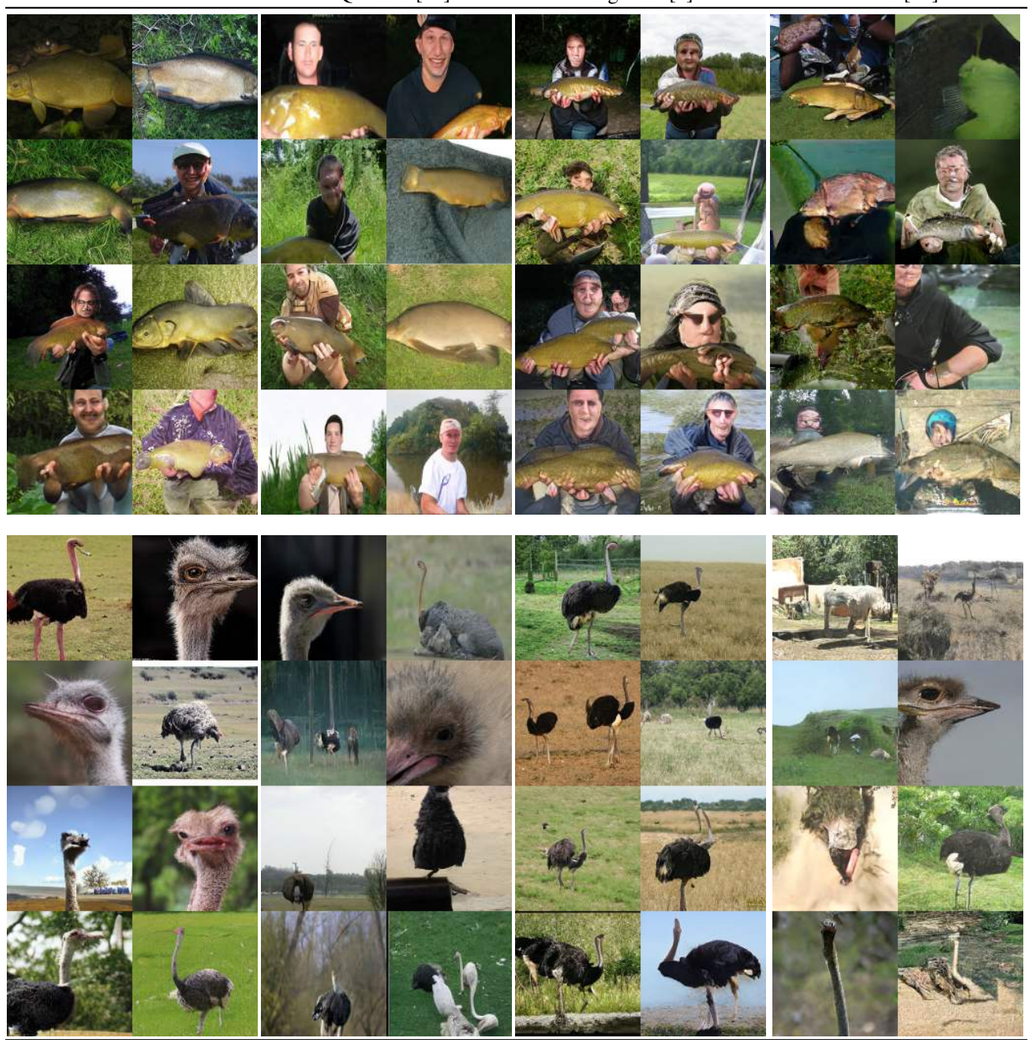


Figure 17. Qualitative assessment of various models for class-conditional image synthesis on ImageNet. Depicted classes: *0: tench* (top) and *9: ostrich* (bottom).

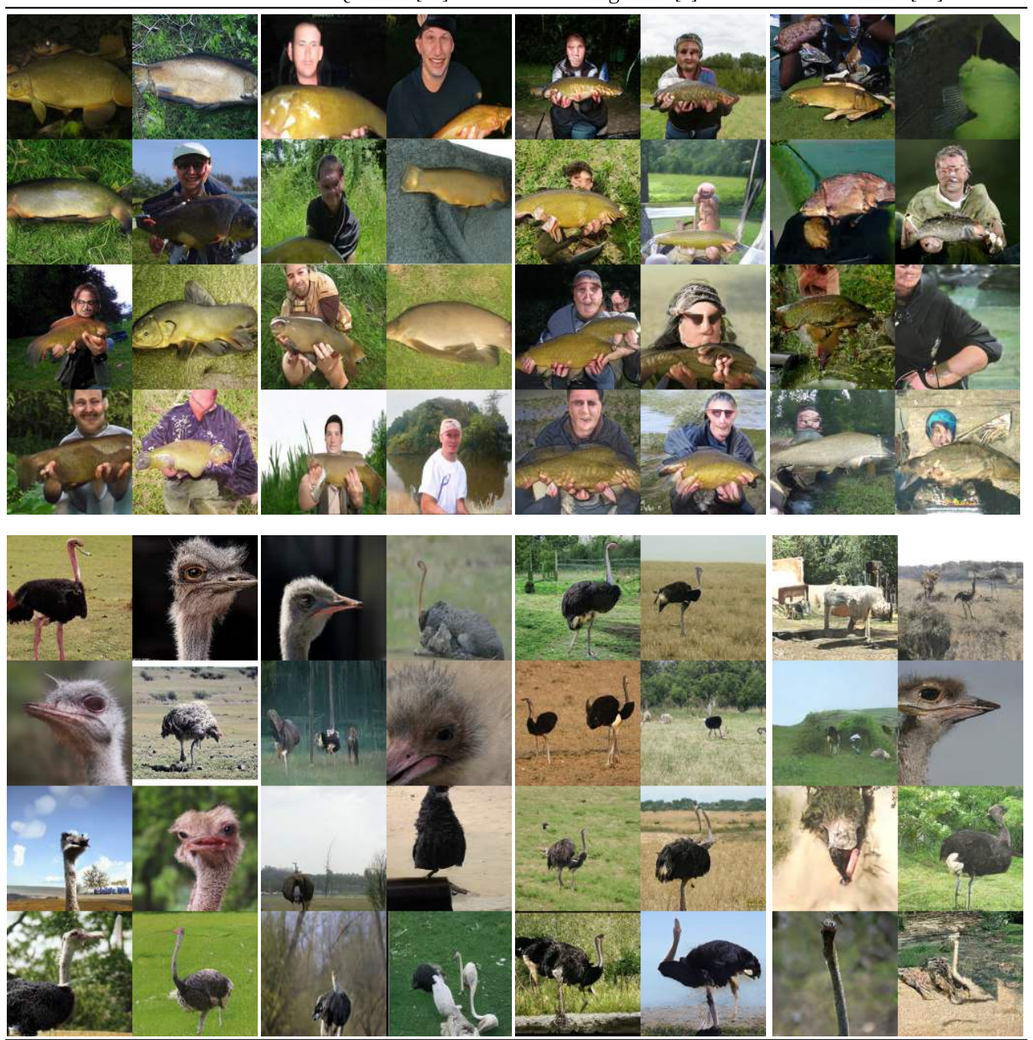


图17. 定性评估 基于ImageNet的各类**有条件图像合成**AR模型。所示类别：*0*：青鳉（至 p) *9*：鸵鸟 （底部）。

933: *cheeseburger*

acc. rate 1.0 | acc. rate 0.5 | acc. rate 0.1



992: *agaric*

acc. rate 1.0 | acc. rate 0.5 | acc. rate 0.1



200: *tibetian terrier*

acc. rate 1.0 | acc. rate 0.5 | acc. rate 0.1

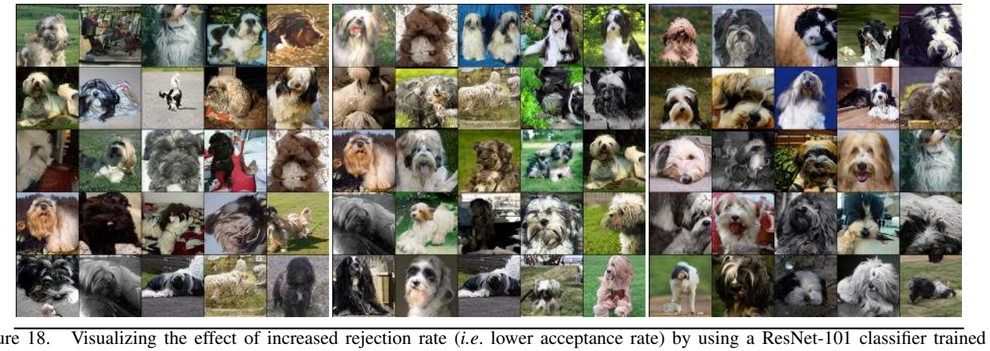


Figure 18. Visualizing the effect of increased rejection rate (*i.e.* lower acceptance rate) by using a ResNet-101 classifier trained on ImageNet and samples from our class-conditional ImageNet model. Higher rejection rates tend to produce images showing more central, recognizable objects compared to the unguided samples. Here, $k = 973$, $p = 1.0$ are fixed for all samples. Note that $k = 973$ is the *effective* size of the *VQGAN*'s codebook, *i.e.* it describes how many entries of the codebook with $\dim \mathcal{Z} = 16384$ are actually used.

933： 芝士汉堡

准确率 1.0 | 准确率 0.5 | 准确率 0.1



992： 蘑菇

准确率 1.0 | 准确率：0.5 | 准确率：0.1



200： 藏獒

准确率：1.0 | 准确率：0.5 | 准确率：0.1



图18：利用在ImageNet数据集以及我们开发的带类别条件的ImageNet模型生成的样本上训练得到的ResNet-101网络，展示了拒绝率上升（即接受率下降）所带来的影响。与无引导的样本相比，较高的拒绝率往往能生成更多包含居中且易于识别的物体的图像。此处，所有样本的 $k = 973$和 $p = 1.0$ 值均为固定值。需注意的是， $k = 973$ 指的是*VQGAN*的码本的有效大小，即它表示实际被使用的、维度为 $\mathcal{Z} = 16384$ 的码本条目数量。

933: *cheeseburger*

$k = 973$ $k = 300$ $k = 100$



992: *agaric*

$k = 973$ $k = 300$ $k = 100$



200: *tibetian terrier*

$k = 973$ $k = 300$ $k = 100$

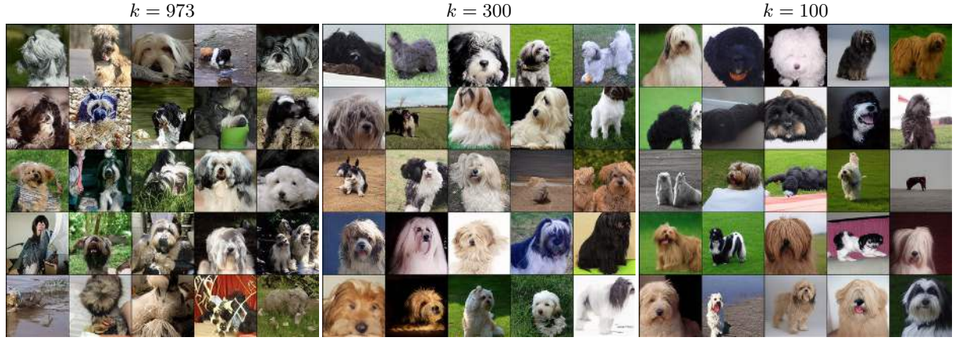


Figure 19. Visualizing the effect of varying k in top-k sampling (*i.e*. truncating the probability distribution per image token) by using a ResNet-101 classifier trained on ImageNet and samples from our class-conditional ImageNet model. Lower values of k produce more uniform, low-entropic images compared to samples obtained with full k. Here, an acceptance rate of 1.0 and $p = 1.0$ are fixed for all samples. Note that $k = 973$ is the *effective* size of the *VQGAN*'s codebook, *i.e*. it describes how many entries of the codebook with $\dim \mathcal{Z} = 16384$ are actually used.

933：芝士汉堡

$k = 973$ $k = 300$ $k = 100$



992: 蘑菇

$k = 973$ $k = 300$ $k = 100$



200：藏獒梗

$k = 973$ $k = 300$ $k = 100$

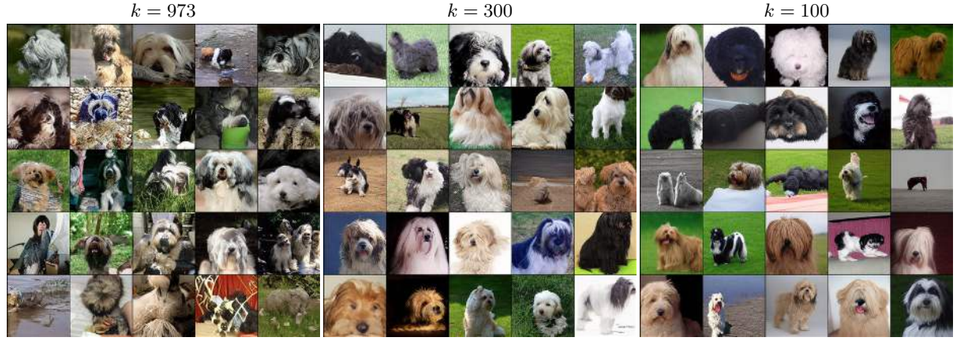


图19. 利用在ImageNet数据集上训练的ResNet-101分类器以及我们基于带类别条件的ImageNet模型生成的样本，可视化top-k采样算法中 k 参数变化（即对每个图像标记的概率分布进行截断）所带来的影响。与使用完整 k值得到的样本相比，较低的 k 值能够生成更加均匀、熵值更低的图像。此处所有样本的接受率均固定为1.0， $p = 1.0$ 值也保持不变。需要说明的是，$k = 973$ 指的是*VQGAN*码本的有效大小，即它表示实际被使用的具有$\dim \mathcal{Z} = 16384$ 维度的码本条目数量。

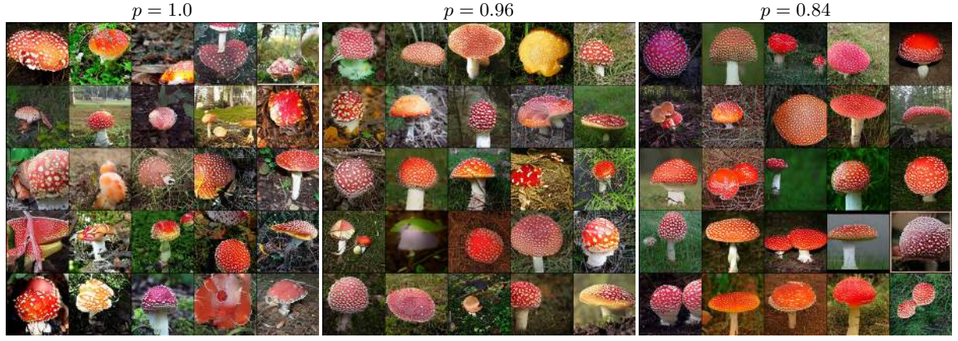
933: *cheeseburger*

$p = 1.0$ $p = 0.96$ $p = 0.84$



992: *agaric*

$p = 1.0$ $p = 0.96$ $p = 0.84$



200: *tibetian terrier*

$p = 1.0$ $p = 0.96$ $p = 0.84$

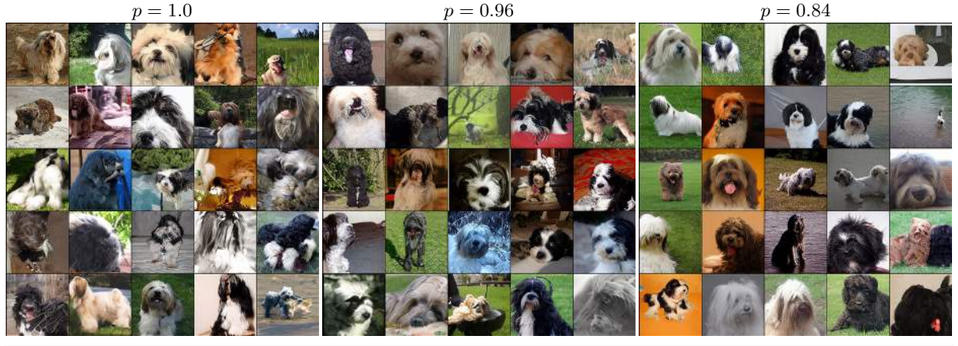


Figure 20. Visualizing the effect of varying p in top-p sampling (or *nucleus sampling* [27]) by using a ResNet-101 classifier trained on ImageNet and samples from our class-conditional ImageNet model. Lowering p has similar effects as decreasing k, see Fig. 19. Here, an acceptance rate of 1.0 and $k = 973$ are fixed for all samples.

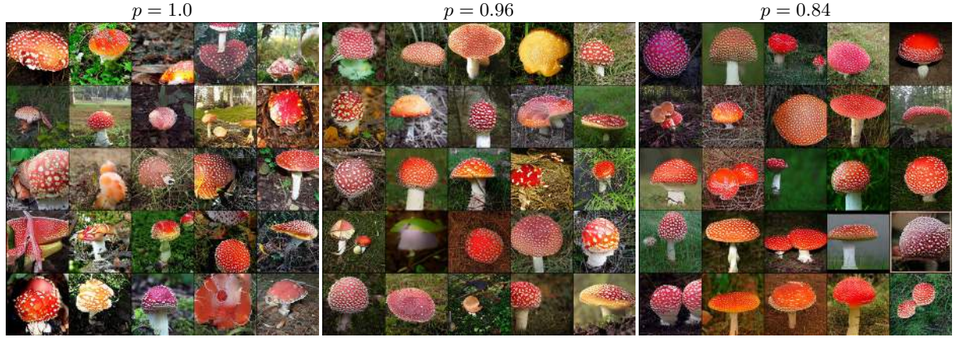
933： 芝士汉堡

$p = 1.0$ $p = 0.96$ $p = 0.84$



992： 蘑菇

$p = 1.0$ $p = 0.96$ $p = 0.84$



200： 藏獒

$p = 1.0$ $p = 0.96$ $p = 0.84$

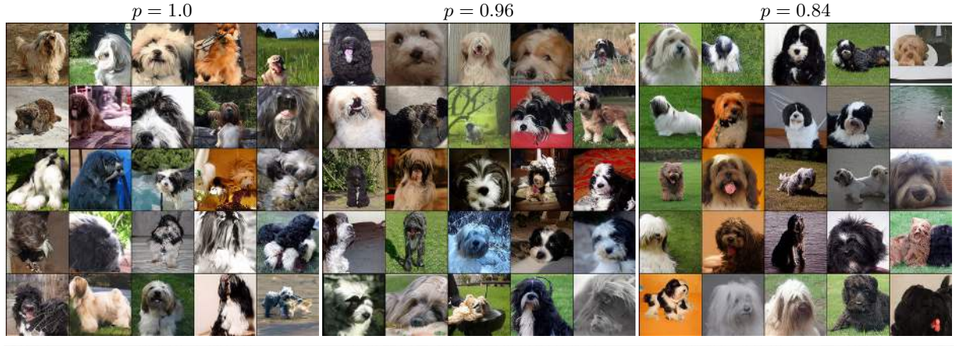


图20. 利用在ImageNet数据集上训练的ResNet-101分类器以及我们基于带类别条件的ImageNet模型生成的样本，展示调整top-p采样（或核采样算法）中的参数p所带来的影响。降低该参数所产生的效果与减少其他相关参数的效果类似，详见图19。在此实验中，所有样本的接受率均固定为1.0。 $k = 973$ 。

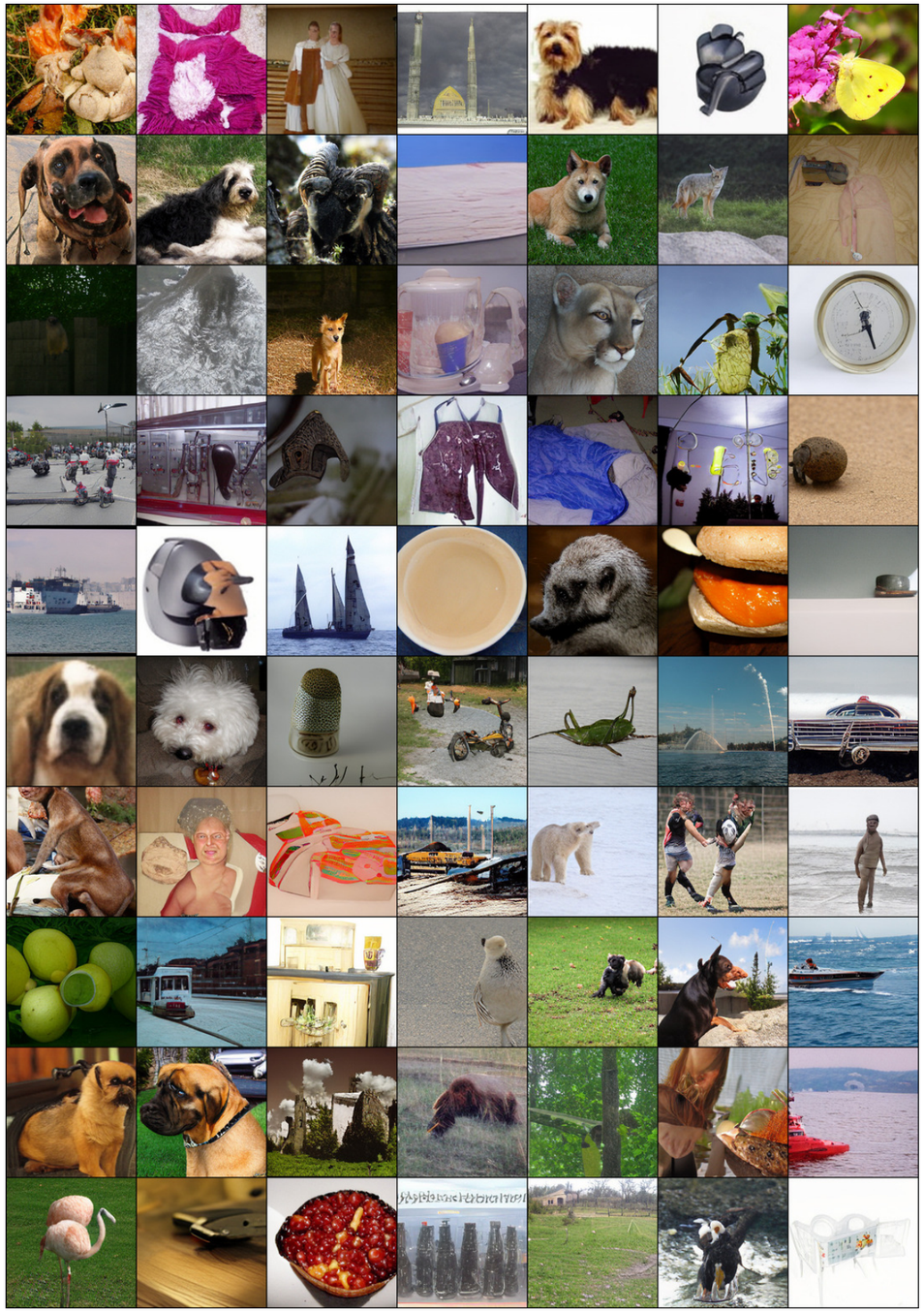


Figure 21. Random samples on 256×256 class-conditional ImageNet with $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$, $p = 1.0$, acceptance rate 1.0. FID: 17.04, IS: 70.6 ± 1.8. Please see https://git.io/JLlvY for an uncompressed version.

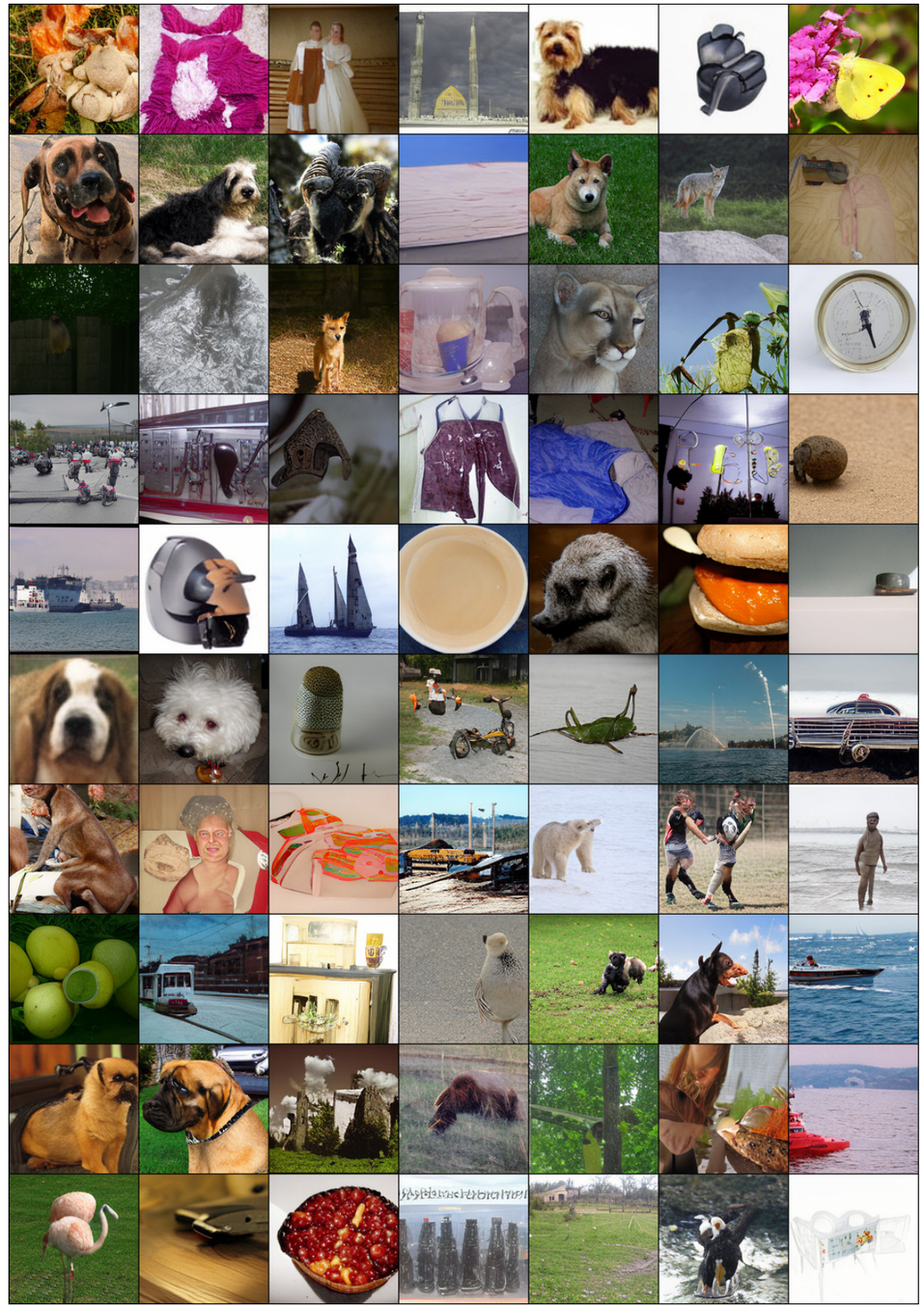


图21. 在带类别条件的ImageNet数据集上抽取的随机样本，其接受率为……FID指标为……IS指标为……详情请参见 256×256 带类别条件的ImageNet数据集 $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$，$p = 1.0$ 1 0 17 04, 70.6 ± 1.8 ，接受率… …FID指标：……IS指标：……完整内容可访问https://git.io/JLlvY查看。23

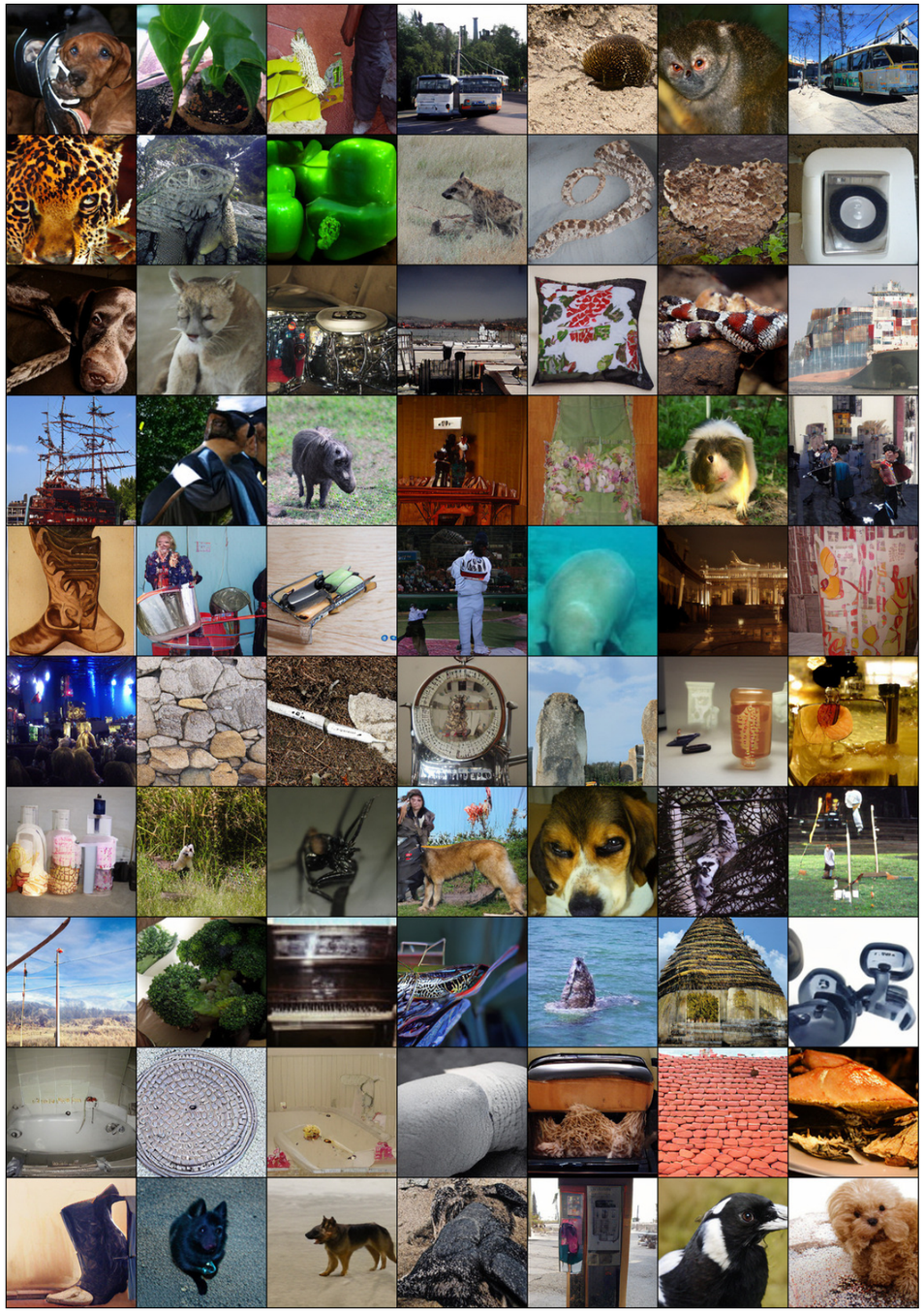


Figure 22. Random samples on 256×256 class-conditional ImageNet with $k = 600$, $p = 1.0$, acceptance rate 0.05. FID: 5.20, IS: 280.3 ± 5.5. Please see https://git.io/JLlvY for an uncompressed version.

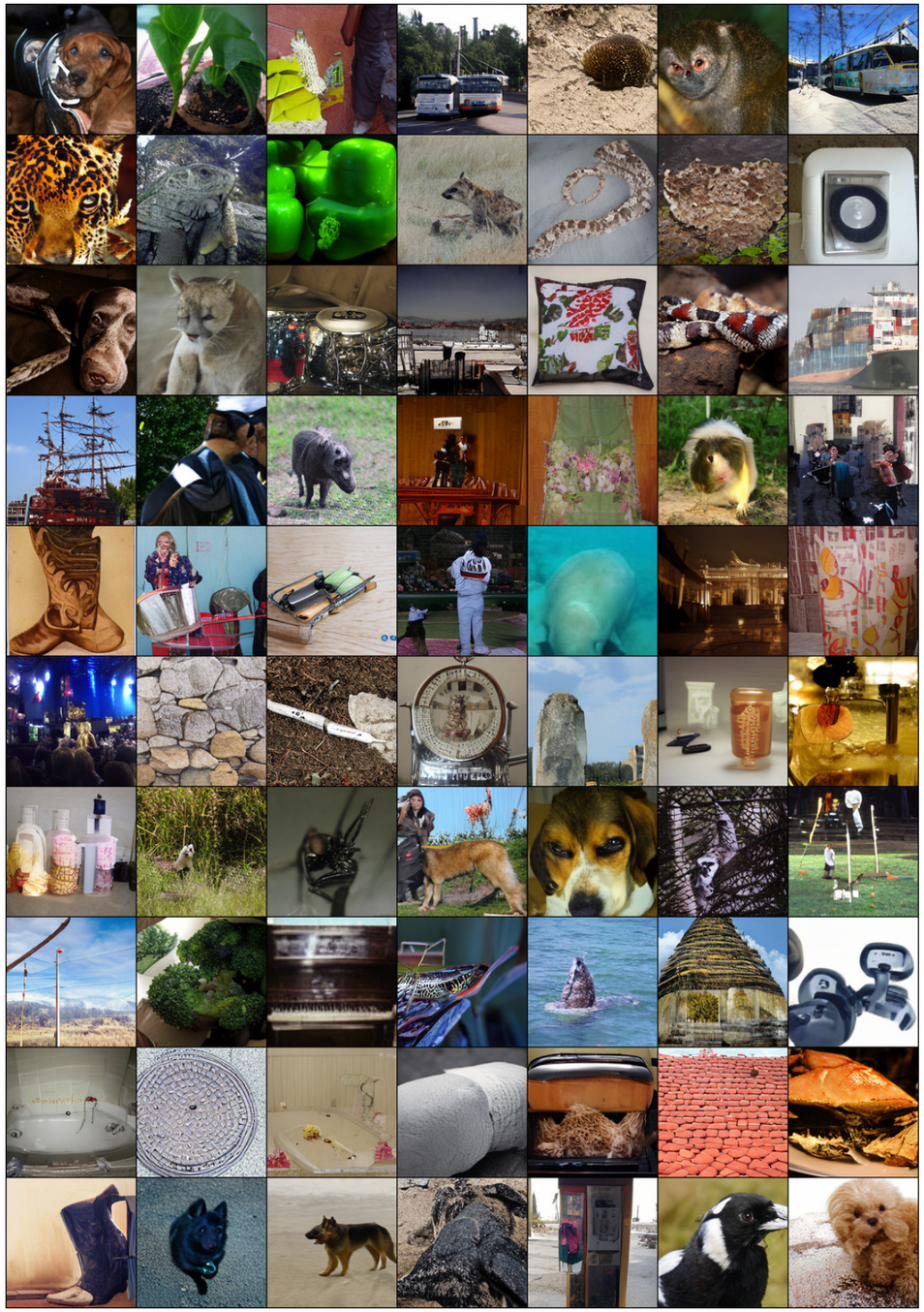


图22. 在带类别条件的ImageNet数据集上抽取的随机样本，接受率为0.05。FID指标为5.20，IS指标为280.3 ± 5.5。如需查看未压缩版本，请访问https://git.io/JLlvY。

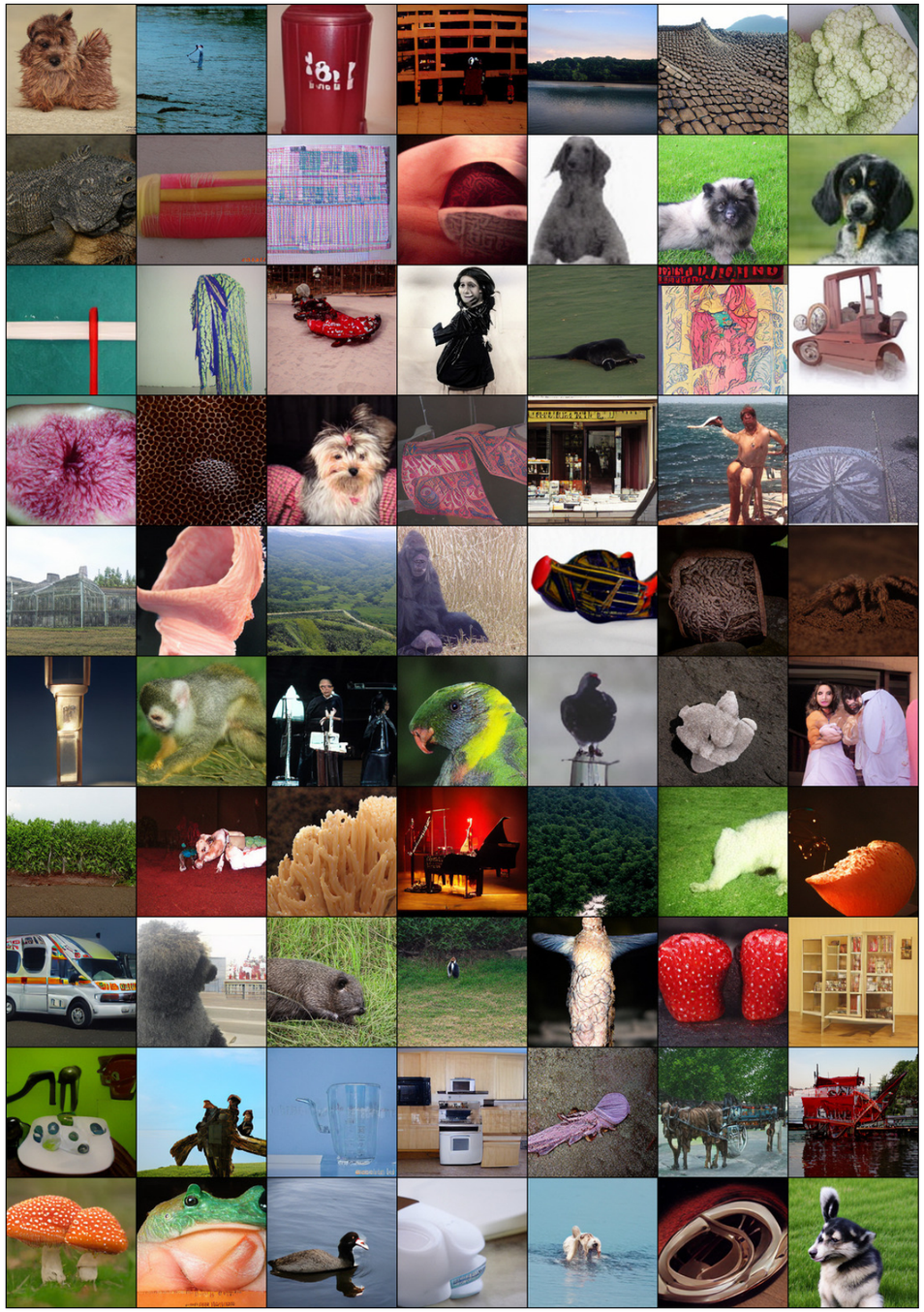


Figure 23. Random samples on 256×256 class-conditional ImageNet with $k = 250$, $p = 1.0$, acceptance rate 1.0. FID: 15.98, IS: 78.6 ± 1.1. Please see https://git.io/JLlvY for an uncompressed version.

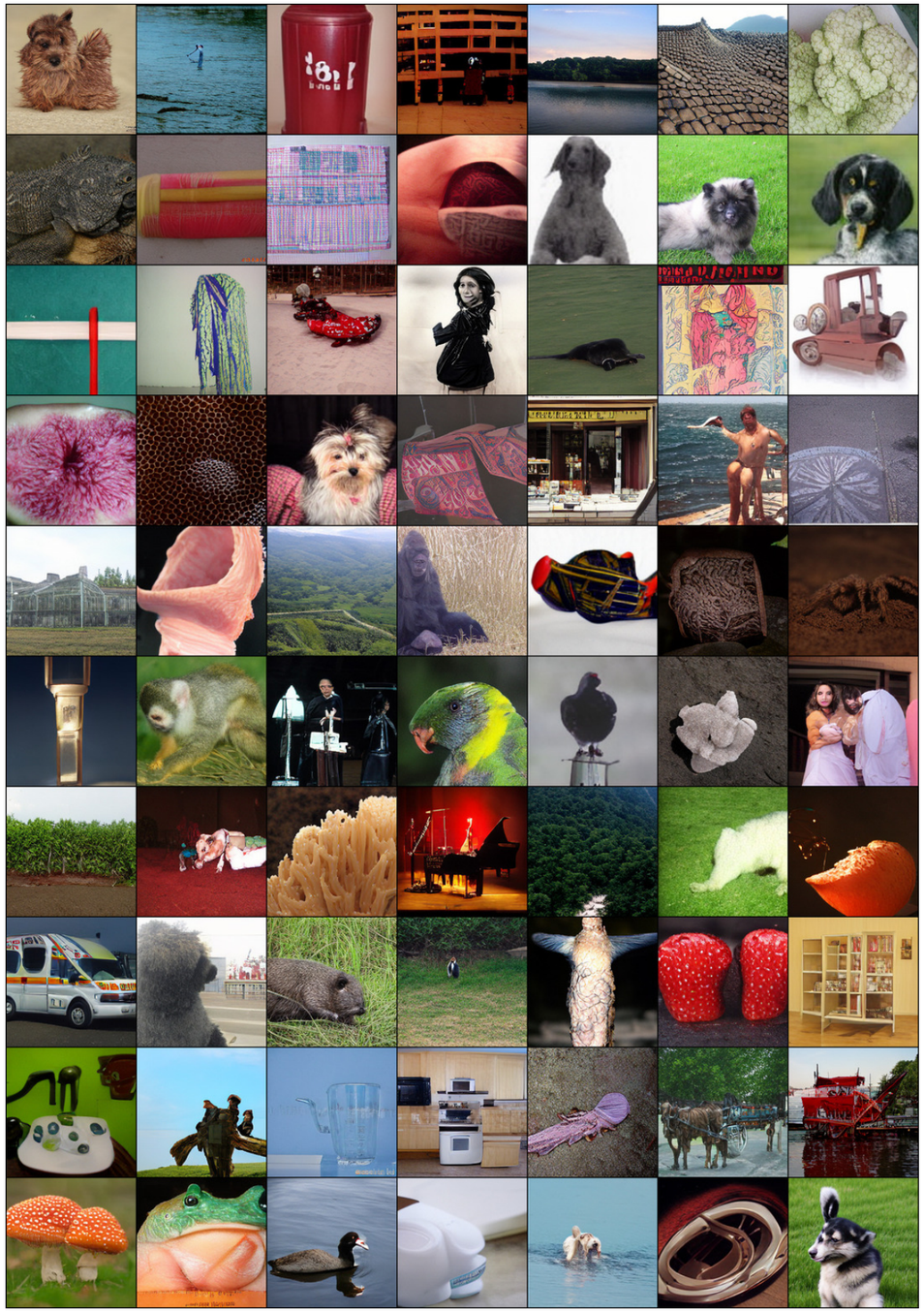


图23. 256×256 带类别条件的ImageNet数据集 $k = 250$上的随机样本， $p = 1.0$，其接受率为1.0。FID指标为15.98，IS指标为78.6 ± 1.1。如需查看未压缩版本，请访问https://git.io/JLlvY



Figure 24. Random samples on 256×256 class-conditional ImageNet with $k = 973$, $p = 0.88$, acceptance rate 1.0. FID: 15.78, IS: 74.3 ± 1.8. Please see https://git.io/JLlvY for an uncompressed version.

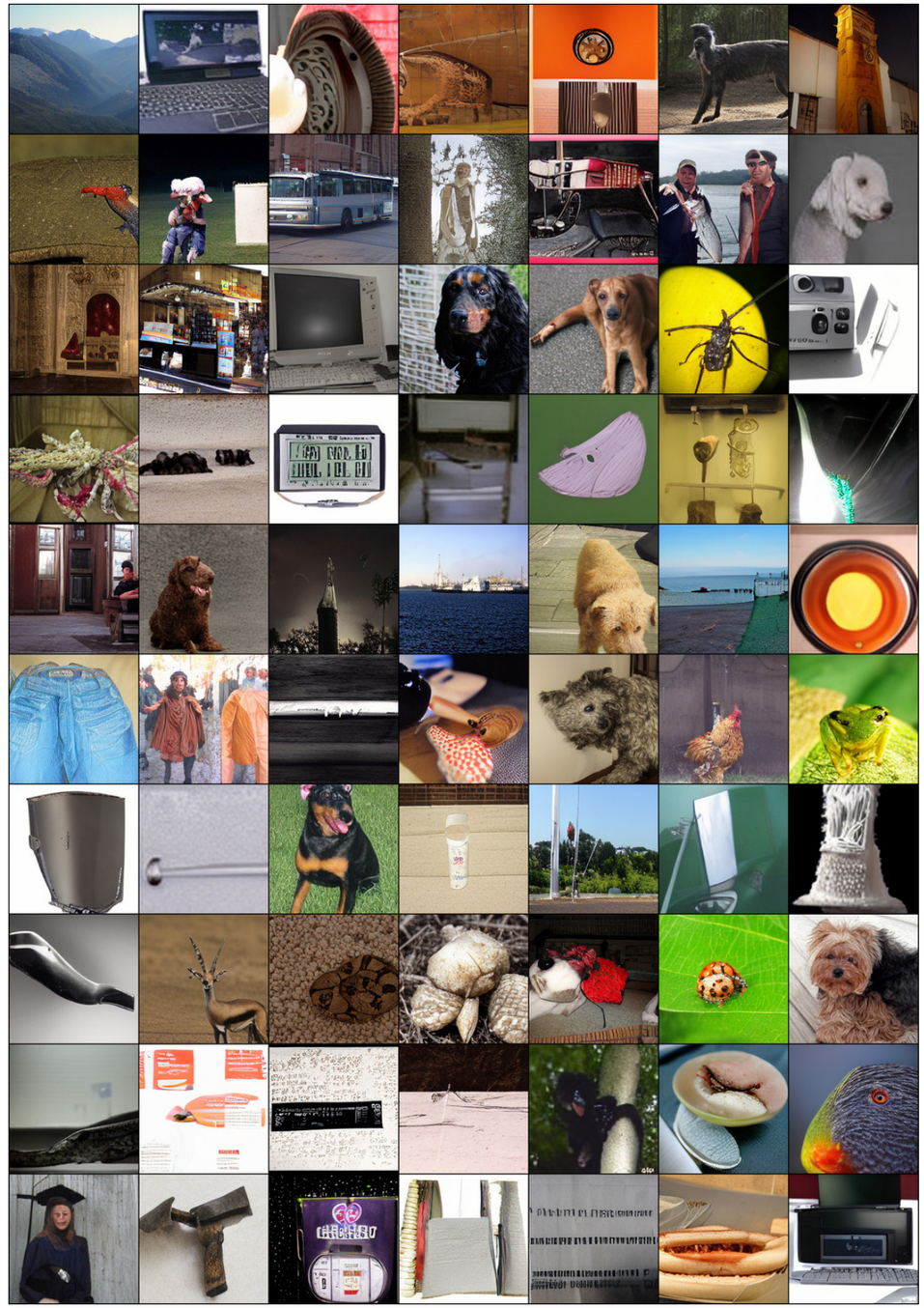


图24. 256×256 带类别条件的ImageNet数据集 $k = 973$上的随机样本， $p = 0.88$，其接受率为1.0。对应的FID指标为15.78，IS指标为74.3 ± 1.8。如需查看未压缩版本，请访问https://git.io/JLlvY 。



Figure 25. Random samples on 256×256 class-conditional ImageNet with $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$, $p = 1.0$, acceptance rate 0.005. FID: 6.59, IS: 402.7 ± 2.9. Please see https://git.io/JLlvY for an uncompressed version.



图25. 256×256 带类别条件的ImageNet数据集 $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$ 上的随机样本，以及对应的 $p = 1.0$ 0 005 6 59, 402.7 ± 2.9 接受率、FID指标和IS指标数值。完整内容可访问https://git.io/JLlvY查看未压缩版本。27

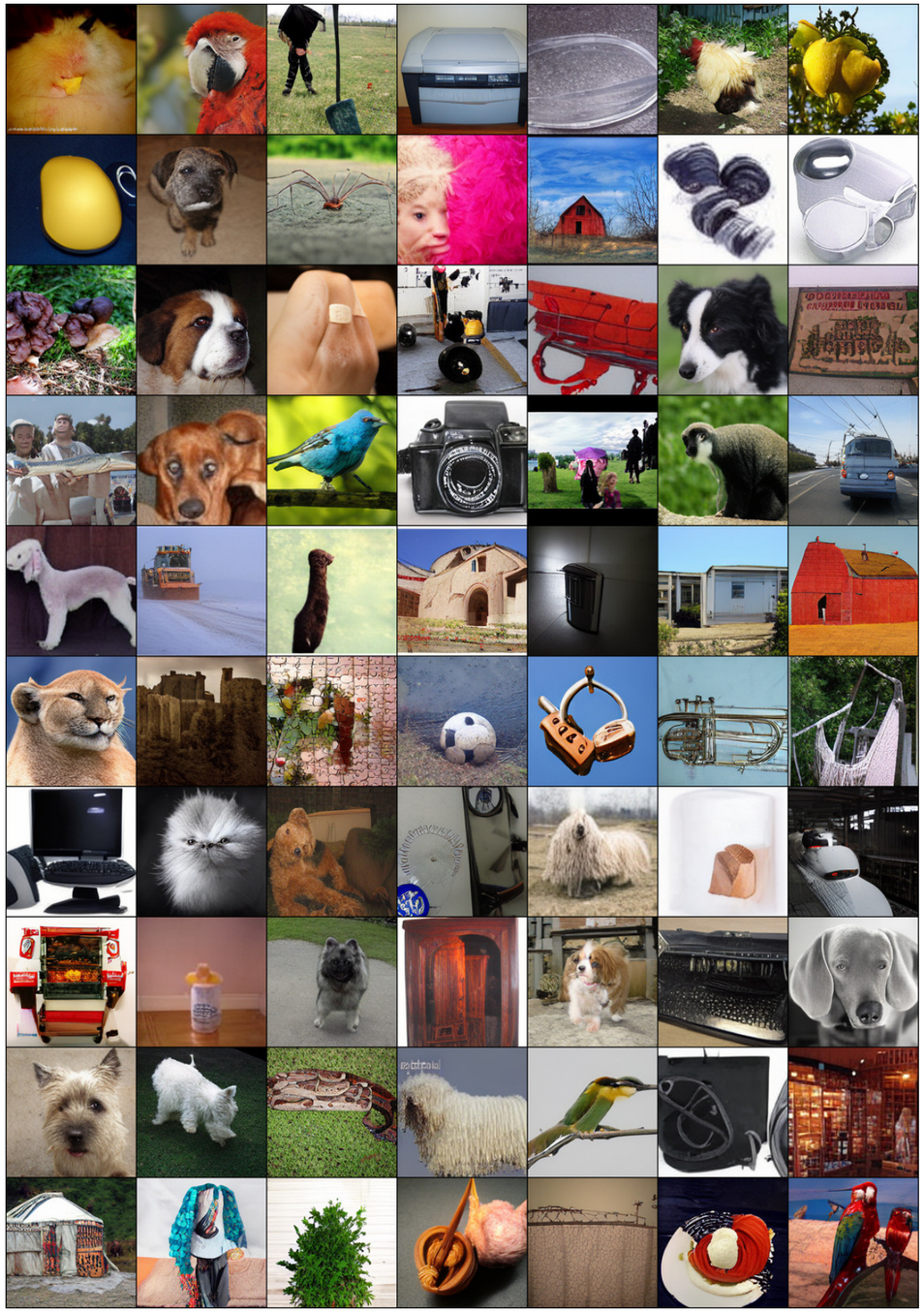


Figure 26. Random samples on 256×256 class-conditional ImageNet with $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$, $p = 1.0$, acceptance rate 0.05. FID: 5.88, IS: 304.8 ± 3.6. Please see https://git.io/JLlvY for an uncompressed version.

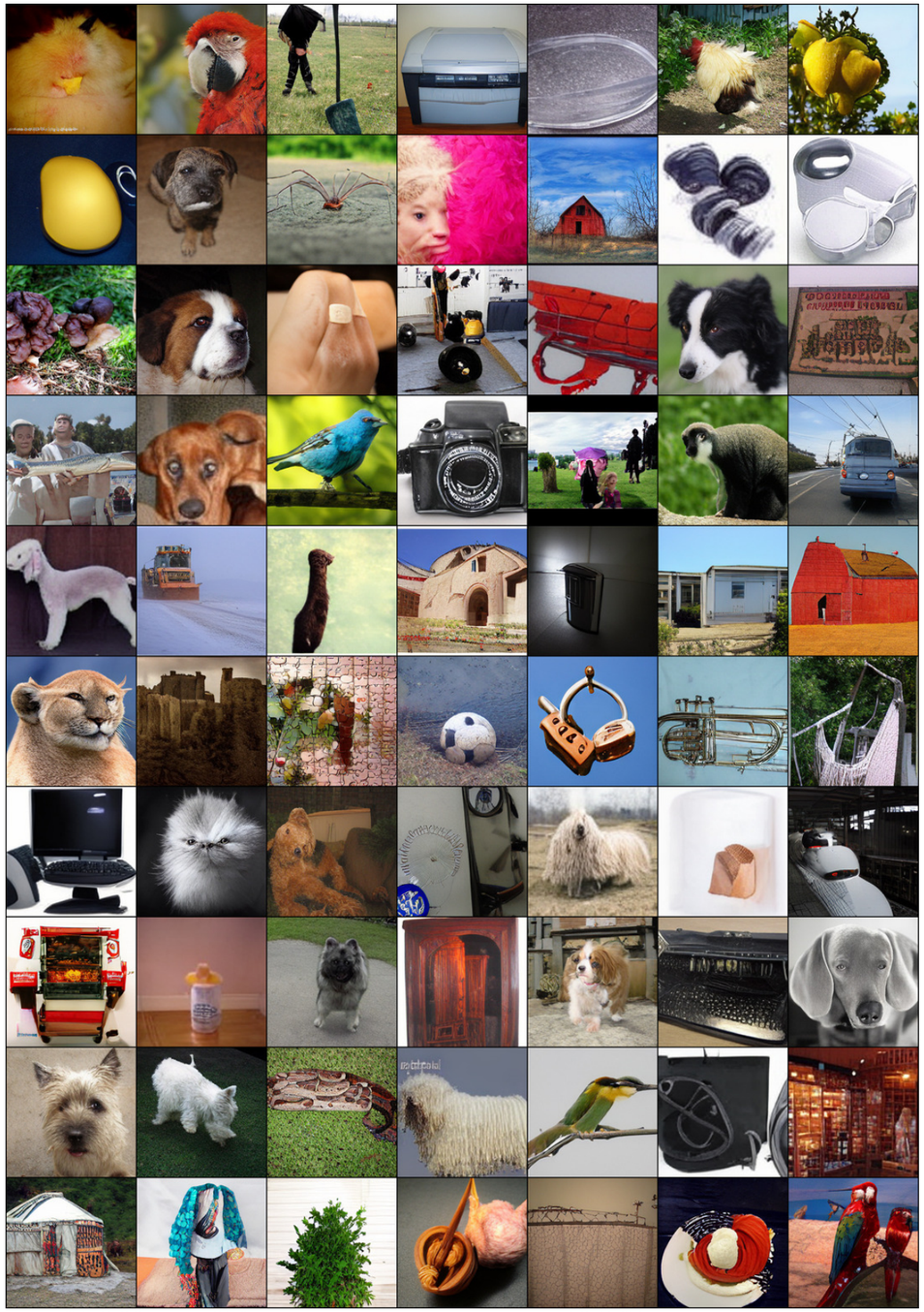


图26. 256×256 带类别条件的ImageNet数据集 $k \in [100, 200, 250, 300, 350, 400, 500, 600, 800, 973]$ 上的随机样本，以及 $p = 1.0$ 0 05 5 88, 304.8 ± 3.6 对应的接受率、FID指标和IS指标数值。
如需未压缩版本，请访问https://git.io/JLlvY，编号为28。

conditioning

ours (top) vs iGPT [8] (bottom)

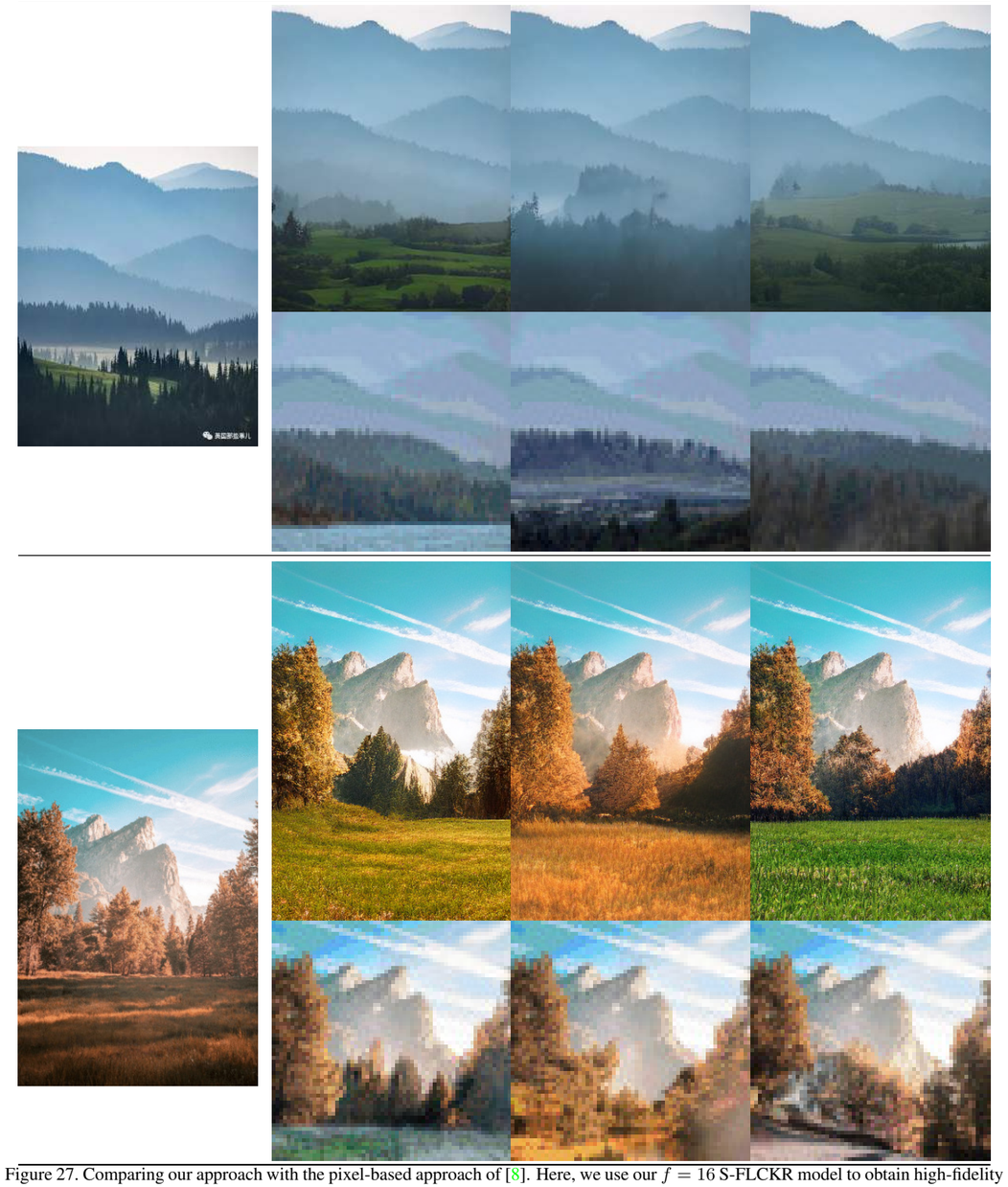


Figure 27. Comparing our approach with the pixel-based approach of [8]. Here, we use our $f = 16$ S-FLCKR model to obtain high-fidelity image completions of the inputs depicted on the left (half completions). For each conditioning, we show three of our samples (top) and three of [8] (bottom).

条件作用

我们的(上方)与iGPT的对比 [8] (下方)

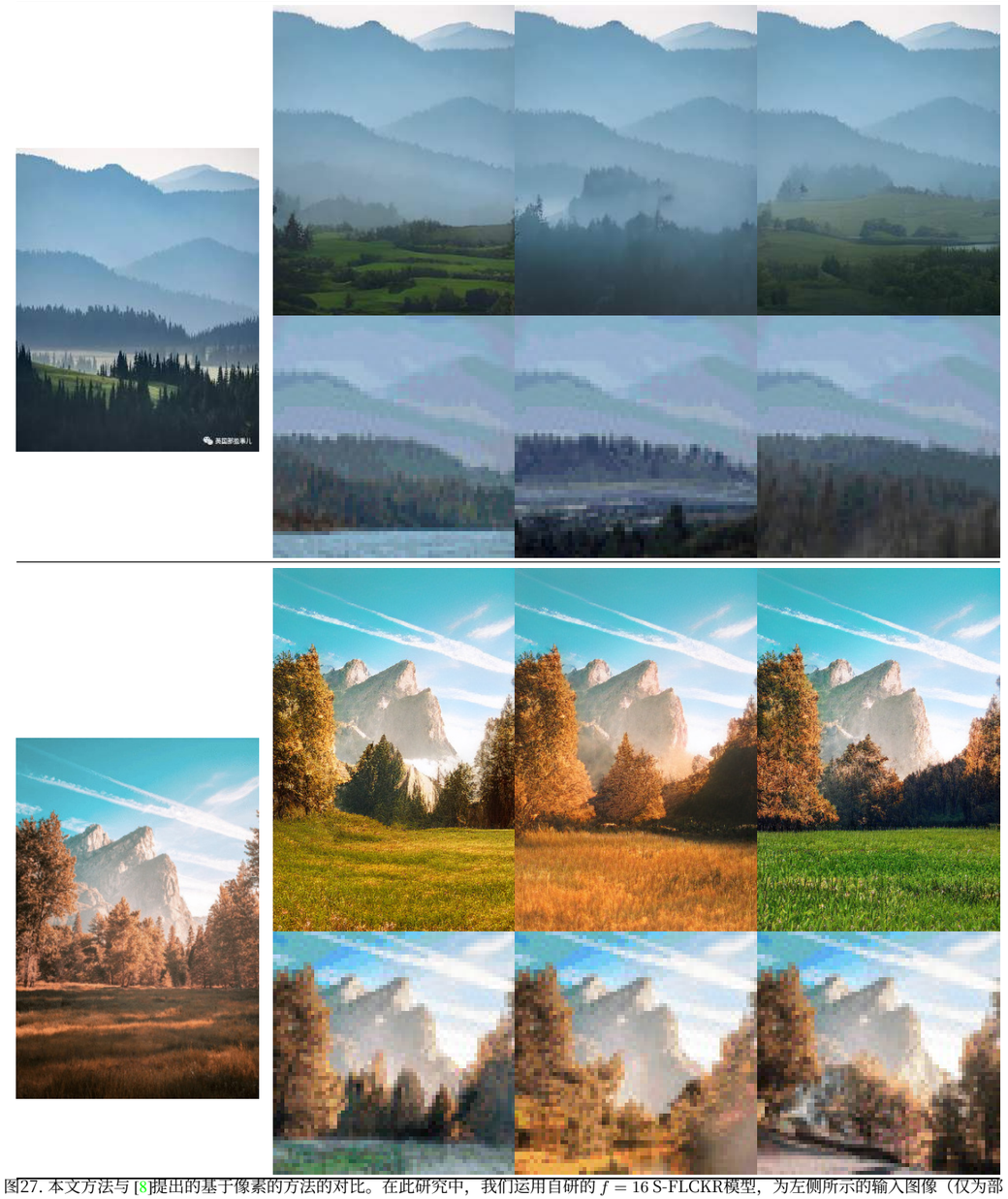


图27. 本文方法与 [8]提出的基于像素的方法的对比。在此研究中，我们运用自研的 $f = 16$ S-FLCKR模型，为左侧所示的输入图像（仅为部分生成内容）生成高保真度的完整图像。对于每一种条件作用设置，我们会展示三个由该模型生成的样本（位于上方），以及三个 [8] 方法生成的样本（位于下方）。

conditioning

ours (top) vs iGPT [8] (bottom)

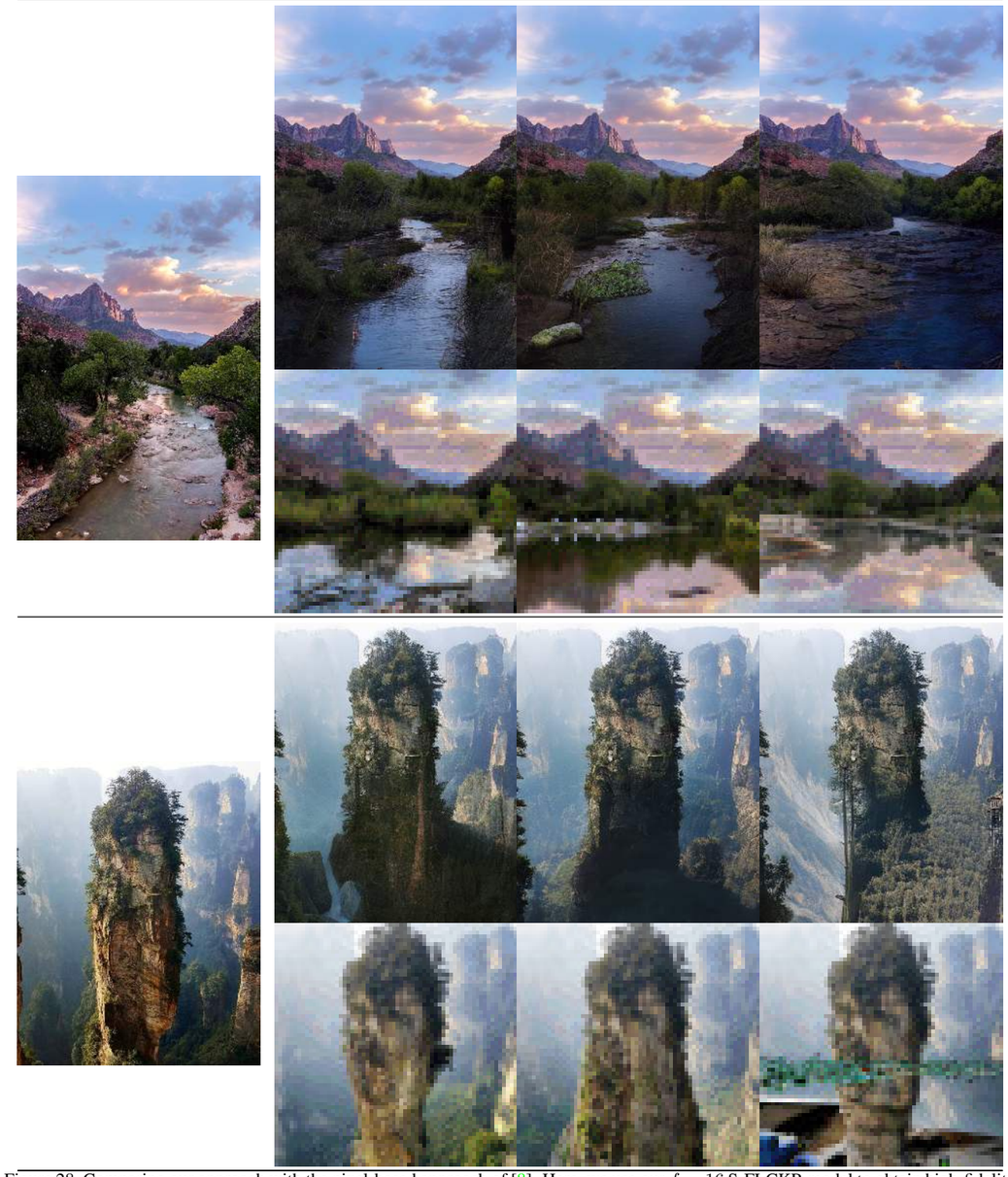


Figure 28. Comparing our approach with the pixel-based approach of [8]. Here, we use our $f = 16$ S-FLCKR model to obtain high-fidelity image completions of the inputs depicted on the left (half completions). For each conditioning, we show three of our samples (top) and three of [8] (bottom).

条件作用

我们的（上方）与iGPT的对比 [8]（下方）

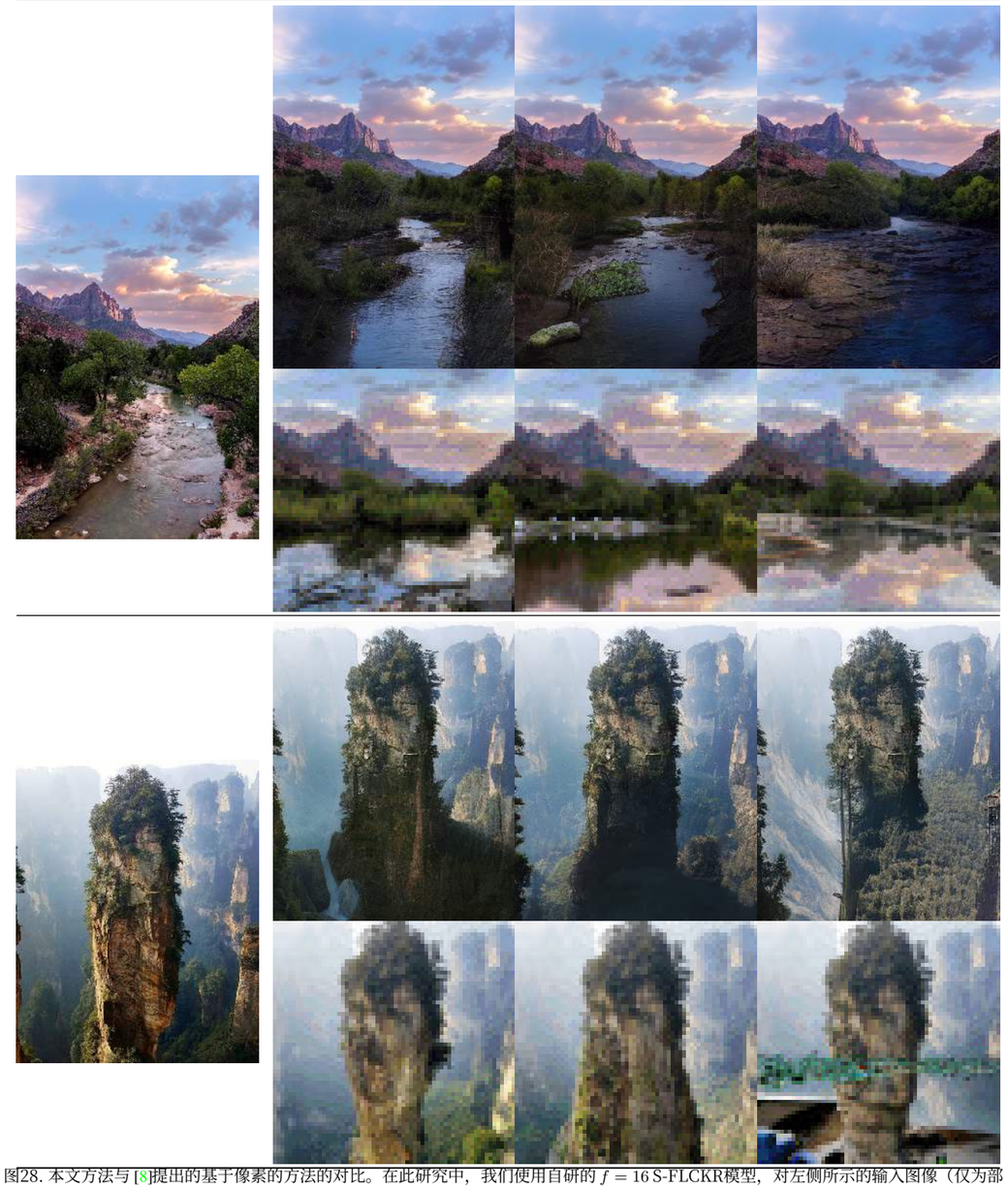


图28. 本文方法与 [8]提出的基于像素的方法的对比。在此研究中，我们使用自研的 $f = 16$ S-FLCKR模型，对左侧所示的输入图像（仅为部分内容）进行高保真度的图像补全。对于每一种条件作用设置，我们会展示三个由该模型生成的样本（位于上方），以及三个 [8] 方法生成的样本（位于下方）。

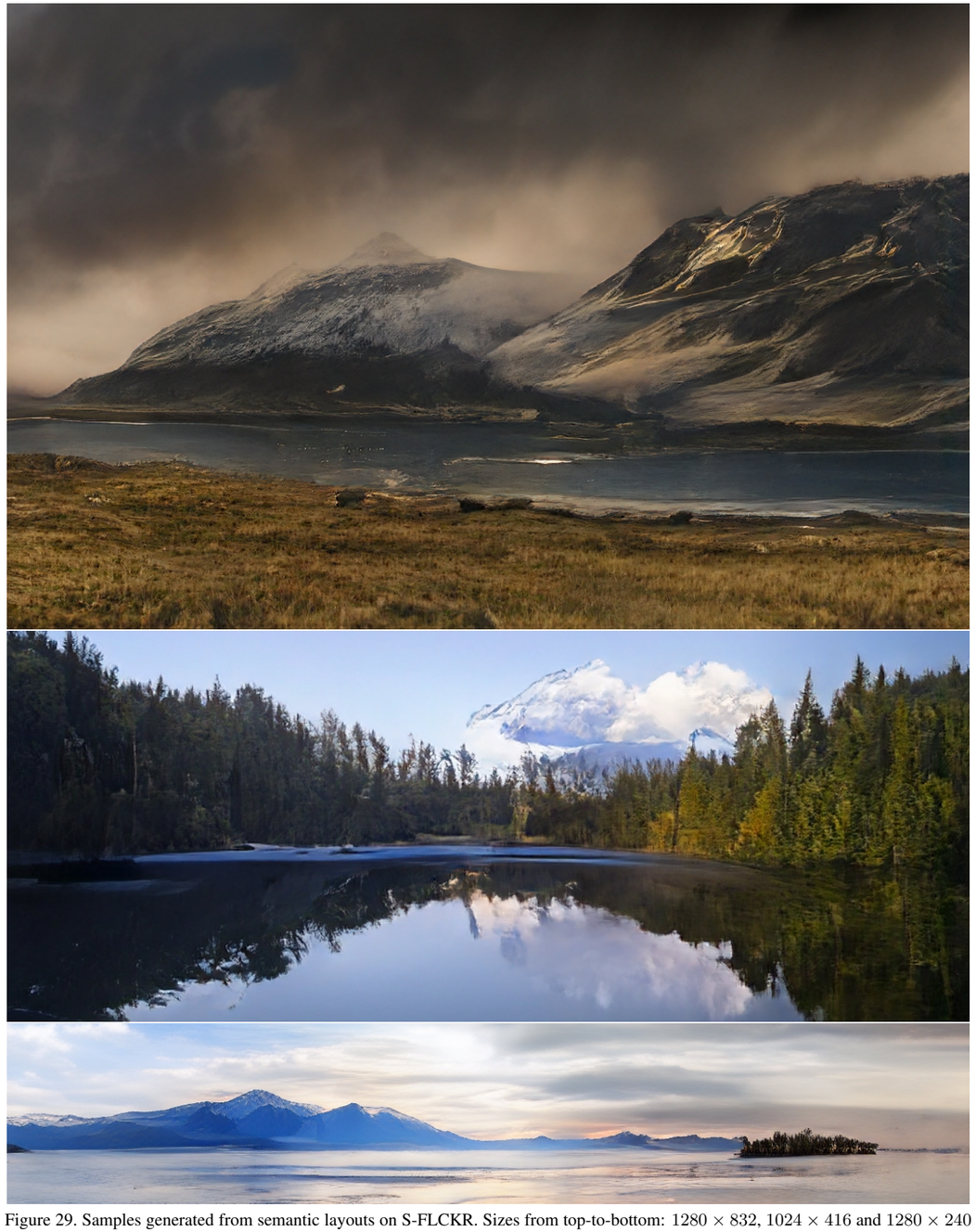


Figure 29. Samples generated from semantic layouts on S-FLCKR. Sizes from top-to-bottom: 1280×832, 1024×416 and 1280×240 pixels.

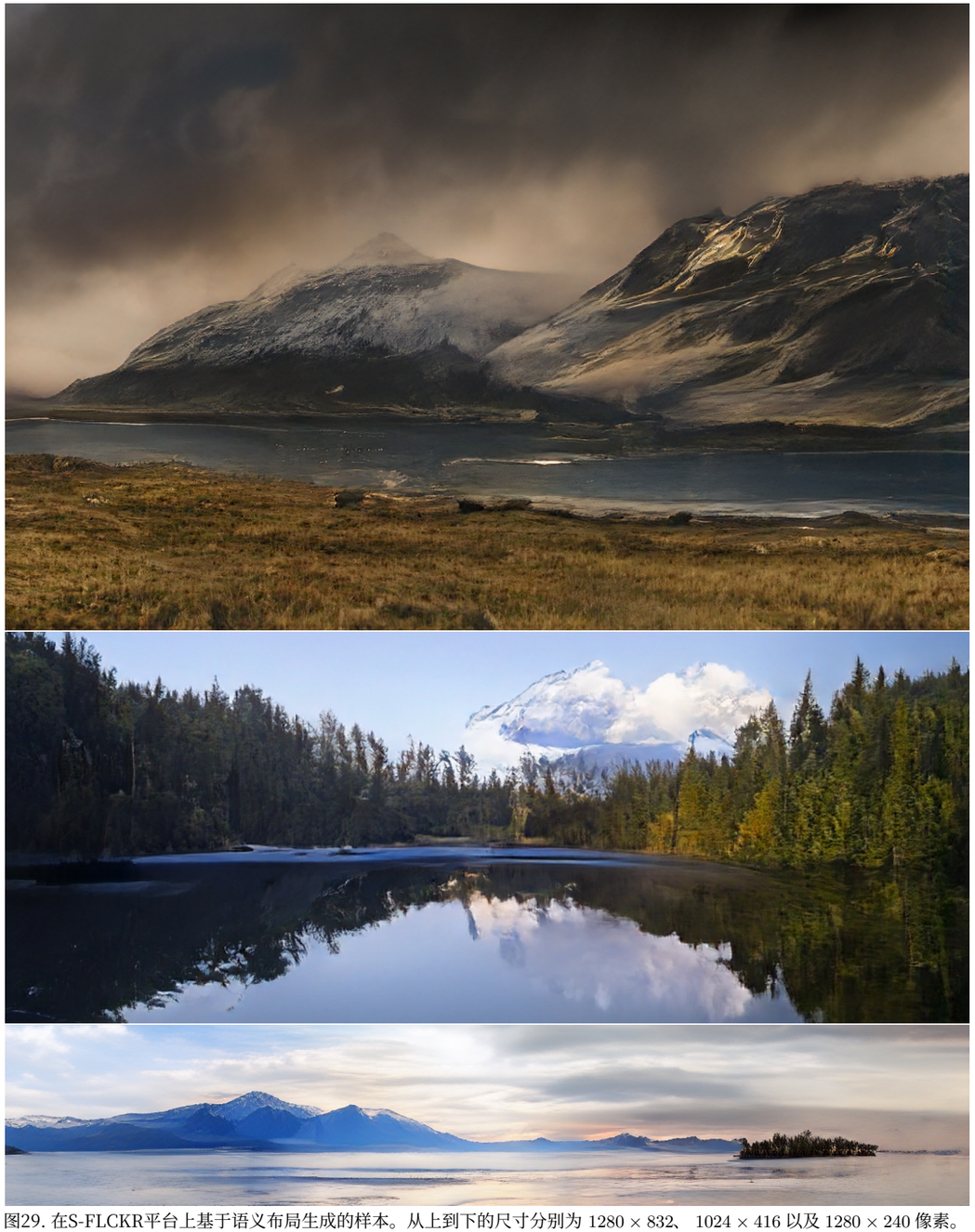


图29. 在S-FLCKR平台上基于语义布局生成的样本。从上到下的尺寸分别为 1280×832、 1024×416 以及 1280×240 像素。

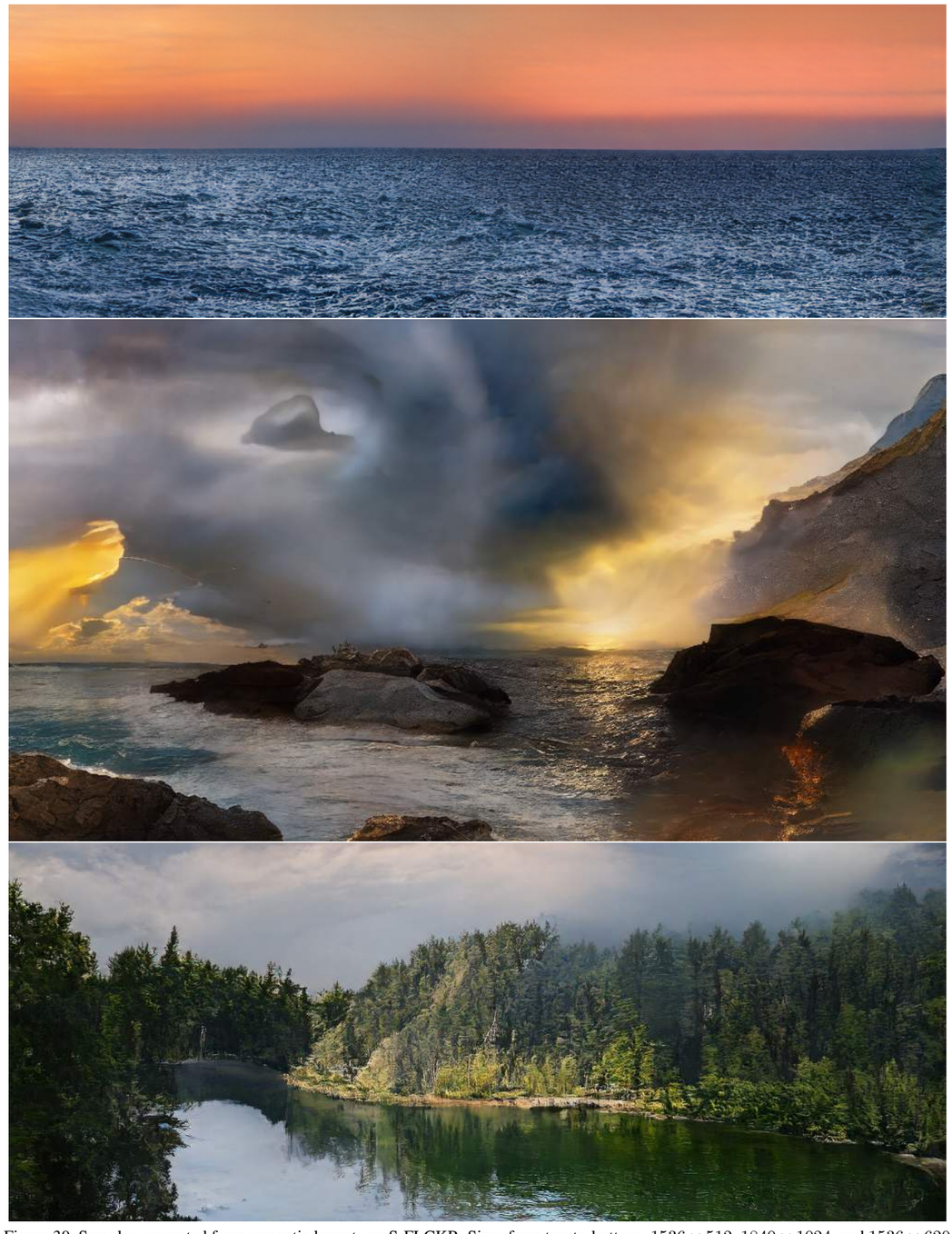


Figure 30. Samples generated from semantic layouts on S-FLCKR. Sizes from top-to-bottom: 1536×512, 1840×1024, and 1536×620 pixels.

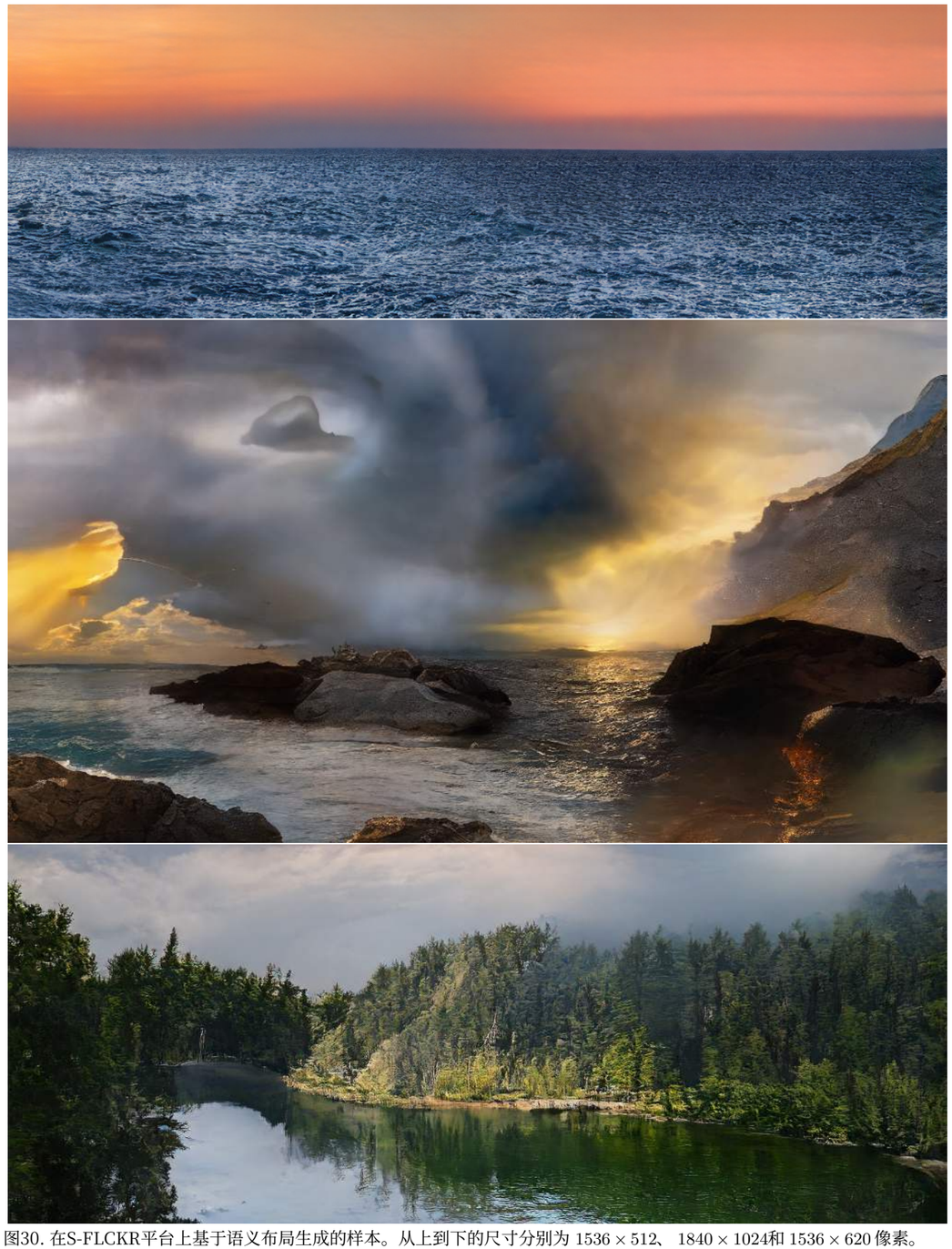


图30. 在S-FLCKR平台上基于语义布局生成的样本。从上到下的尺寸分别为 1536×512、 1840×1024和 1536×620 像素。

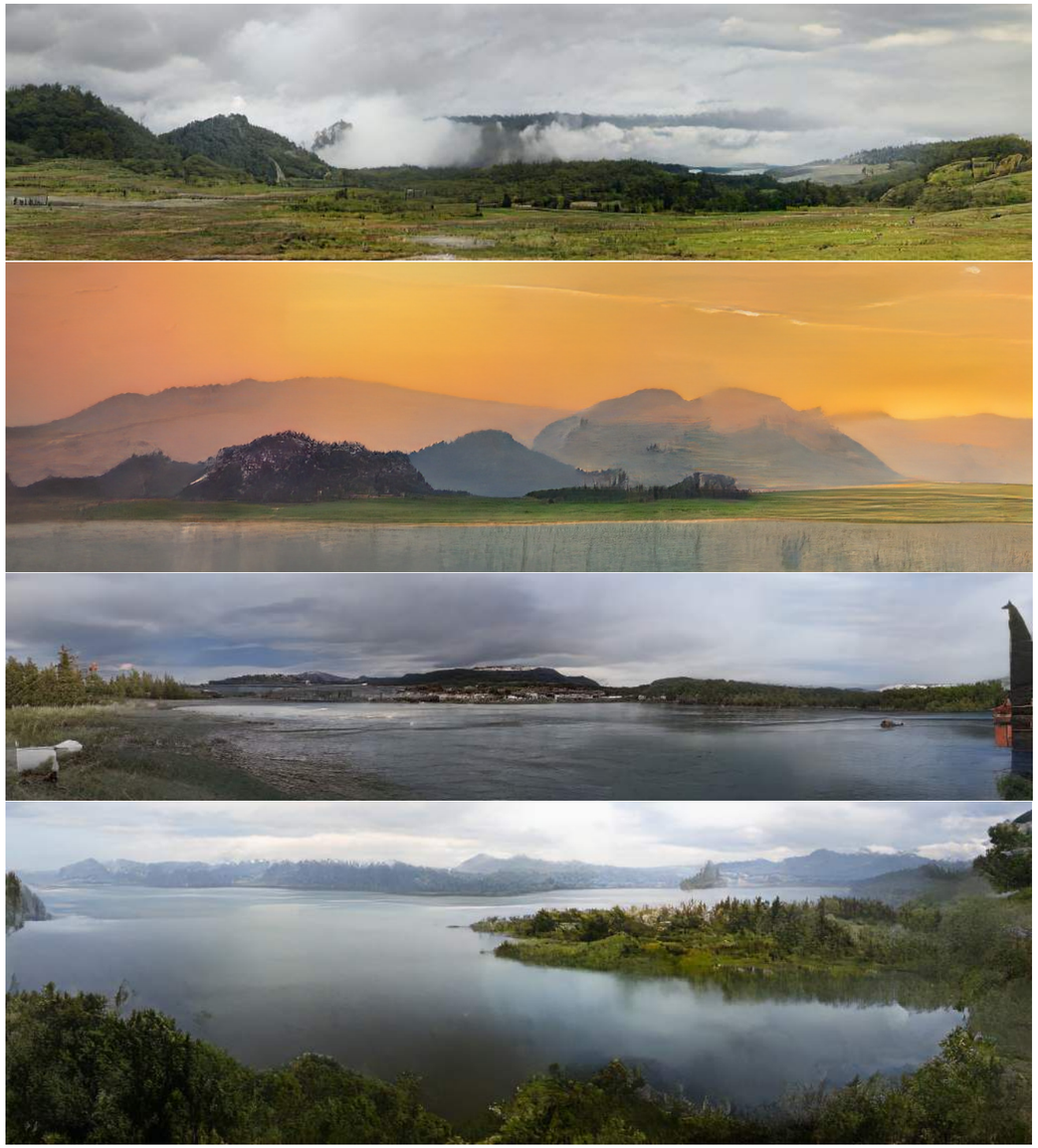


Figure 31. Samples generated from semantic layouts on S-FLCKR. Sizes from top-to-bottom: 2048×512, 1460×440, 2032×448 and 2016×672 pixels.

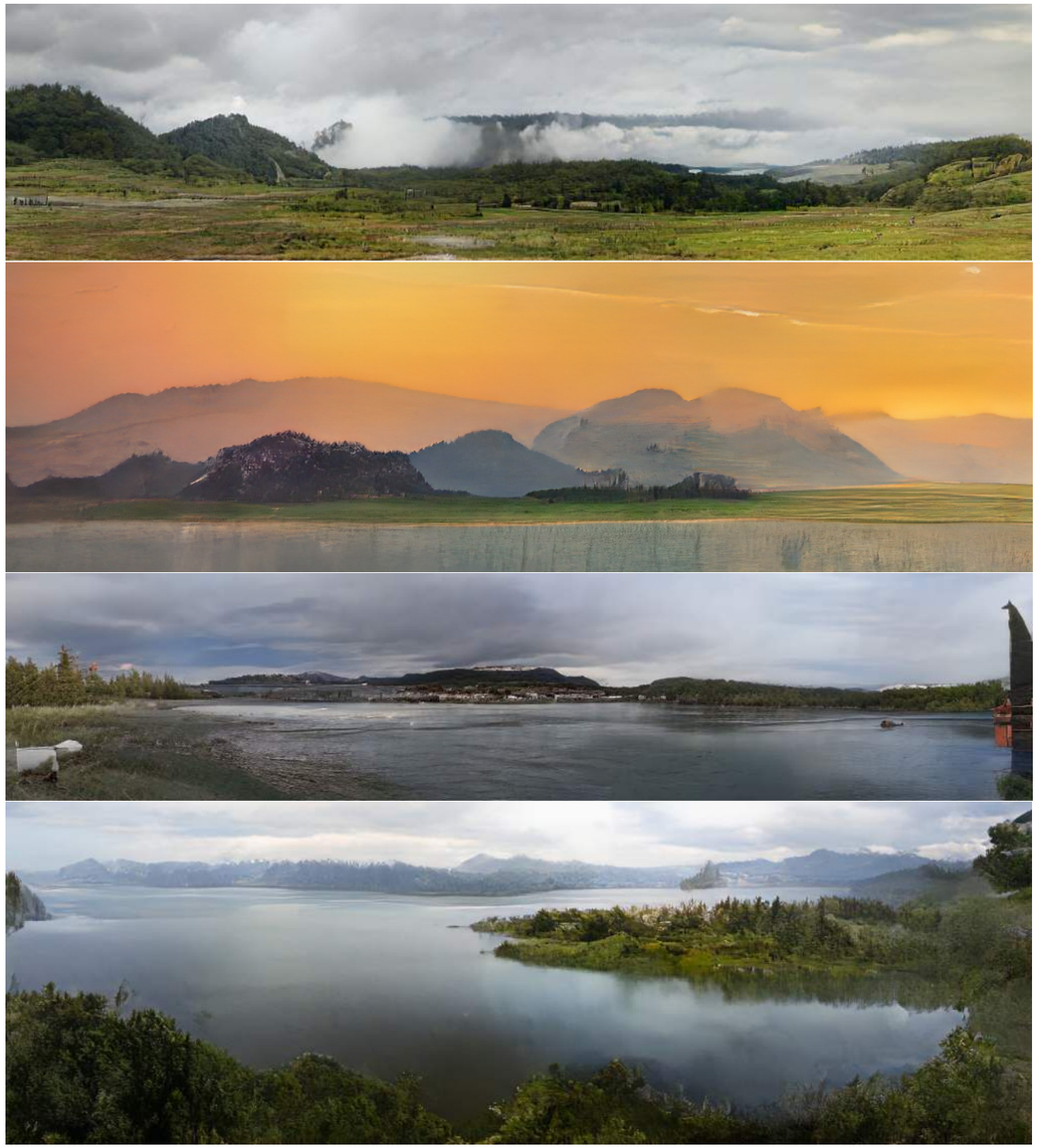


图31. 在S-FLCKR平台上基于语义布局生成的样本。从上到下的尺寸分别为 2048×512、 1460×440、 2032×448 以及2016×672像素。

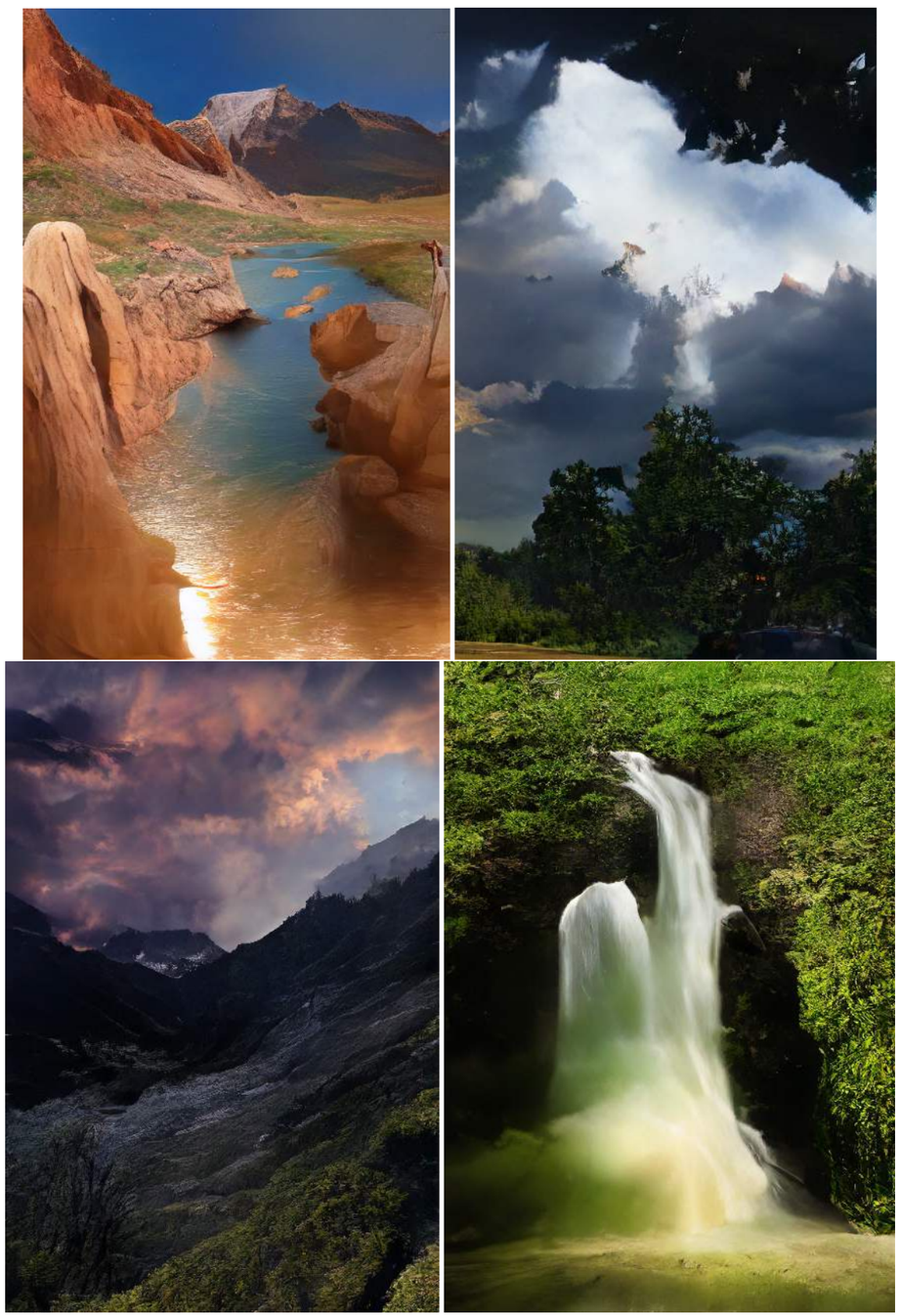


Figure 32. Samples generated from semantic layouts on S-FLCKR. Sizes from top-to-bottom: 1280×832, 1024×416 and 1280×240 pixels.

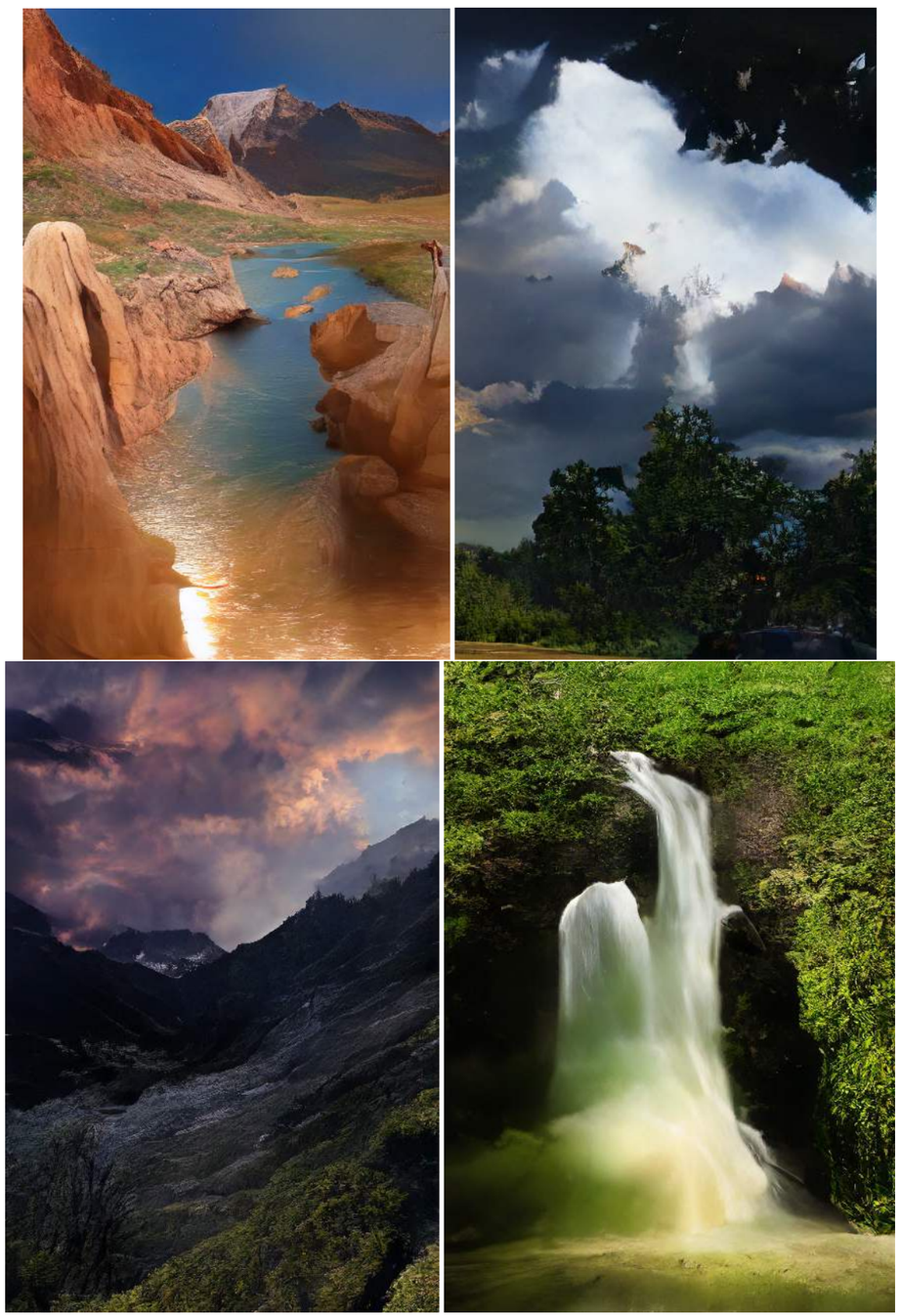


图32. 在S-FLCKR平台上基于语义布局生成的样本。从上到下的尺寸分别为 1280×832、 1024×416 和 1280×240 像素。

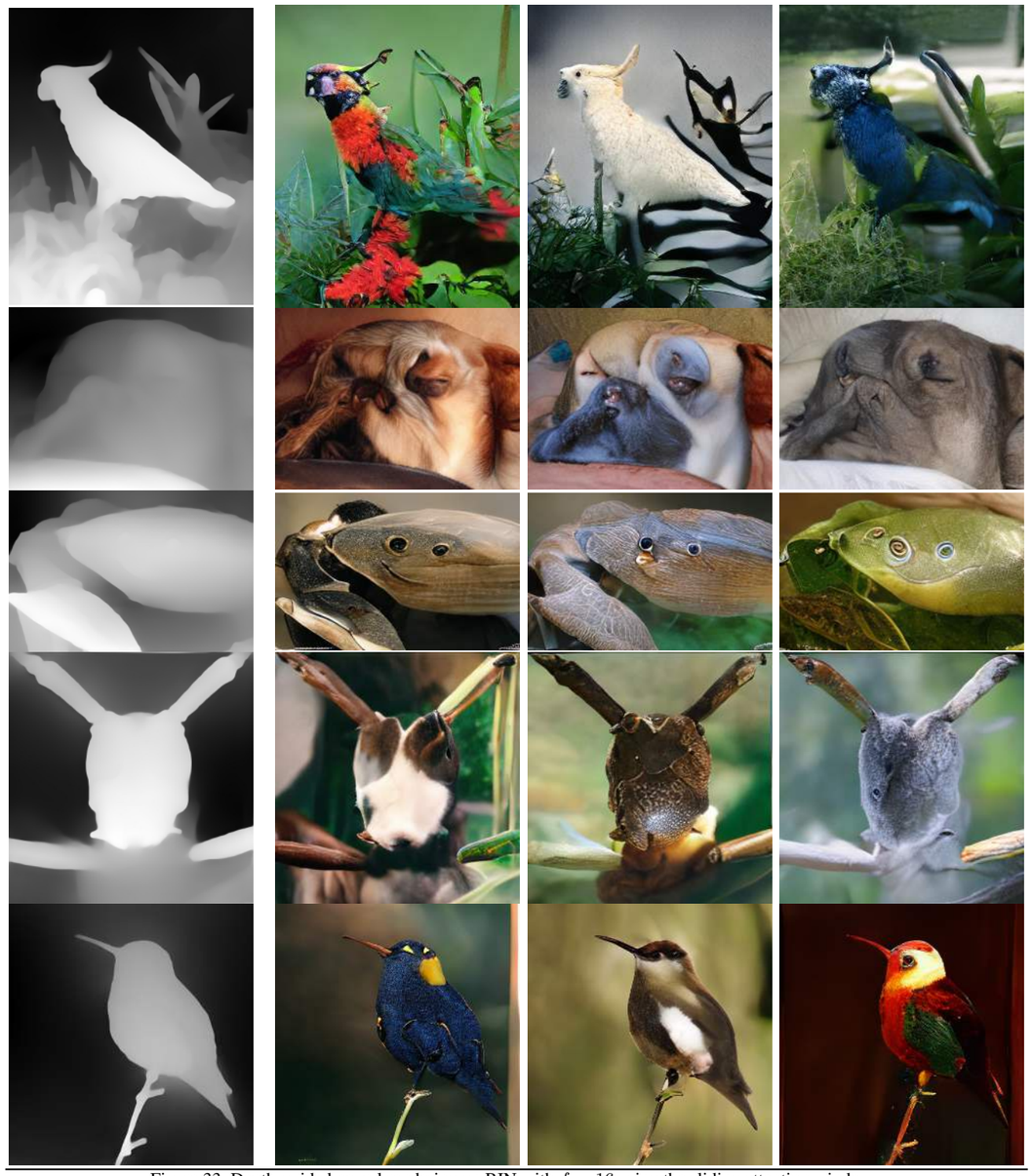


Figure 33. Depth-guided neural rendering on RIN with $f = 16$ using the sliding attention window.

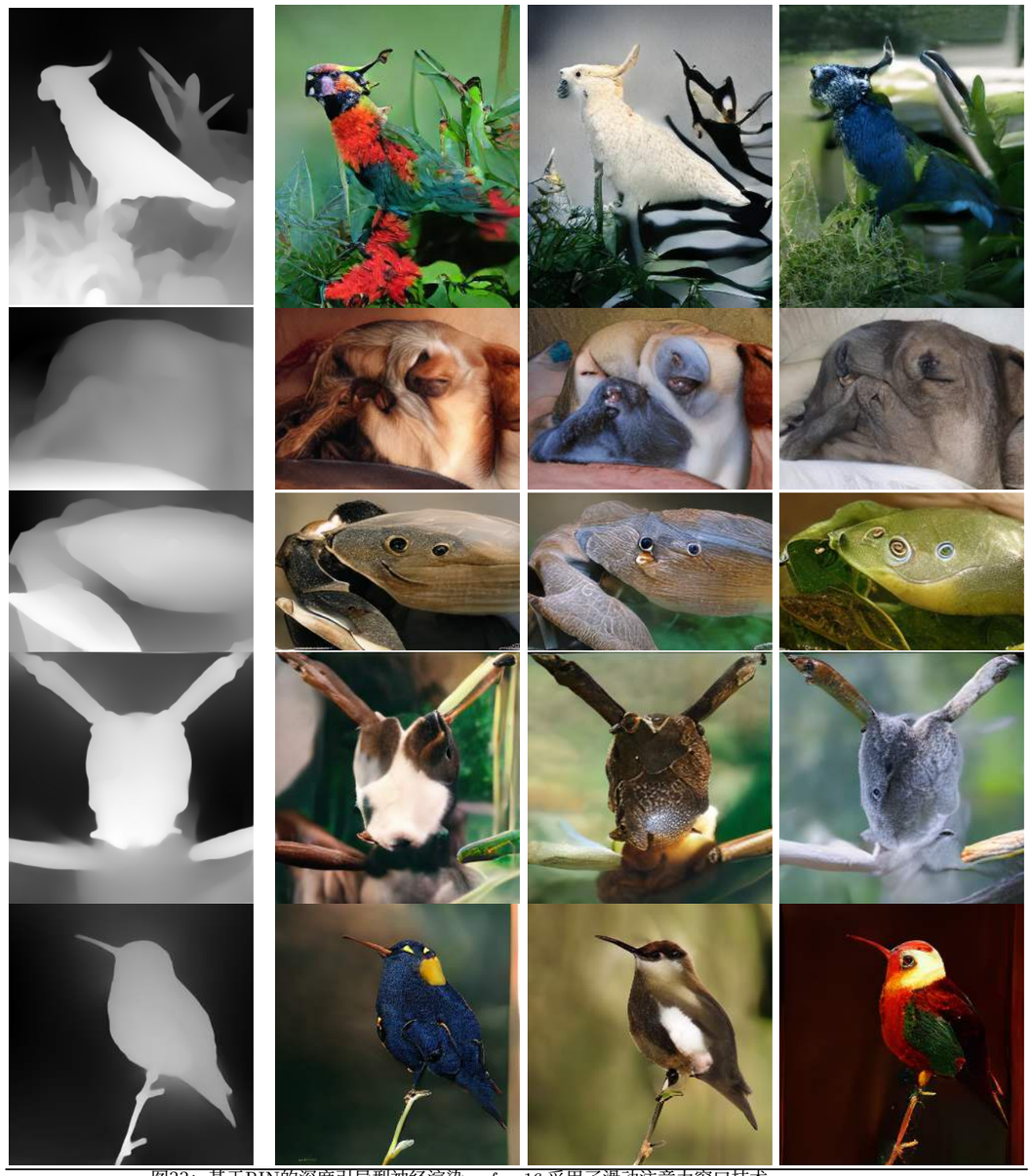


图33：基于RIN的深度引导型神经渲染， $f = 16$ 采用了滑动注意力窗口技术。

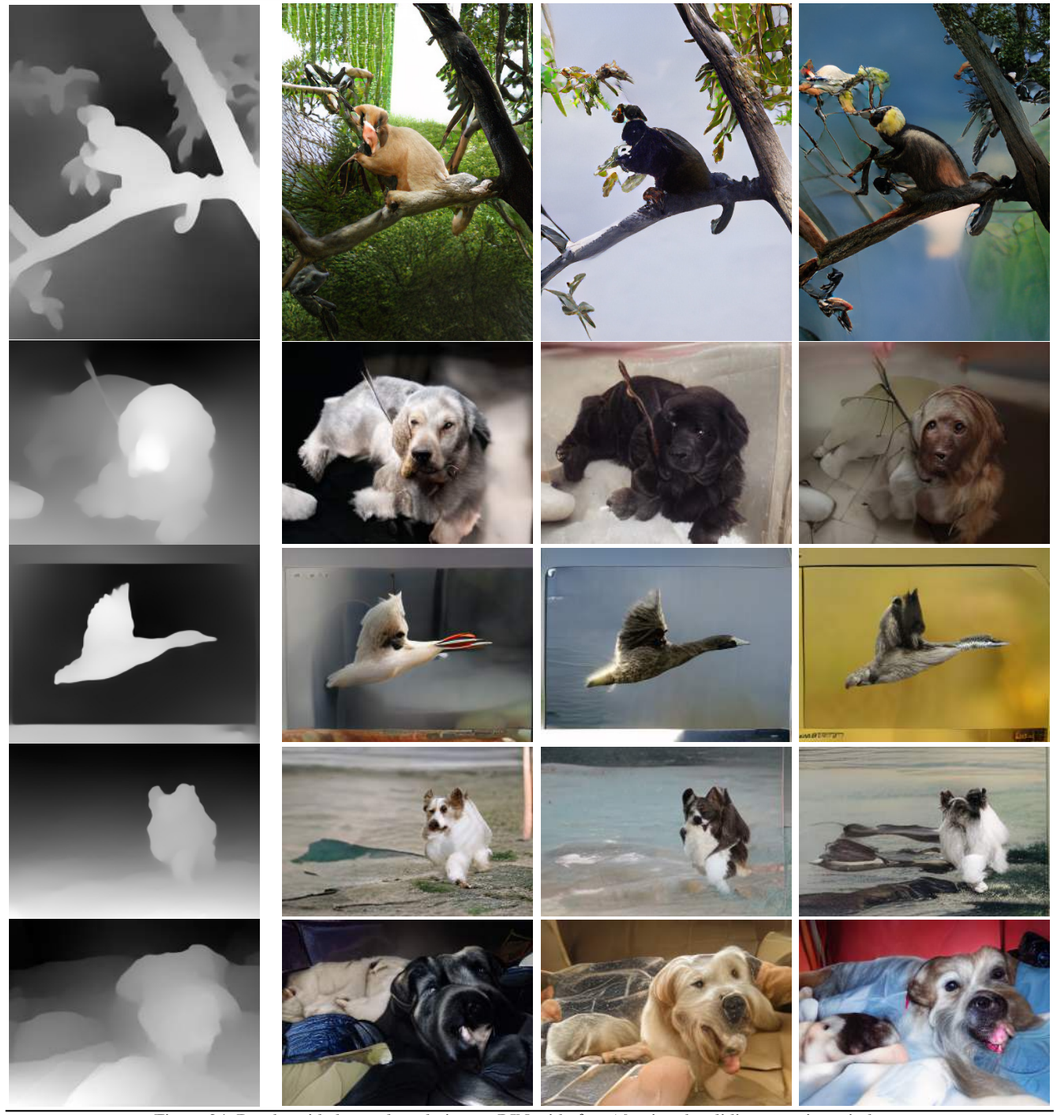


Figure 34. Depth-guided neural rendering on RIN with $f = 16$ using the sliding attention window.

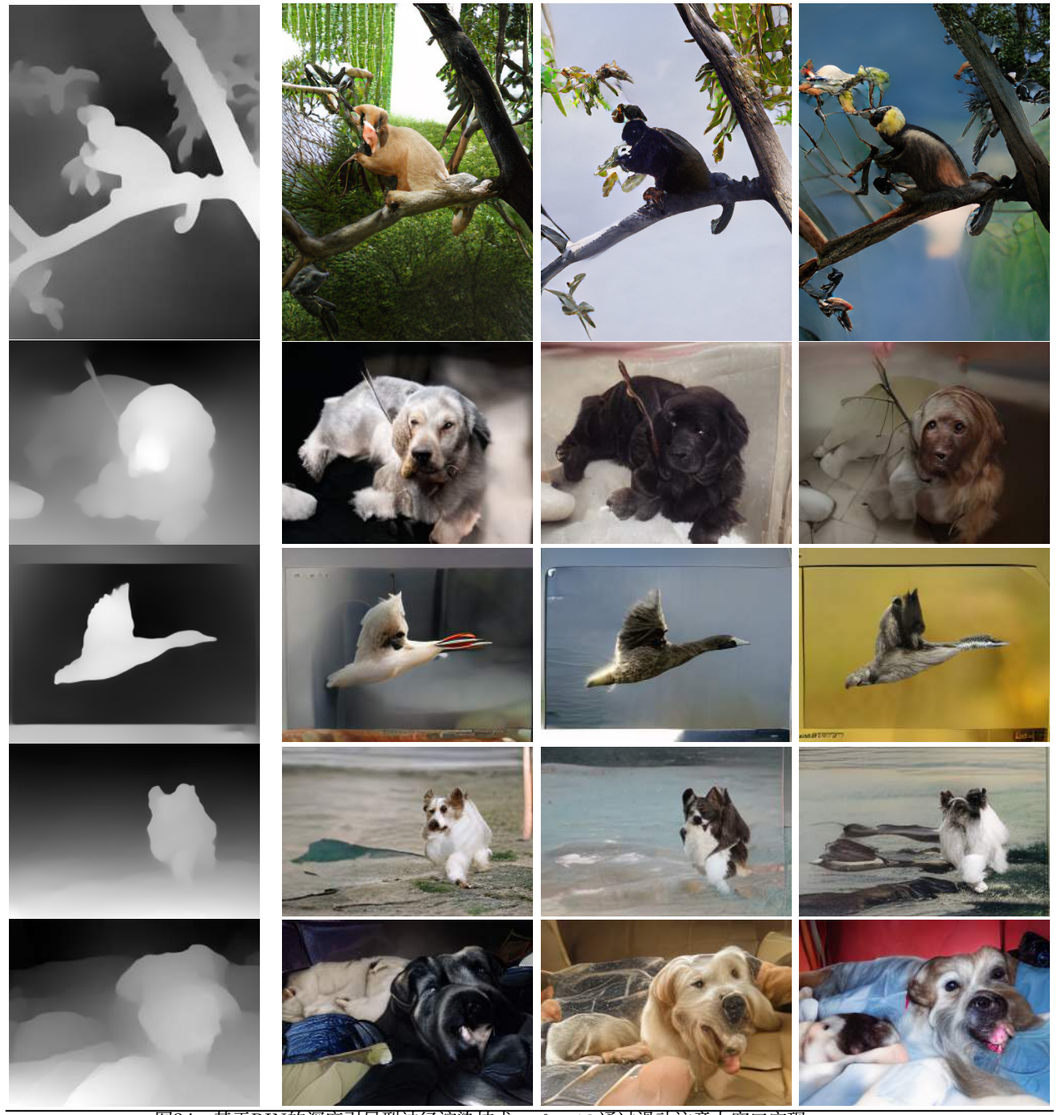


图34：基于RIN的深度引导型神经渲染技术， $f = 16$ 通过滑动注意力窗口实现。

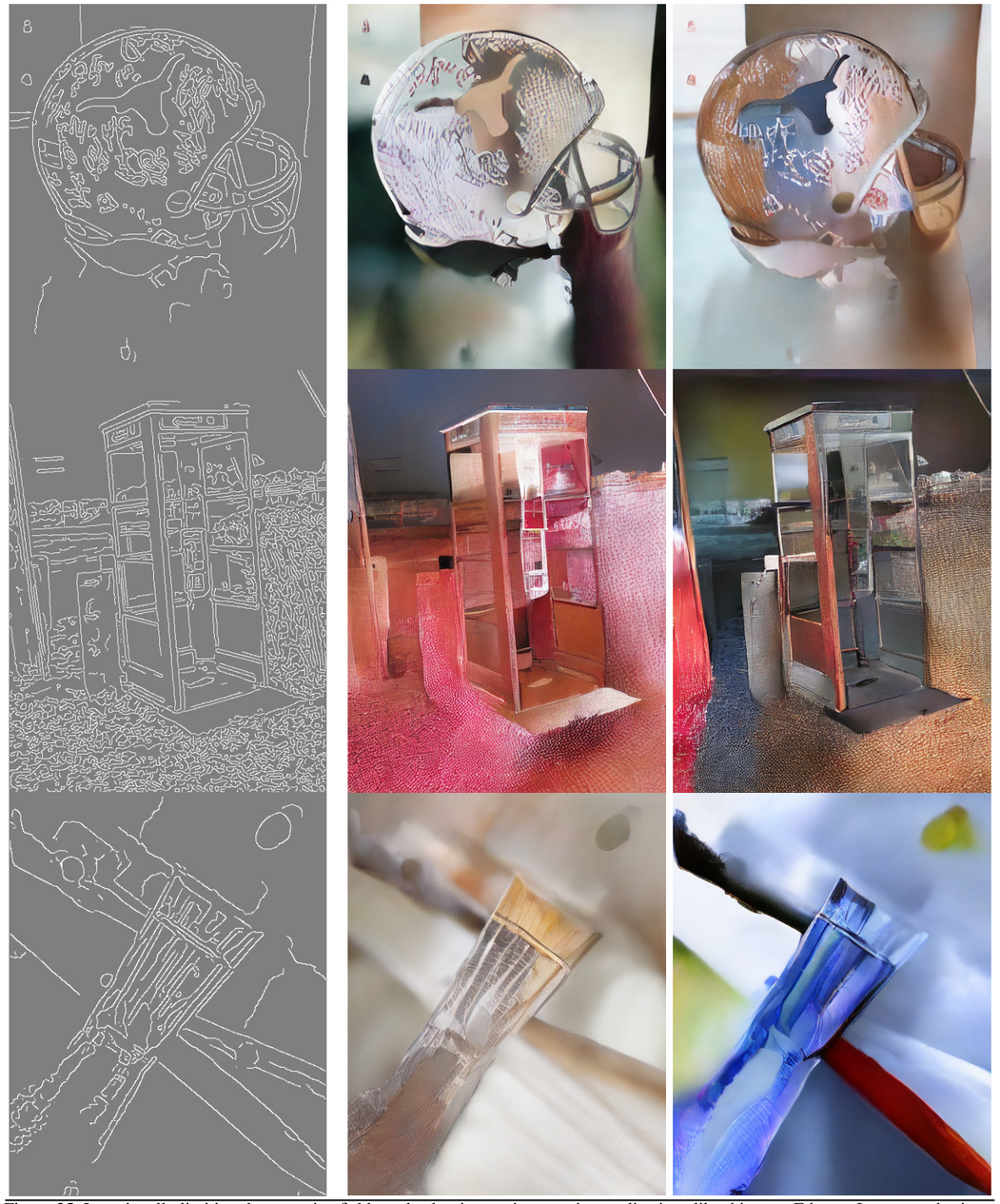


Figure 35. Intentionally limiting the receptive field can lead to interesting creative applications like this one: Edge-to-Image synthesis on IN with $f = 8$, using the sliding attention window.

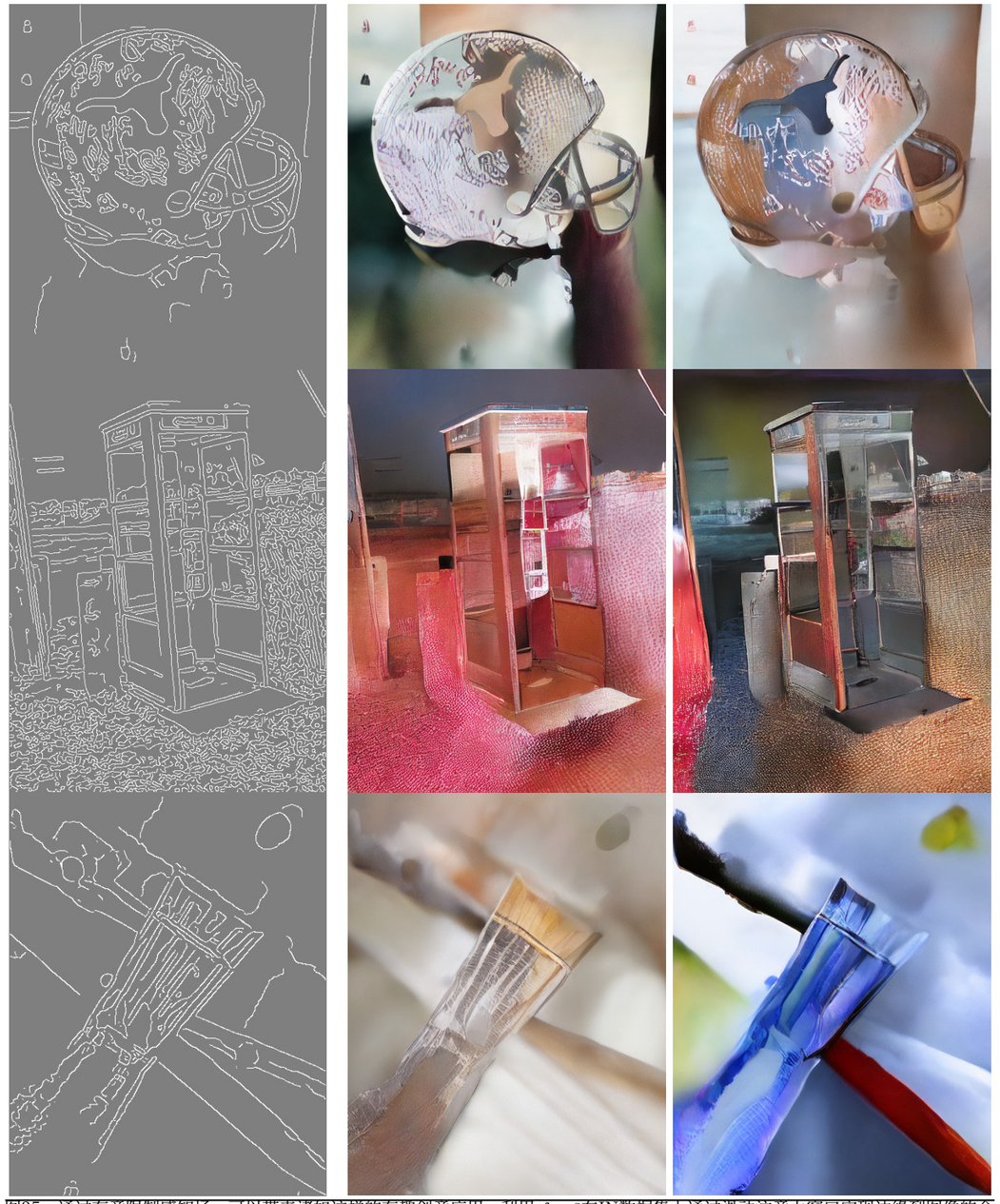


图35。通过有意限制感知场，可以带来诸如这样的有趣创意应用：利用 $f = 8$在IN数据集上通过滑动注意力窗口实现边缘到图像的合成。



Figure 36. Additional results for stochastic superresolution with an $f = 16$ model on IN, using the sliding attention window.



图36. 加法运算 图注：在IN数据集上运用R算法及滑动注意力窗口，针对随机超分辨率任务所获得的参数f相关结果 w.

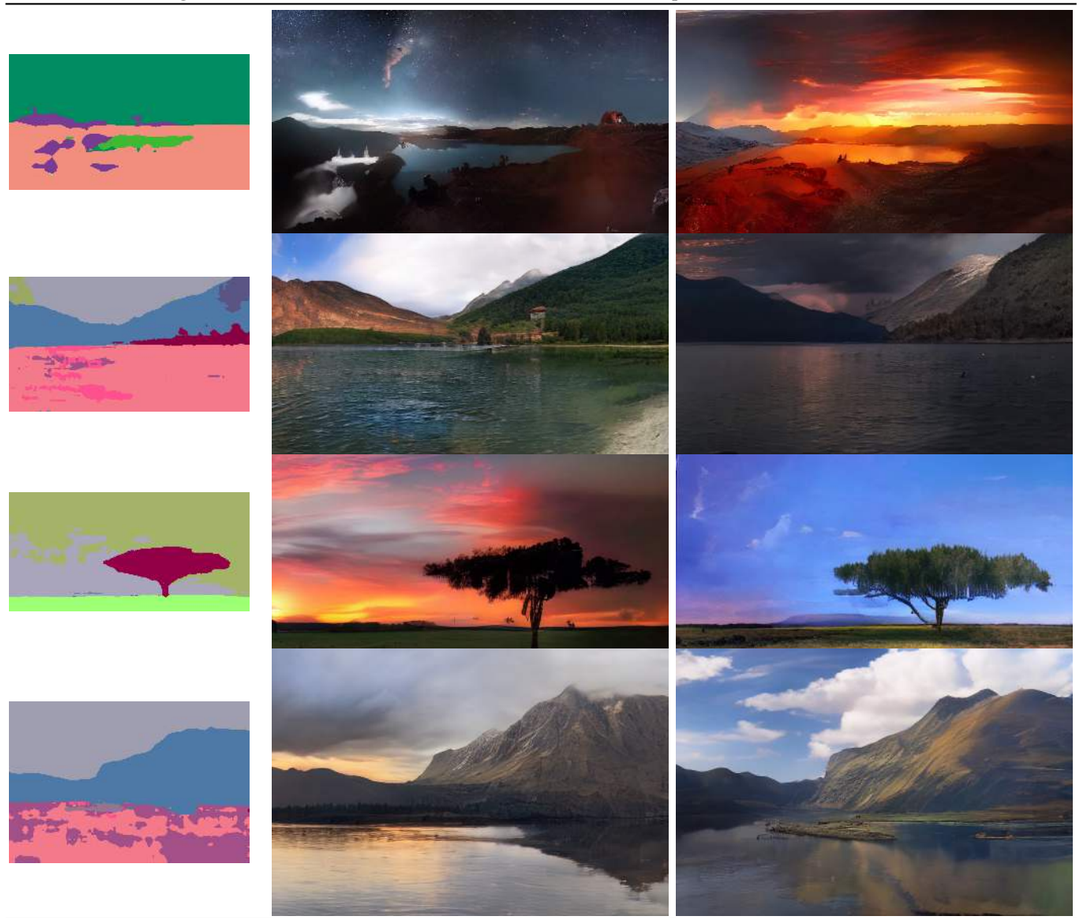


Figure 37. Samples generated from semantic layouts on S-FLCKR with $f = 16$, using the sliding attention window.

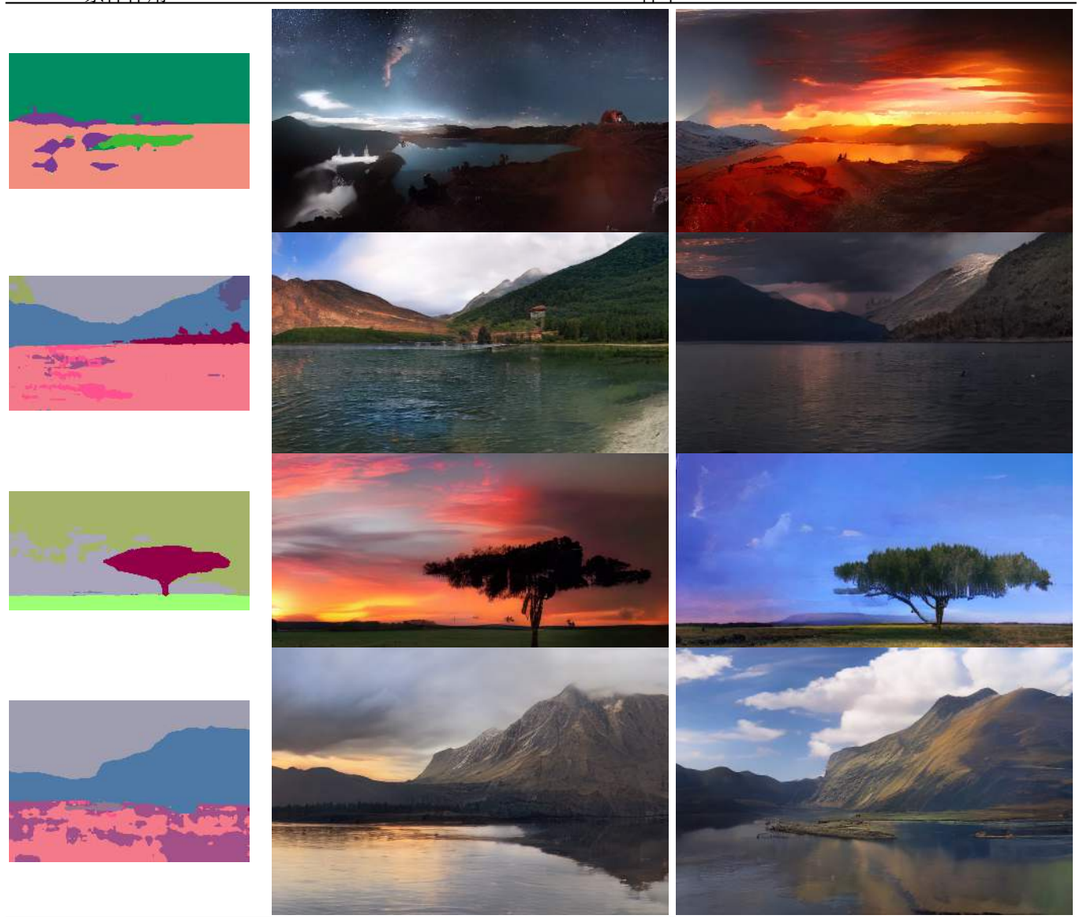


图37 。这些样本是通过在S-FLCKR平台上运用语义布局，并借助滑动注意力窗口以及 $f = 16$生成的。.

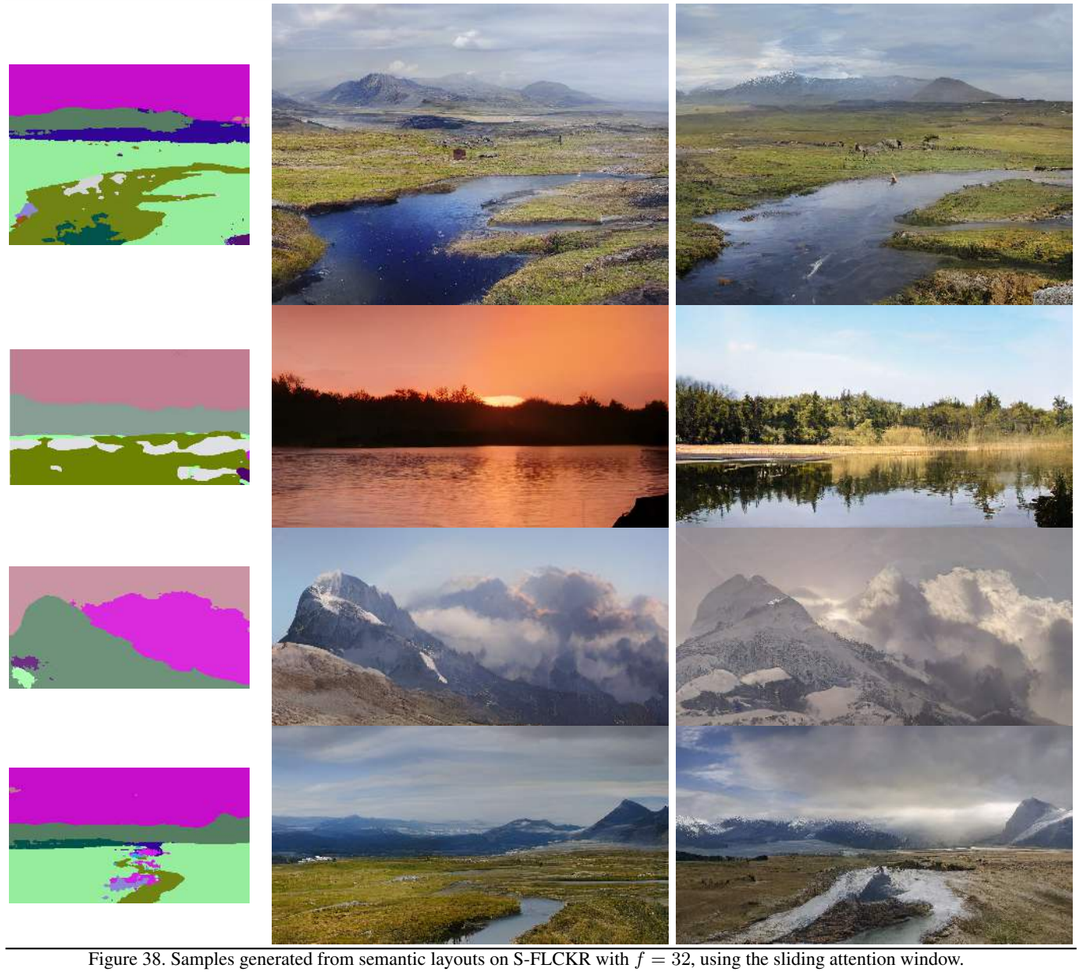


Figure 38. Samples generated from semantic layouts on S-FLCKR with $f = 32$, using the sliding attention window.

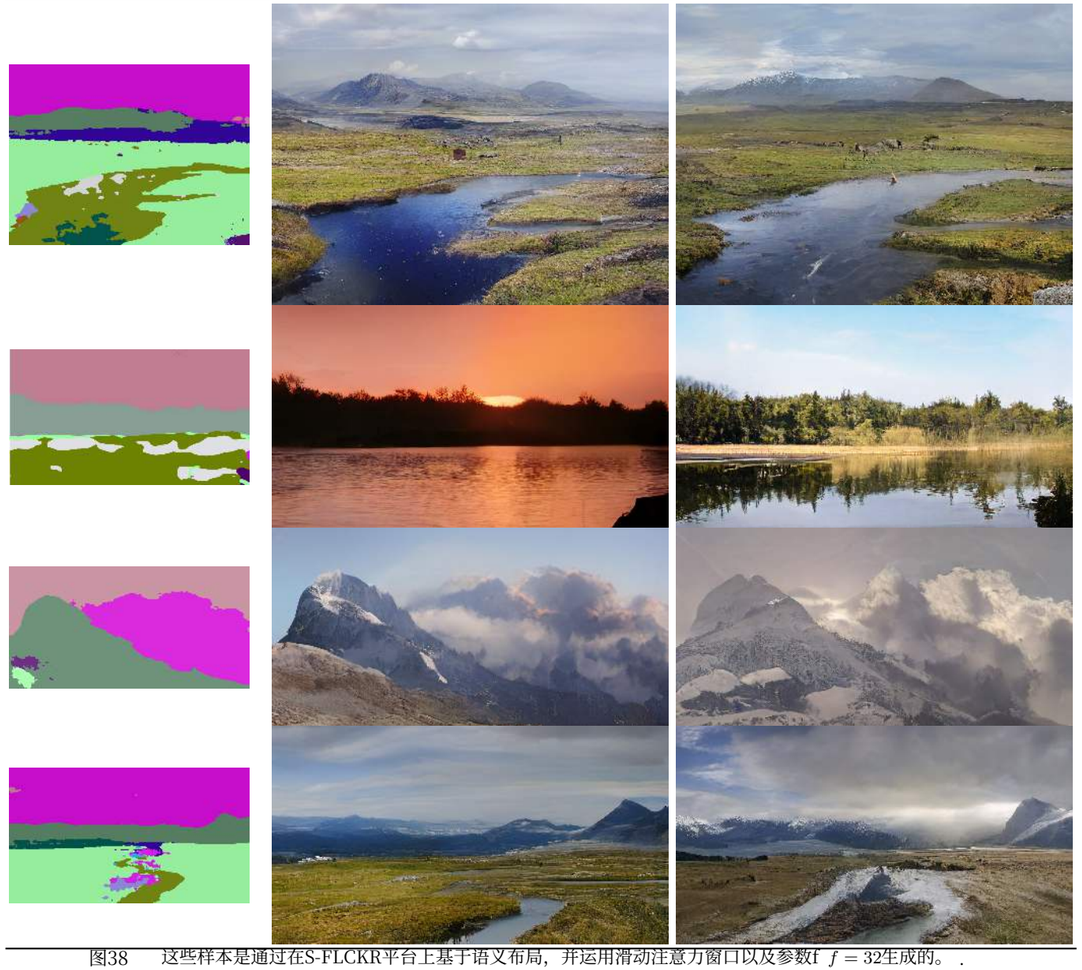


图38　这些样本是通过在S-FLCKR平台上基于语义布局，并运用滑动注意力窗口以及参数f $f = 32$生成的。 .



Figure 39. Unconditional samples from a model trained on LSUN Churches & Towers, using the sliding attention window.



图39. 未压缩状态 “图注”： 图39展示的是通过滑动注意力窗口技术，在LSUN教堂与塔楼数据集上训练的模型所输出的传统风格样本。 .

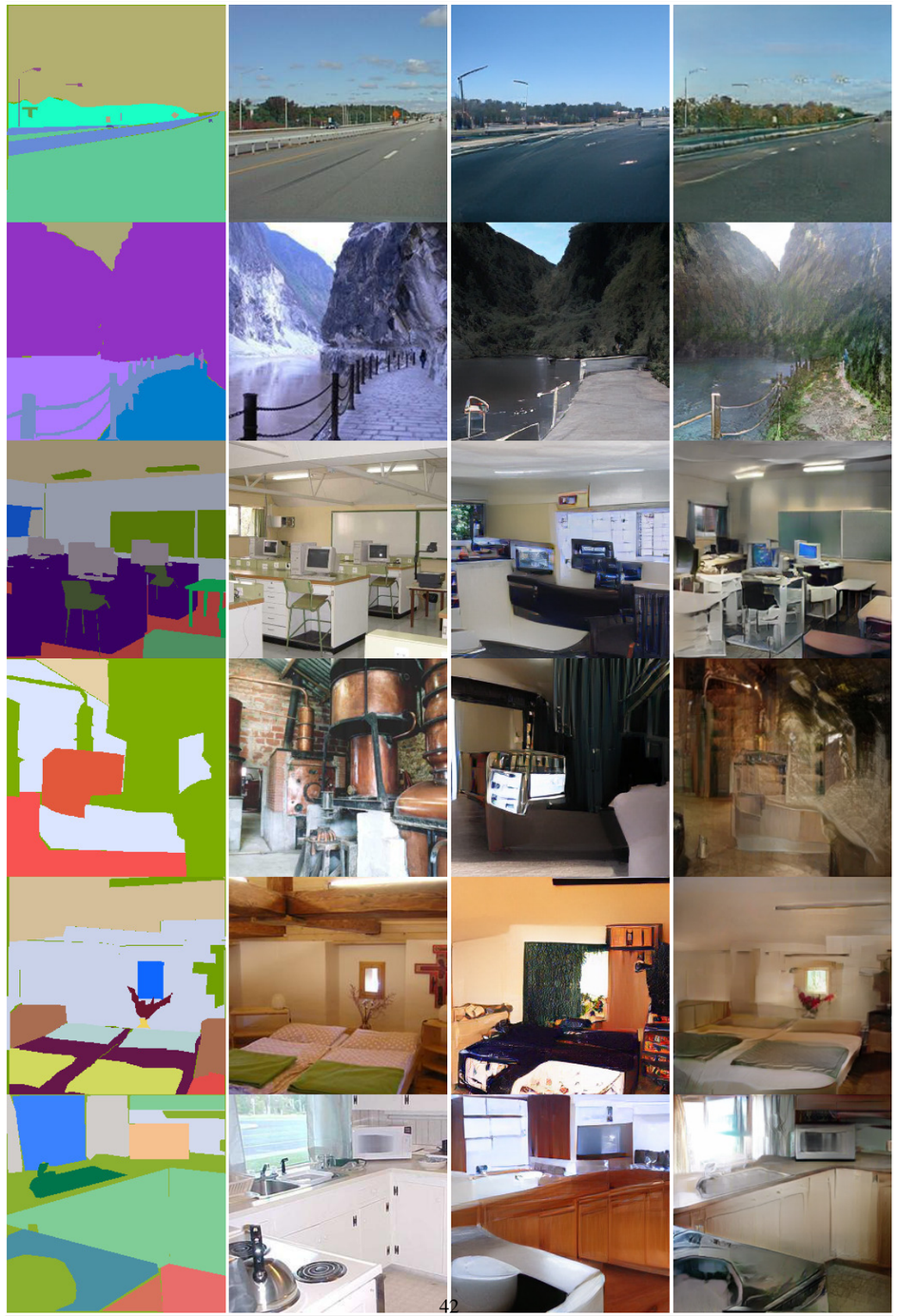


Figure 40. Qualitative comparison to [53] on 256×256 images from the ADE20K dataset.

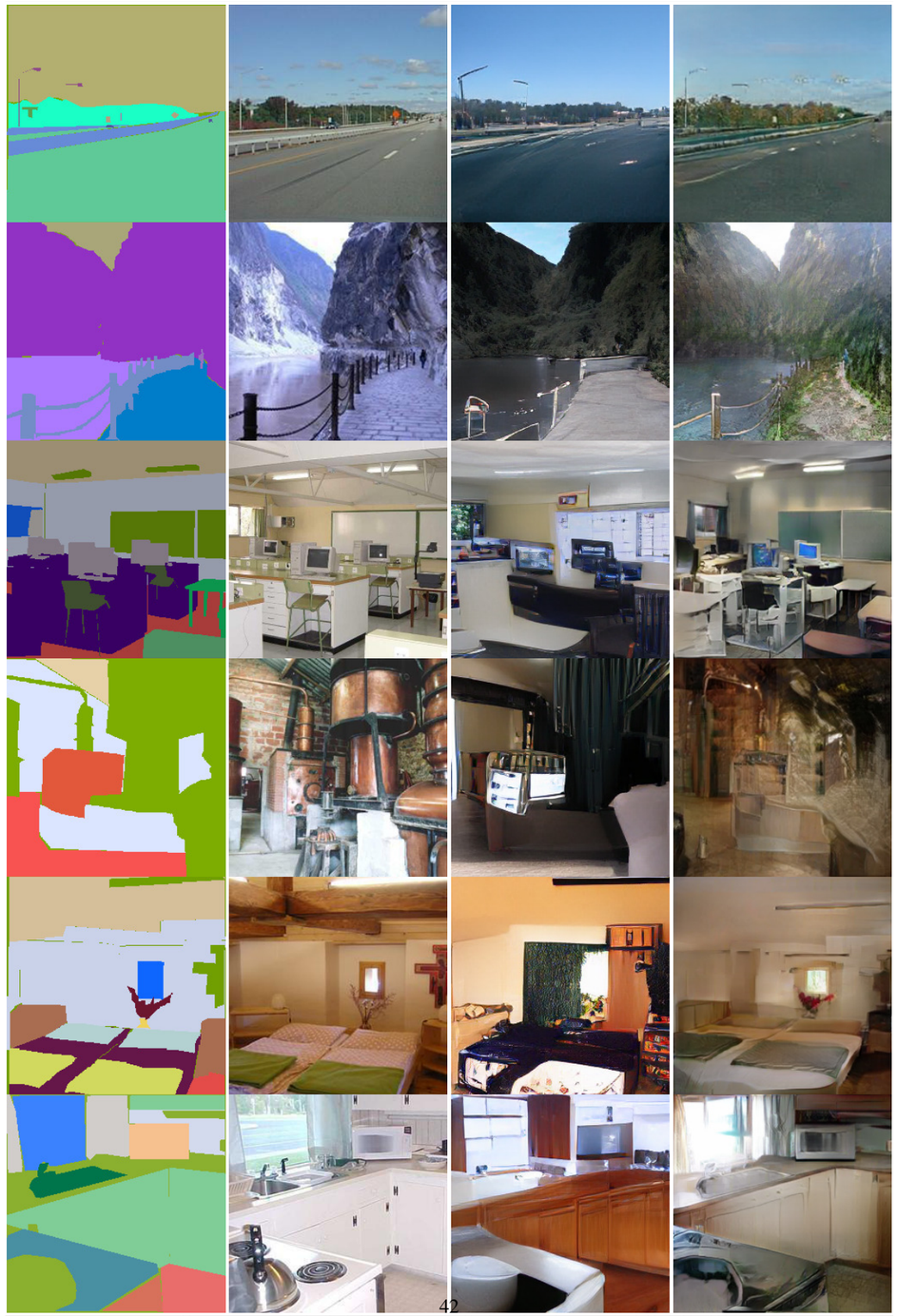


图40：基于ADE20K数据集中的 256×256 格式图像，与 [53] 方法的定性对比。

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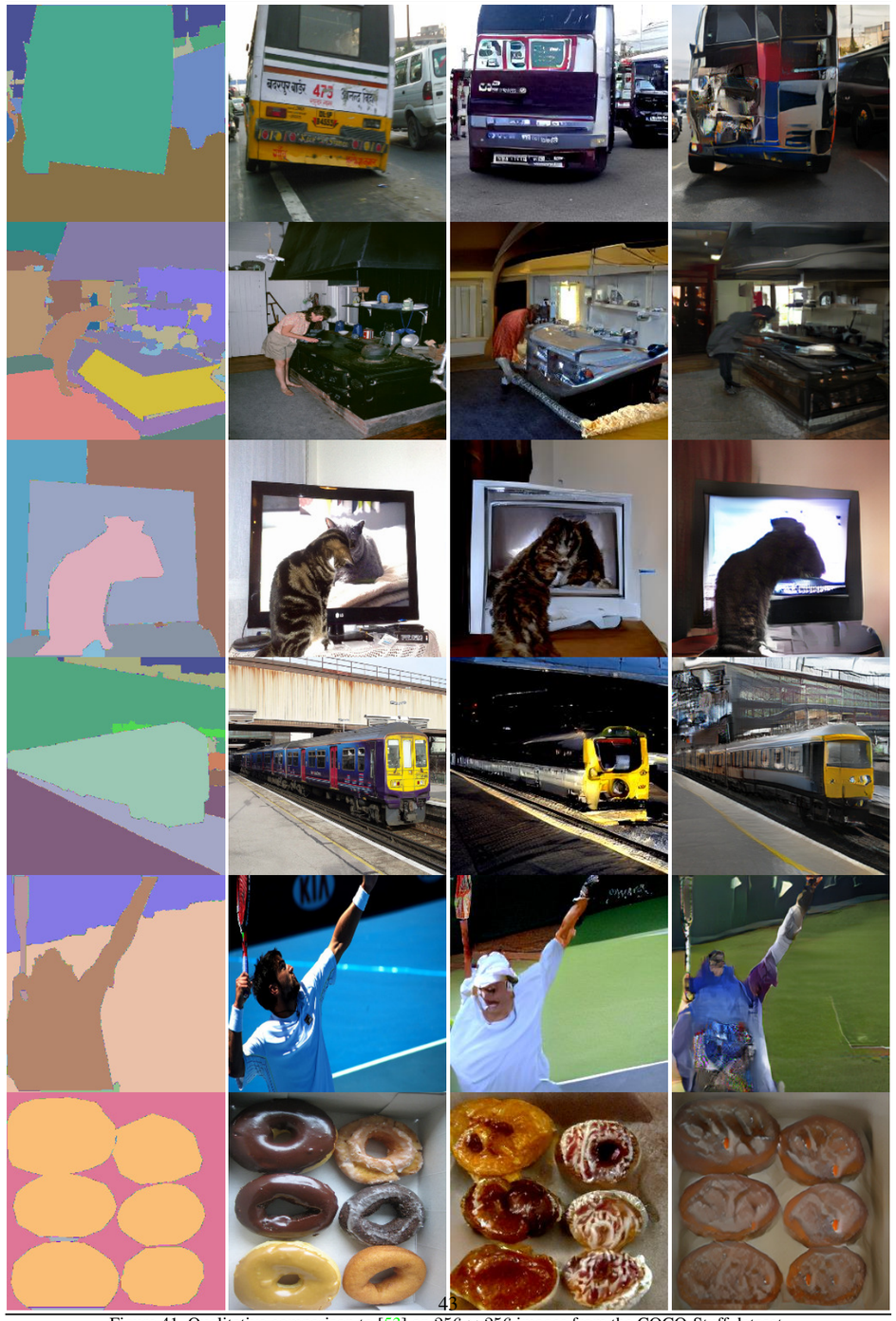


Figure 41. Qualitative comparison to [53] on 256×256 images from the COCO-Stuff dataset.



图41：基于COCO-Stuff数据集中的 256×256 格式图像，与 [53] 进行的定性对比。

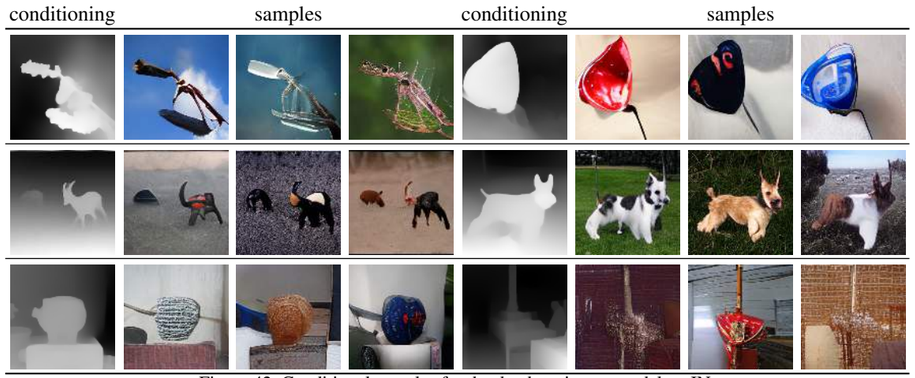


Figure 42. Conditional samples for the depth-to-image model on IN.

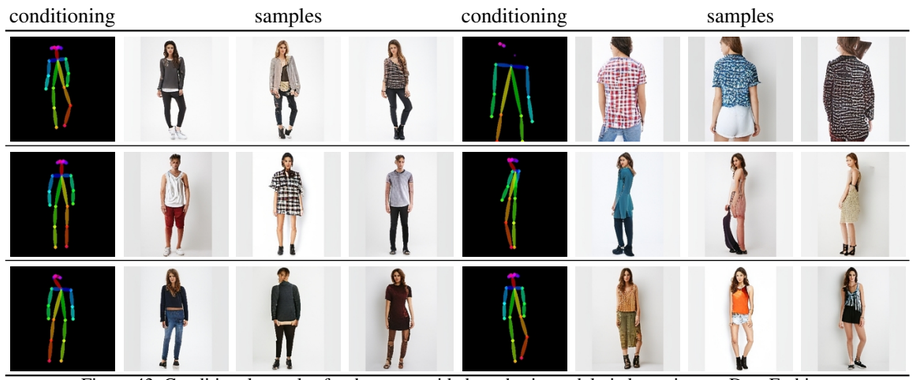


Figure 43. Conditional samples for the pose-guided synthesis model via keypoints on DeepFashion.

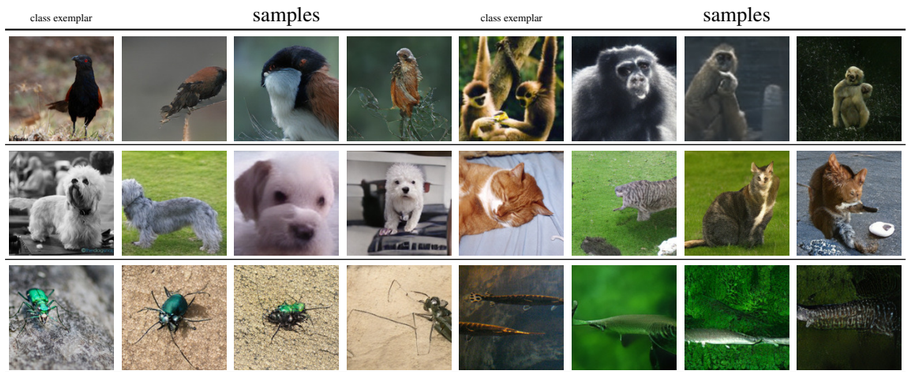


Figure 44. Samples produced by the class-conditional model trained on RIN.



图42：IN数据集上深度到图像模型的条件样本。

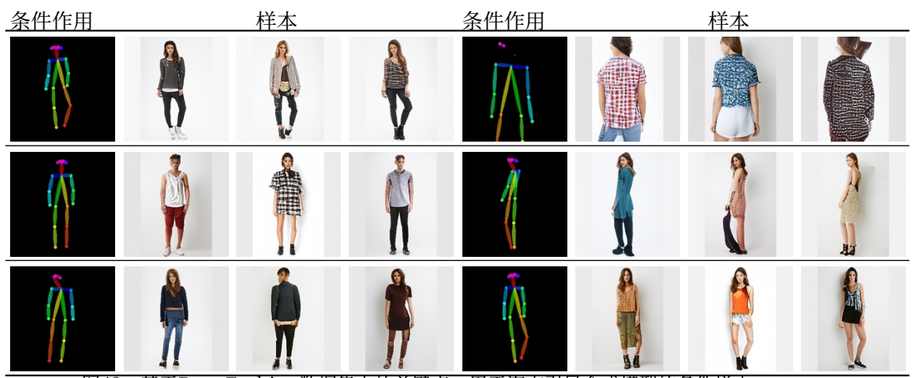


图43：基于DeepFashion数据集上的关键点，用于姿态引导合成模型的条件样本。

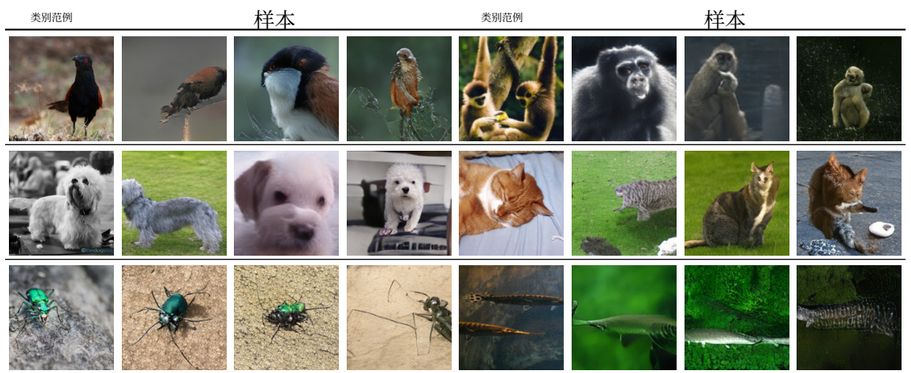


图44：基于RIN训练得到的条件分类模型所生成的样本。

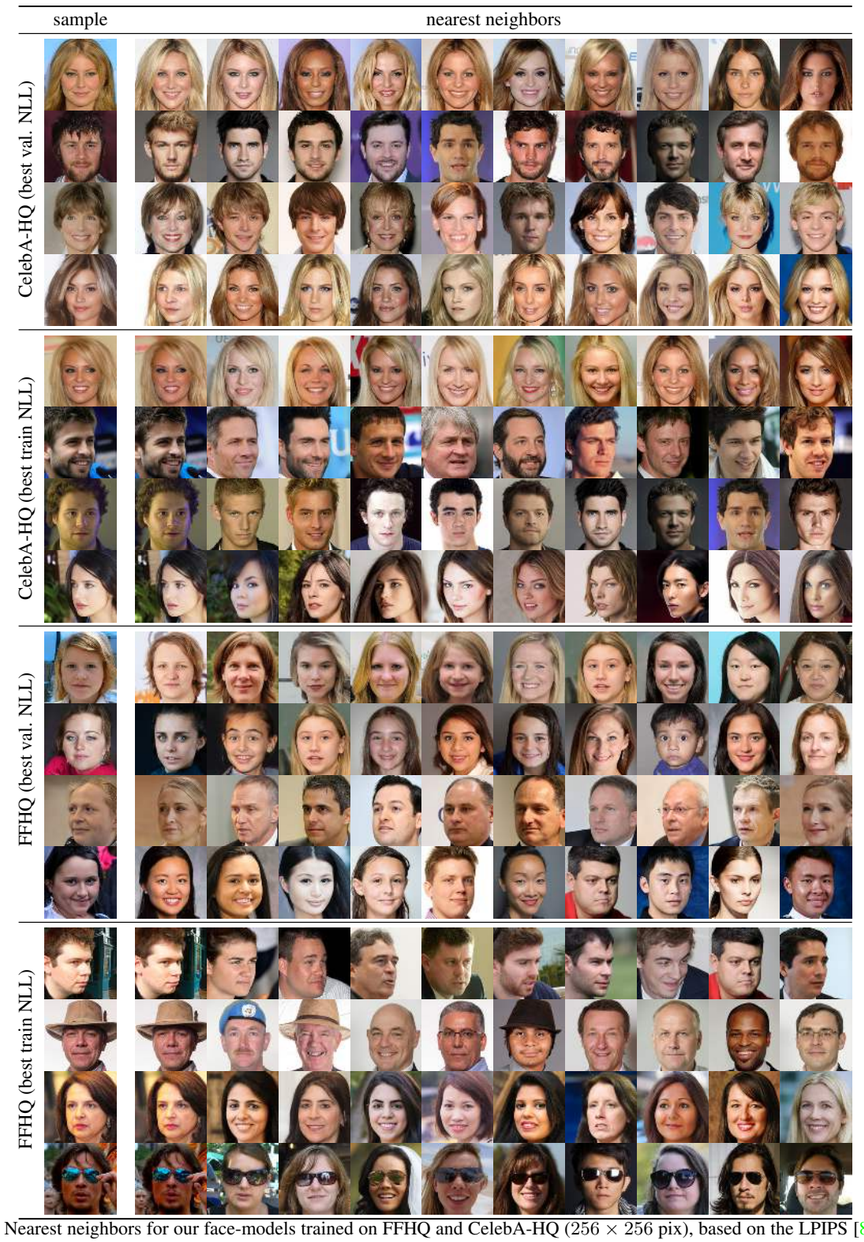


Figure 45. Nearest neighbors for our face-models trained on FFHQ and CelebA-HQ (256×256 pix), based on the LPIPS [82] distance. The left column shows a sample from our model, while the 10 examples to the right show the nearest neighbors from the corresponding class (increasing distance) in the training dataset. We evaluate two different model checkpoints for each dataset: Best val. NLL denotes the minimal NLL over the course of training, evaluated on unseen testdata. For this checkpoint, both models generate crisp, high-quality samples not present in the training data. However, when drastically overfitting the model, it reproduces samples from the training data (best train NLL). Although not an ideal measure of image quality, NLL thus provides a proxy on model selection, whereas FID does not. See also Sec. E.

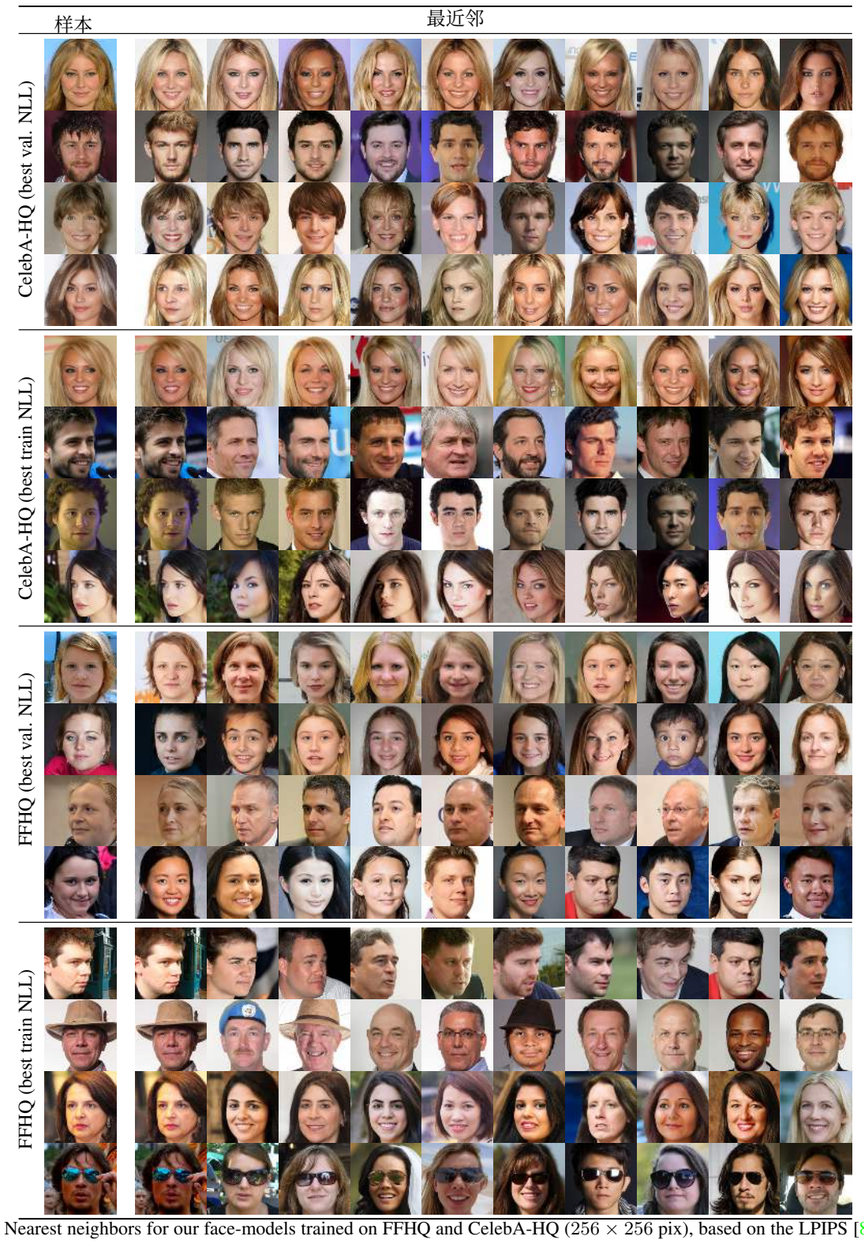


Figure 45. Nearest neighbors for our face-models trained on FFHQ and CelebA-HQ (256×256 pix), based on the LPIPS [82] distance. The left column shows a sample from our model, while the 10 examples to the right show the nearest neighbors from the corresponding class (increasing distance) in the training dataset. We evaluate two different model checkpoints for each dataset: Best val. NLL denotes the minimal NLL over the course of training, evaluated on unseen testdata. For this checkpoint, both models generate crisp, high-quality samples not present in the training data. However, when drastically overfitting the model, it reproduces samples from the training data (best train NLL). Although not an ideal measure of image quality, NLL thus provides a proxy on model selection, whereas FID does not. See also Sec. E.

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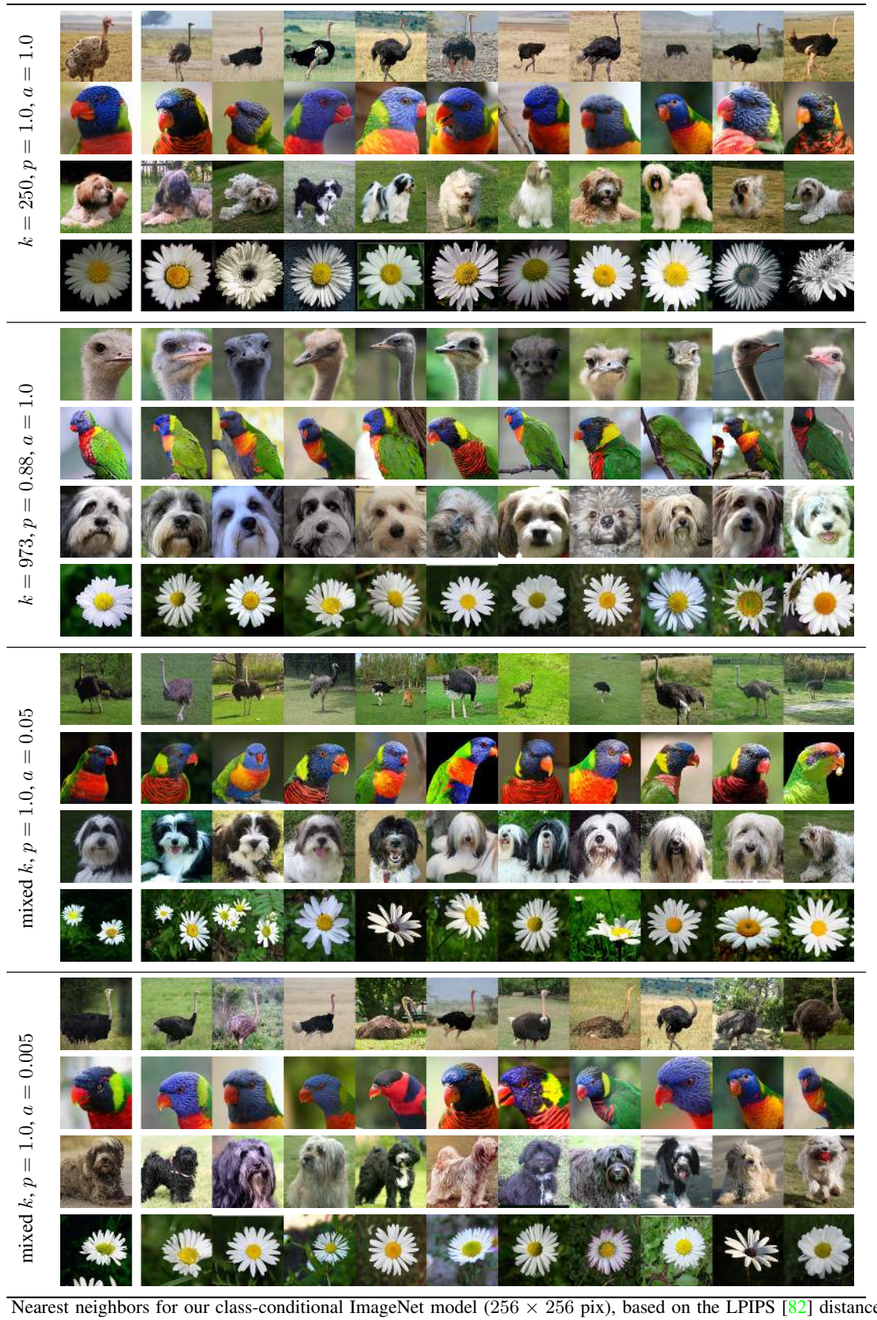


Figure 46. Nearest neighbors for our class-conditional ImageNet model (256 × 256 pix), based on the LPIPS [82] distance. The left column shows a sample from our model, while the 10 examples to the right show the nearest neighbors from the corresponding class (increasing distance) in the training dataset. Our model produces new, unseen high-quality images, not present in the training data.

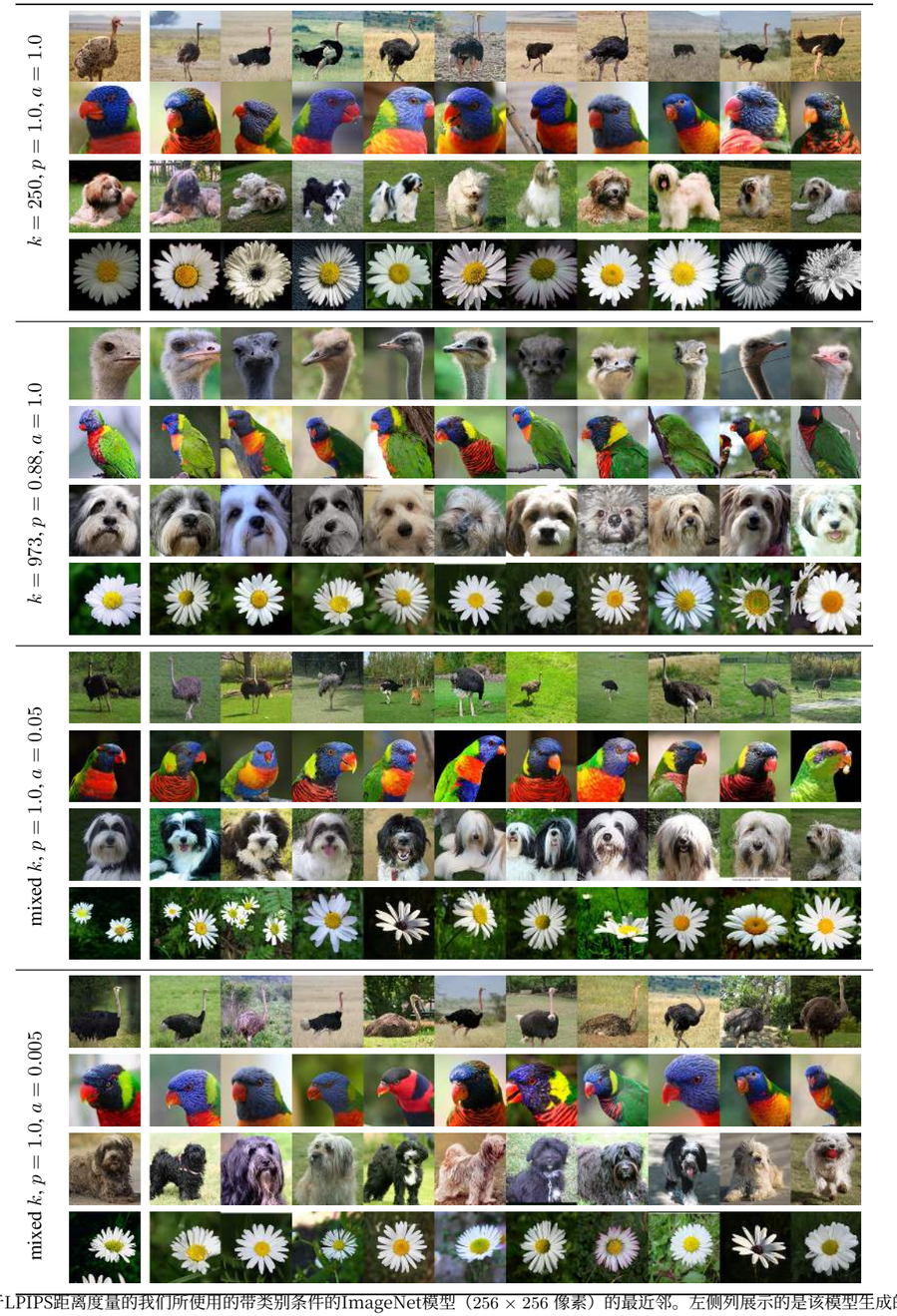


图46. 基于LPIPS距离度量的我们所使用的带类别条件的ImageNet模型（256 × 256 像素）的最近邻。左侧列展示的是该模型生成的样本，而右侧的10个示例则分别是训练数据集中同一类别中距离最近的样本（按距离由近及远排序）。我们的模型能够生成训练数据中不存在的、全新的高质量图像。

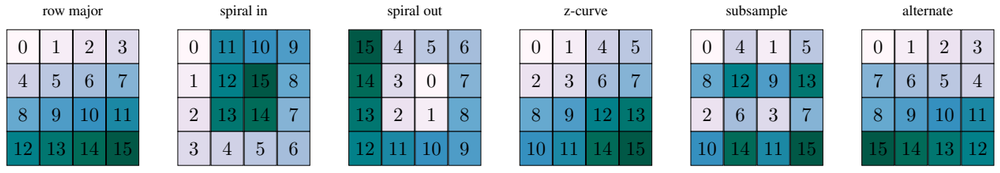


Figure 47. Top: All sequence permutations we investigate, illustrated on a 4×4 grid. Bottom: The transformer architecture is permutation invariant but next-token prediction is not: The average loss on the validation split of ImageNet, corresponding to the negative log-likelihood, differs significantly between different prediction orderings. Among our choices, the commonly used row-major order performs best.

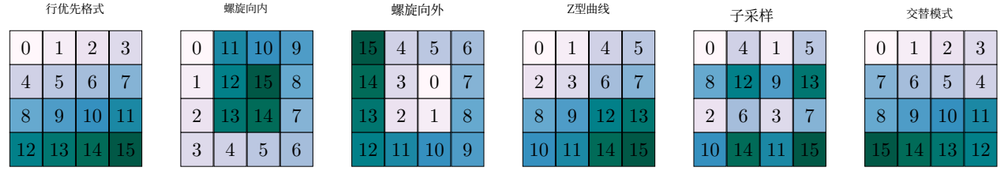


图47。上方：我们研究的所有序列排列方式，展示在 4×4 网格上；下方：变换器模型本身对排列具有不变性，但下一个token的预测则不具备此特性——在ImageNet数据集的验证集上，对应于负对数似然的平均损失会因不同的预测顺序而出现显著差异。在各种可选顺序中，常用的行主序表现最佳。

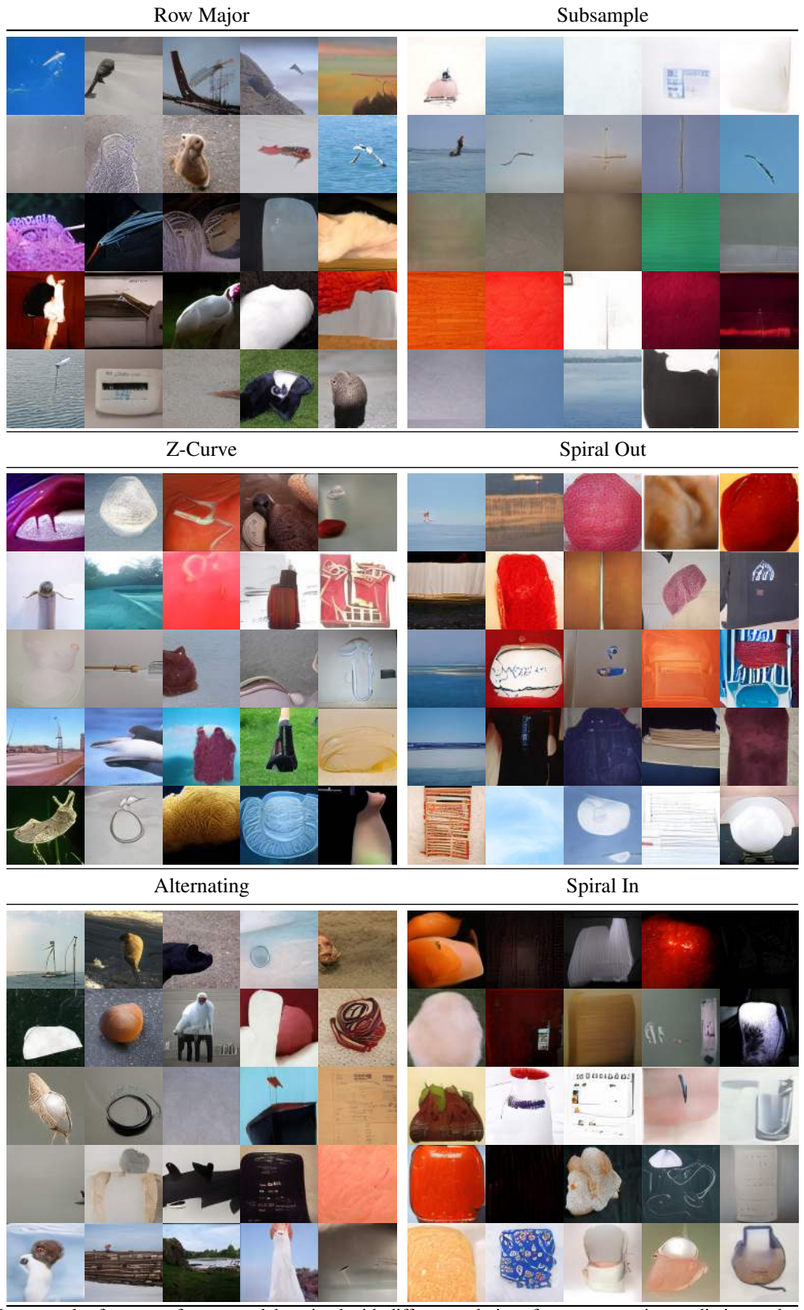


Figure 48. Random samples from transformer models trained with different orderings for autoregressive prediction as described in Sec. F.

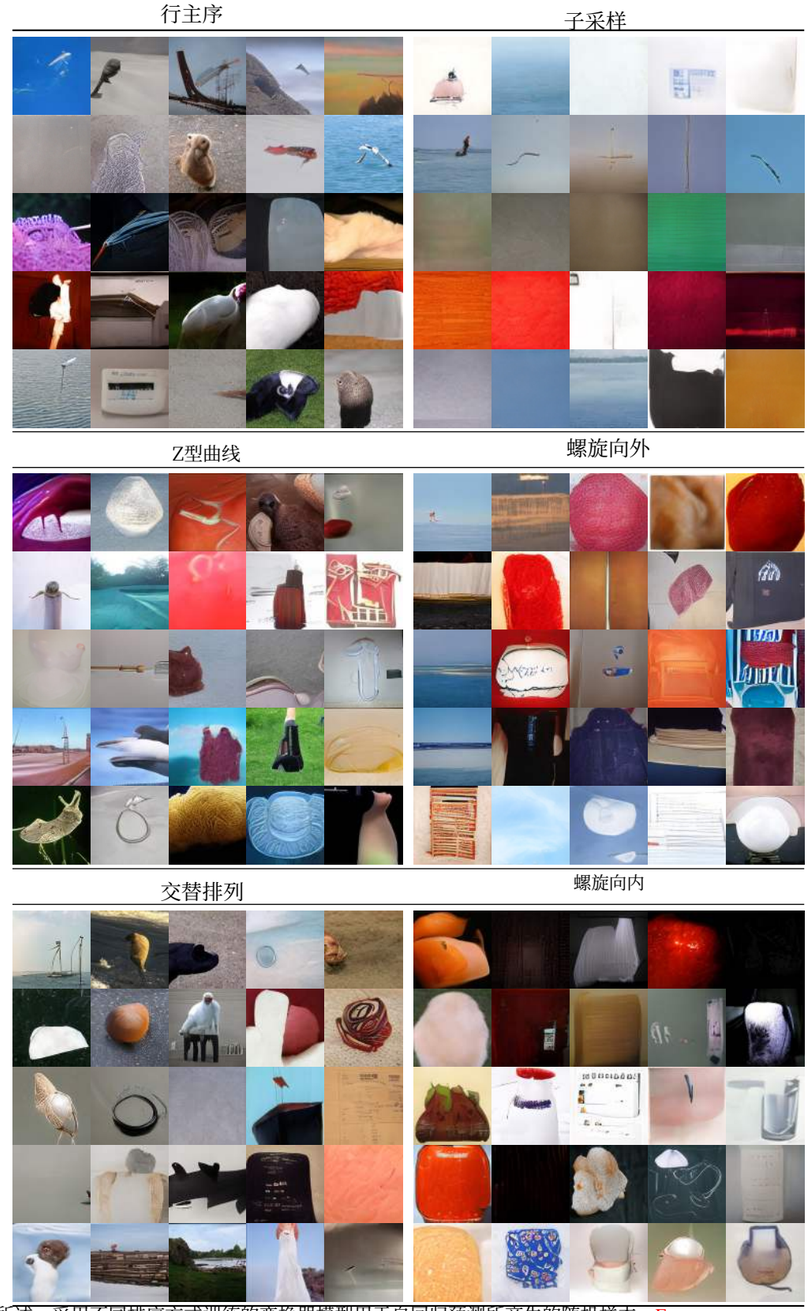


图48. 如第节所述，采用不同排序方式训练的变换器模型用于自回归预测所产生的随机样本。F。

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